

North American Bat Monitoring Program in British Columbia

Data Summary and Activity Trend Analyses: 2017 - 2025

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1 Executive Summary

The North American Bat Monitoring (NABat) program is a multi-agency initiative administered by the US Geological Survey (USGS). In 2024, the North by Northwest (NNW) Bat Hub was formed by combining the former Alberta Hub and the BC and Southeast Alaska (BC-SE_AK) Hub. The NNW Bat Hub leads the NABat program within British Columbia, with key support from provincial and state representatives and WCSC.

In 2025, 57 grid cells throughout British Columbia were monitored. Two statistical analysis methods were used complimentary to evaluate the trends of bat species across the province. Results indicate that bat populations across British Columbia remain stable, with some key species showing small positive trends, including Little Brown Myotis and Eastern Red Bat. Given growing threats to bat populations in BC, continued monitoring is essential to detect any significant changes.

2 Land Acknowledgement

Biodiversity Pathways respectfully acknowledges that our work takes place on Treaty 8 and Douglas Treaties Territories as well as the traditional and unceded territories of First Nations and Métis Peoples across all regions of British Columbia, whose histories, languages, and cultures are deeply connected to the biodiversity we monitor. We acknowledge the traditional teachings of the lands that we work on, and that reciprocal, meaningful, and respectful relationships with Indigenous peoples make our work possible. We are deeply grateful for their stewardship of these lands, and we are committed to supporting Indigenous-led monitoring programs, while learning Indigenous ways of knowing, being, and doing.

3 Introduction

3.1 Overview of NABat and the NNW Bat Hub

North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species to address gaps in knowledge and lack of long-term studies across North America (Loeb et al. 2015). The program is administered by the US Geological Survey (USGS) and implemented by the North by Northwest Bat (NNW) Hub in British Columbia, Alberta and S.E Alaska.

NNW Bat Hub was established in 2024 with collaboration from Government of British Columbia, Government of Alberta, and Government of Alaska. The hub was formed by combining the previous Alberta Hub and the BC and southeast Alaska (BC-SE_AK) Hub. The implementation of the program within British Columbia has key support from Wildlife Conservation Society Canada (WCSC), government staff, other wildlife biologists, Indigenous Nations, community members and naturalists across the province.

The centralization of efforts for the hub will expand exciting programs to coordinate more efficiently across jurisdictions, ensuring consistent and long-term bat monitoring through standardized data collection, and streamlined bioacoustic data management. It also seeks to enhance engagement with NABat, foster collaboration among stakeholders, and produce annual reports on regional monitoring.

3.2 Threats to bat populations in BC

Bat populations in British Columbia face a wide range of threats including forestry practices, anthropogenic expansion, and climate change among others. A comprehensive review of these threats and their relative impacts on individual bat species can be found in the BC Bat Action Plan (Team 2024). Of these threats, White-Nose Syndrome and wind energy development have emerged as two of the highest-priority concerns.

3.2.1 White-Nose Syndrome

White-nose Syndrome (WNS) is a deadly fungal disease caused by the fungus *Pseudogymnoascus destructans* (*Pd*) that affects hibernating bats. As of 2026, WNS has been confirmed in nine Canadian provinces including Alberta, and *Pd* has been detected within British Columbia at two locations (Figure 1: CWHC-RCSF (2024)). Continued monitoring of populations in BC is critical as *Pd* and WNS continue to spread throughout the province.

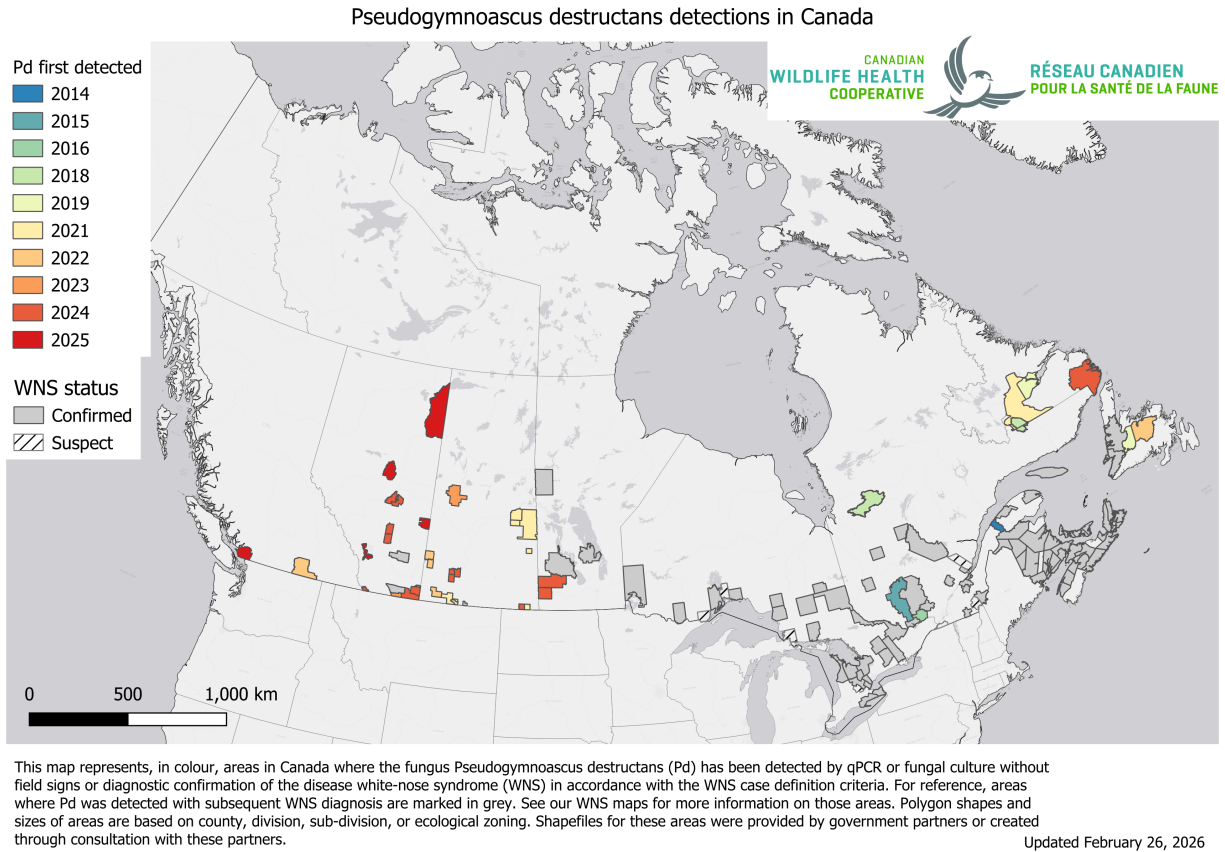


Figure 1: White-nose syndrome (WNS) and *Pseudogymnoascus destructans* (Pd) occurrence in Canada. Source: CWHC-RCSF (2026)

3.2.2 Wind Energy Development

Wind energy farms has been linked to high mortality rates across all bat species in Canada, with long distance migratory species accounting for 73% of recorded fatalities (Zimmerling and Francis 2016). In British Columbia alone, one estimate predicts an annual mortality rate of 680 bats per wind turbine (Zimmerling and Francis 2016). There are currently close to 300 turbines active in the province across 10 different projects with the most turbines found in the Peace region (Figure 2). In late 2024, B.C announced nine new wind projects which would require the installation of over 300 new wind turbines across the province (Energy and Climate Solutions 2024). As of 2023, COSEWIC designated all three of B.C.'s long distance migratory species as endangered due in part to fatalities at wind turbine facilities (Status of Endangered Wildlife in Canada 2023).

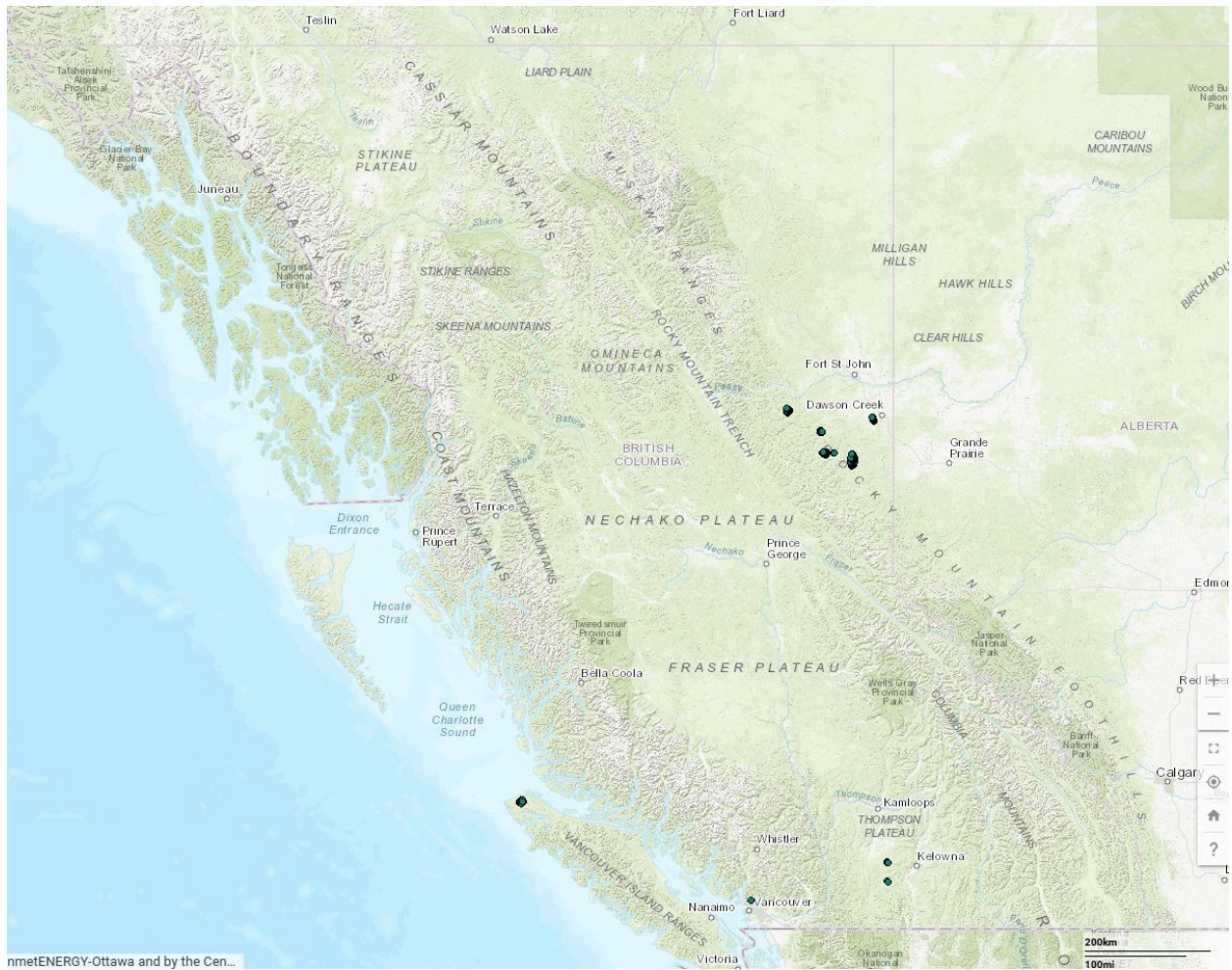


Figure 2: Current wind turbines in British Columbia based on the Canadian Wind Turbine Database (Source: <https://search.open.canada.ca/openmap/79fdad93-9025-49ad-ba16-c26d718cc070>)

3.3 Acoustic Monitoring of Bats

Acoustic recording is a cost-effective method for monitoring bat populations across large geographic and temporal scales. However, several limitations affect the conclusions that can be drawn from these data:

1. **Imperfect species identification** — Unlike birds, bats rely on echolocation rather than song, and species with similar ecology and physiology often produce overlapping calls. While detectors are deployed to maximize recordings of diagnostic Search Phase calls, species identifiability will always vary depending on the local assemblage.
2. **Variable detectability** — Bat species are not equally detectable acoustically. Larger, wide-ranging species produce louder calls that travel farther, while smaller species with quieter calls or specialized habitat use are less reliably recorded. As a result, trend estimates cannot be produced for all species using acoustic methods alone.
3. **Data variability** — Many factors influence recording quality and bat activity, including wind speed, humidity, and local habitat. Additionally, stationary detectors cannot account for individual recapture, meaning acoustic data can only inform relative activity and occupancy.
4. **Processing limitations** — Auto-ID software enables efficient processing of large datasets but has known high misclassification rates for bats, making manual verification necessary to ensure accurate results.

3.4 Updates in Modelling Approaches

Bat monitoring data can be analyzed using different statistical approaches, each with distinct assumptions and outputs. Activity trend models examine changes in relative acoustic activity over time, providing a straightforward measure of whether bat detections are increasing, decreasing, or remaining stable across the monitoring network.

Dynamic occupancy models take a different approach, estimating the probability that a species occupies a given site while explicitly accounting for imperfect detection. Rather than treating a non-detection as an absence, these models distinguish between a species being truly absent and simply going undetected, an important consideration given the variable detectability of bats discussed above. They also track colonization and extinction probabilities across sites over time, providing insight into range shifts and local population dynamics.

The two approaches differ in sensitivity and therefore serve complementary roles. Activity data are inherently noisy, influenced by weather, local foraging conditions, and the number of individuals passing a detector, but they respond quickly to change and can provide an early indication of decline. Occupancy, in contrast, changes only when a species is lost from (or gains) sites, so a stronger and more sustained signal is needed before a trend becomes apparent. Occupancy models can therefore act as a second, more conservative test of patterns first suggested by activity data. When both approaches point in the same direction, the evidence for a real population change is stronger, and confidence in the results is correspondingly higher.

In 2025, both model types were run to examine differences in results and assess the complementary information each approach provides.

4 Methods

4.1 Data Collection

The NABat monitoring program was established in BC in 2016 by WCSC and has sampled 65 grid cells in total. Cells were selected using the GRTSID system (Loeb et al. 2015), based on accessibility, presence of bat habitat, suitable recorder sites, and where possible, a road suitable for a driving transect (30–45 km, relatively straight).

Table 1: Total number active grid cells and number of site nights sampled summarized by calendar year in British Columbia. Site nights are defined as the cumulative number of nights during which recorders were operational

Year	Number of Grid Cells Sampled	Number of BC Parks Grids	Site Nights
2016	22	2	566
2017	27	1	900
2018	47	15	939
2019	49	16	1489
2020	53	17	1530
2021	51	16	1448
2022	56	16	1635
2023	51	16	1436
2024	58	18	1410
2025	57	15	1826

In 2025 a total of 58 grid cells were monitored for a total of 18 site nights (Table 1). Of the cells sampled in 2025 a total of 15 were sampled by BC Parks staff (Table 2). The network is well-distributed across the province, with a slight bias toward southern BC and population centres (Figure 3).

Table 2: Summary of yearly sampling of grid cells in BC Parks.

GRTS Cell ID	Park	Year							
		2016	2017	2018	2019	2020	2021	2022	2023
243562	Kimberley								
6202	Goldstream								
986	Bowron								
38442	Churn Creek								
115322	Tweedsmuir								
145578	Premier Lake								
212906	Horne Lake								
265749	Kettle/Conkle								
283586	Lakelse Lake								

317428	Manning
16426	Carp
37274	Mount Robson
132650	Junction Sheep
172090	Mount Seymour
175354	Liard Hot Springs
230186	Skagit
17706	Wells Gray
26170	Yakoun
206394	Naikoon

4.1.1 Stationary Surveys

Two to four stationary detectors were deployed across at least two quadrants per grid cell for a minimum of seven nights. Sites are selected to maximize species detection and recording quality across different habitat types (see (Rae and Lausen 2024) for full site selection criteria). Deployment occurs between mid-May and mid-July, after bats have emerged from hibernation but before young bats are volant. Sites and timing are kept consistent between years to minimize variability in conditions affecting bat activity.

4.1.2 Transect Surveys

Where possible, driving transects were conducted following Loeb et al. (2015). Two replicates of a 30–35 km route are driven at 30–35 km/h with a rooftop detector, using straight roads that minimize switchbacks. Transects are conducted concurrently with stationary deployments.

4.1.3 Equipment

A wide range of equipment is used across sampling locations in BC. Stationary units are primarily Wildlife Acoustics SM4Bat-FS or SM2Bat units (Maynard, MA), alongside an increasing number of Titley Scientific Swift and Express units (Brisbane, AU). Mobile transects are conducted using either the Wildlife Acoustics EchoMeter Touch2Pro or Titley Scientific Anabat units (SD1/2 zero-cross or Walkabout models).

2025 represents a transition year for equipment standardization. All SM2Bat units are being replaced with Titley Scientific Chorus units, and Anabat units are being phased out in favour of the EchoMeter Touch2Pro for mobile transects. To account for detection differences between outgoing and incoming equipment, paired deployments were conducted at a number of grid cells this year, both stationary (SM2Bat and Chorus) and mobile (Anabat SD1 and EchoMeter Touch2Pro). These paired deployments will provide correction factors to ensure data comparability across the transition period.

4.1.4 Landscape and Weather Covariates

We used the same covariates for the activity trend models as Rae and Lausen (2022). Where possible, temperature and relative humidity were recorded in each grid cell with onsite data loggers. Where loggers were unavailable, values came from the nearest climate station in the Pacific Climate Impacts Consortium (PCIC) [Provincial Climate Data Set \(PCDS\)](#), which aggregates and quality-controls observations from multiple provincial and federal networks. From these station records we also derived nightly mean wind speed (km/h) and nightly total precipitation (mm). Degree days above 10 °C since January 1 were calculated from daily temperature records for each site-night.

Sun and moon positions were calculated with the `astral` package (v3.2) in Python. Sunset and sunrise times defined the night period for each site-night. Moon illumination was then calculated as the proportion of the lunar disc illuminated multiplied by the proportion of the night the moon spent above the horizon. Site elevation was extracted from the [Open-Meteo Elevation API](#). Distance to clutter and percent clutter were recorded by grid leaders at deployment, and distance to water and water body type were derived from the [BC Freshwater Atlas](#).

For the dynamic occupancy models, we explored additional factors that might influence colonization and extinction: distance to the nearest road and distance to the nearest timber harvest, each measured from the survey location to the closest feature. Road features came from the Human Footprint Index (HFI) layer under development by the Geospatial Centre at Biodiversity Pathways (unpublished data). Harvest features came from the Province of British Columbia's [consolidated cutblocks layer](#). All continuous covariates were standardized by centering on the mean and dividing by the standard deviation. Missing values were imputed with the global mean before standardization.

Since 2022, insect activity has also been sampled along transect routes using vehicle-mounted sticky traps. Samples are identified to order, and we will evaluate insect abundance as a potential covariate of bat activity in future years.

4.2 Acoustic Processing

In 2025 acoustic analysis mostly followed the standards set in previous years (Rae and Lausen 2022, 2024). This included running all files through two automatic classifiers, Kaleidoscope's Bats of North America 5.4.0 classifier and Sonobat 3.0's Northwestern British Columbia classifier, to remove files that do not contain bat pulses and giving a preliminary bat species identification.

Manual vetting largely follows the protocol of Reichert et al. (2018a) and previous analysis standards developed by WCSC (Rae and Lausen 2022). The key difference in the 2024 and 2025 acoustic analysis compared to previous years is that not all automatic species identifications were manually verified. Instead, based on findings from Stratton et al. (2022), at least 50% of recordings were selected for manual verification.

4.2.1 Subsetting for manual verification

Starting in 2024, the NNW Bat Hub implemented manual vetting of only a subset of all the bioacoustic data collected under the NABat program. Given the project's large scale, manually vetting all recordings is not feasible long-term. To address this, the NNW Bat Hub, in collaboration with WCSC and contractor Brian Paterson, developed procedures to ensure that selecting a subset

of recordings for manual review minimizes biases in analytical results. These standards prioritize the accurate identification of rare and difficult-to-recognize species to maintain the integrity of models and analyses.

To ensure data accuracy, manual verification was conducted based on the number of tagged recordings at each site. For sites with 200 or fewer tagged recordings, all non-noise recordings were manually reviewed. For sites with more than 200 tagged recordings, verification depended on the proportion of recordings assigned a species label. All tags for rare species (comprising 10% of prior years' identifications) and commonly misidentified species were fully vetted. Specifically, all Eastern Red Bat (*Lasiurus borealis*) tags were manually reviewed due to frequent misclassification as Little Brown Bats (*Myotis lucifugus*). Additionally, recordings of common species were randomly selected for verification until 50% of total recordings were reviewed. If fewer than 50% of recordings had species tags, all classifier-assigned tags were manually verified, and "NoID" recordings were randomly selected until 50% of all recordings were reviewed.

4.3 Statistical Analyses

4.3.1 Stationary Activity Trends

Stationary activity trends followed closely what was established by WCSC and Dr. Carl Schwarz in 2021 (Rae and Lausen 2022). This included conducting trend analyses examining activity patterns in the data collected to date using negative binomial GLMM models with a set of species-specific covariates (Table 3). Species-specific covariates for stationary models were selected using AIC-based stepwise selection (Table 3).

Table 3: Top-performing stationary models selected by AIC-based stepwise selection. Definitions of covariates are: Temperature = nightly temperature (°C); Wind speed = nightly mean wind speed (km/h); RH = nightly mean relative humidity (%); dd = cumulative degree days above 10 °C since January 1; Moon = proportion of the moon illuminated $\times$ proportion of the night the moon was above the horizon; Clutter = distance (m) to the nearest clutter object; Nearby water = whether the detector was within 100 m of water (1 = yes, 0 = no); $I(\text{YearS}^2)$ = quadratic year term allowing for non-linear temporal trends. $A * B$ denotes the main effects of A and B plus their interaction.

Species	Species Code	Top Performing Model Covariates
Pallid Bat	ANPA	Nightly Mean Temperature
Townsend's Big-eared Bat	COTO	Water Nearby, Nightly Min Temperature, dd
Big Brown Bat	EPFU	Nightly Mean Wind Speed, Moon*Distance to Clutter, Nightly Min Temperature
Spotted Bat	EUMA	Nightly Min Temperature
Eastern Red Bat	LABO	Nightly Mean Wind Speed* Distance to Clutter, Water Nearby, Nightly Mean Temperature
Hoary Bat	LACI	Moon* Distance to Clutter, dd, Nightly Mean Temperature, $I(\text{YearS}^2)$

Silver-haired Bat	LANO	Nightly Mean RH, Nightly Mean Wind Speed* Distance to Clutter, Water Nearby, Nightly Min Temperature
California Myotis	MYCA	Nightly Mean Wind Speed* Distance to Clutter, Water Nearby, Nightly Mean Temperature
Western Small-footed Myotis	MYCI	Nightly Mean Wind Speed* Distance to Clutter, Water Nearby, Nightly Mean Temperature
Long-eared Myotis	MYEV	Nightly Mean Wind Speed* Distance to Clutter, Water Nearby, dd, Nightly Min Temperature
Little Brown Myotis	MYLU	Nightly Mean Wind Speed* Distance to Clutter, Water Nearby, Nightly Mean Temperature
Northern Myotis	MYSE	Nightly Mean RH, Moon* Distance to Clutter, Water Nearby
Fringed Myotis	MYTH	dd
Long-legged Myotis	MYVO	Nightly Mean Wind Speed* Distance to Clutter, Water Nearby, Nightly Mean Temperature
Yuma Myotis	MYYU	Nightly Mean Wind Speed* Distance to Clutter, Water Nearby, dd

All models followed the same basic structure, modelling the total number of detected calls as a negative binomial response with log link:

```
Total detections ~ YearS + log(Sample Nights) + (1|Grid Cell ID) + (1|Quadrant)
+ species specific covariates
```

Where total detections is the total number of detected calls of a given species at a site in a given year. YearS is the standardized survey year (years since 2016) and provides the estimate of the temporal activity trend. $\log(\text{Sample Nights})$ is the natural logarithm of the number of detector-nights and accounts for variation in sampling effort among site-years. The random intercepts (1|Grid Cell ID) and (1|Quadrant) allow baseline activity to vary among grid cells and among quadrants within them, respectively.

Post-hoc tests were run on the final models to ensure proper model fit. Species were excluded from the analysis if model fit was not appropriate due to issues including under-dispersion or standard errors that were more than 3 times the estimate. Model results were also excluded if there was less than 150 recordings across all sample years. We conducted our statistical analyses on provincial and regional scales. For the regional analyses, we assigned all sampled grid cells into four geographic regions (Southwestern, Southcentral, Southeastern, and Northern), each roughly corresponding to different species sets (Figure 3).

4.3.1.1 Use of Automated Identification Results in Stationary Trend Analyses

Stationary trend analyses used AutoID classifications from Kaleidoscope Pro. Manual verification was applied only to flag non-bat noise; where manual review suggested a different species than the AutoID label, the original AutoID classification was retained.

This approach was first identified as necessary in 2023 (Rae and Lausen 2022), when analyses of manually verified data revealed a consistent positive trend across all species, this was interpreted as an artifact of reviewer learning over time rather than a true biological signal. As verifier experience increased across years, calls were increasingly assigned to specific species rather than left unidentified, producing an apparent increase in detections independent of actual bat activity. Because this observer effect is difficult to control for statistically, AutoID classifications are now used as the standard for all activity trend analyses, as automated classifiers apply consistent criteria across all years. While AutoID is subject to some misclassification, the error rate is consistent across years and does not confound detection of genuine temporal trends.

4.3.2 Transect Abundance Models

The province-wide transect abundance trend models used only manually verified detections. Transects are surveyed from a moving vehicle, so each bat is likely recorded only once as the vehicle passes. Call counts therefore serve as an index of relative abundance rather than of activity. When a detection was identified as a species couplet, it was counted toward each species in the couplet.

All models had the same basic structure: total detections were modelled as a negative binomial response with a log link:

```
Total detections ~ YearS + log(transect length) + (1|Grid Cell ID)
+ species specific covariates
```

Total detections is the number of calls of a given species detected on a transect in a given year. YearS is survey year expressed as years since 2016, and provides the estimate of the temporal activity trend. $\log(\text{transect length})$ is the natural logarithm of the total length of the transect sampled and accounts for variation in sampling effort among transects. The random intercept (1|Grid Cell ID) allow baseline activity to vary among grid cells.

Covariates were limited to weather variables, because landscape-level covariates cannot be represented meaningfully for a survey that moves through the landscape. Species-specific covariates for transect models were selected based on those previously established in Rae and Lausen (2022) and were selected using AIC-based stepwise selection (Table 4).

Since transects are only conducted in roughly 50% of our grid cells on a typical year and each transect records fewer bats per night than stationary monitoring, we have elected not to conduct regional-scale transect trend analyses due to low sample sizes.

Table 4: Top-performing transect models selected by AIC-based stepwise selection. Definitions of covariates are: Temperature = nightly temperature (°C); Wind speed = nightly mean wind speed (km/h); RH = nightly mean relative humidity (%); Moon = proportion of the moon illuminated $\times$ proportion of the night the moon was above the horizon.

Species	Species Code	Top Performing Model Covariates
Big Brown Bat	EPFU	Nightly Mean Wind Speed
Hoary Bat	LACI	Nightly Mean Temperature, Nightly Mean Wind Speed
Silver-haired Bat	LANO	Nightly Mean Wind Speed, Moon
California Myotis	MYCA	Nightly Mean Temperature, Nightly Mean Wind Speed, Moon
Western Small-footed Myotis	MYCI	Nightly Mean Wind Speed
Long-eared Myotis	MYEV	Nightly Mean Wind Speed
Little Brown Myotis	MYLU	Nightly Mean Wind Speed, Moon
Long-legged Myotis	MYVO	Nightly Mean RH, Nightly Mean Wind Speed
Yuma Myotis	MYYU	Nightly Mean Wind Speed, Moon

4.3.3 Dynamic Occupancy

We fitted multi-season (dynamic) occupancy models (MacKenzie et al. 2003) to estimate trends in occupancy, colonization, and persistence while accounting for imperfect detection. This analysis used nightly acoustic detection/non-detection data from passive bat detectors deployed at sites in British Columbia (BC).

Initial occupancy (ψ) was modeled as a function of distance to water and elevation. Colonization (γ) and persistence (φ) were modeled as a function of distance to water, elevation, year-varying distance to harvest, and distance to road. Detection probability (p) was modeled as a function of percent clutter, nightly mean temperature, and Julian night.

We excluded 2016 because few sites were surveyed that year. Starting the time series in 2016 would have required the model to estimate initial occupancy () for the many sites not yet sampled, and a poorly informed can propagate through the colonization and extinction estimates and produce false trends.

To reduce temporal autocorrelation between consecutive nights, we subsampled every other night, using a maximum of four nights per site-year as repeat surveys. Sites were included if they had at least two nights of data in a given year and data from at least two years. After filtering, 185 sites in 63 grid cells in British Columbia were retained. Models were fit separately for each of the 15 bat species.

Models were fit in a Bayesian framework using JAGS (Version 4.3.2) via the jagsUI package (Version 1.6.3) in R (Version 4.5.1). We ran 3 chains in parallel for 30,000 iterations each, discarding the first 15,000 as burn-in and thinning the remainder by 10, yielding 9,000 posterior samples for inference. Priors on all regression coefficients were normal with mean 0 and precision 0.368 (equivalent to a standard deviation of approximately 1.65), and the standard deviation of the region-level random effect was assigned a Uniform(0, 5) prior. Convergence was assessed using Rhat diagnostics, with values below 1.1 taken to indicate convergence. Results are summarized using posterior means and 95% Bayesian credible intervals. We calculated an overall occupancy trend metric () as the ratio of estimated occupancy in the final year to that in the initial year; values of > 1 with 95% credible intervals entirely above 1 indicate an increase in occupancy, values < 1 with intervals entirely below 1 indicate a decline, and intervals overlapping 1 indicate an uncertain or stable.

5 Results

2025 marked the tenth year of coordinated NABat sampling done in BC. A total of `r[samplingE$`Number of Grids`[samplingE$Year==2025]` grid cells were sampled with the support of over 60 individuals including contractors, volunteers and staff members from BC Parks, BC WLRS, and BC Ministry of Forests. Grid leaders have also collected a total of 16 transect insect sticky trap samples in 2025, which will be added to previous years of data to explore the addition of them as an added covariate to transects (to be done in 2026?).

5.1 Species Detection

In 2025, all species previously documented in BC were detected. Total acoustic activity varied substantially among grid cells and regions (Figure 3). Figure 3 is an interactive map. Readers can zoom in to see the individual stationary monitoring sites (blue circles), and click a site to open a pop-up with a summary pie chart of total detections by species and year.

5.2 Activity and Abundance Trend Models

We estimated yearly trends in nightly acoustic activity for 11 of BC's 15 resident species. Transect data was sufficient to estimate trends for 9 species in BC (Table 5). Of the stationary models, seven species showed statistically significant positive trends ($p < 0.05$):

- Townsend's Big-eared Bat (0.11 ± 0.05)
- Big Brown Bat (0.07 ± 0.05)
- Long-legged Myotis (-0.00 ± 0.04).
- Yuma Myotis (0.08 ± 0.05).

No significant negative trends were detected in the stationary models. The remaining species with sufficient data had non-significant trend estimates.

Transect models yielded results for fewer species, with only Little Brown Myotis showing a significant positive trend (0.16 ± 0.08). Trend estimates for the remaining species with transect data were small and non-significant.

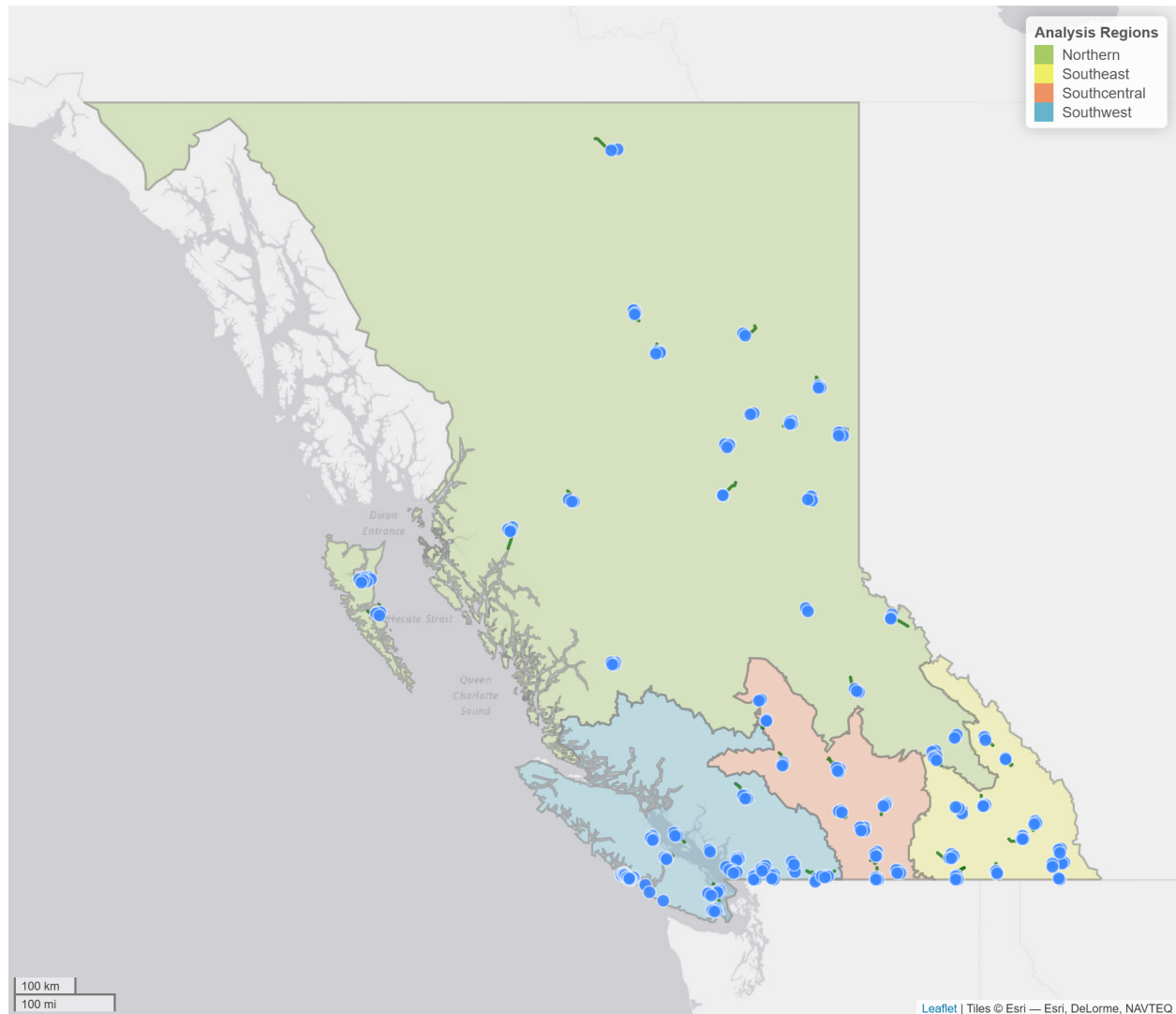


Figure 3: NABat survey efforts in 2025 within British Columbia. Analysis regions were defined by Wildlife Conservation Society Canada (WCSC) based on species ranges maps to group areas with similar species compositions.

Table 5: Provincial stationary and transect trend estimates from 2017-2025 NABat monitoring in BC adjusted for covariates. Trends estimated on a logarithmic scale, thus a trend estimate of 0.05 corresponds to an annual change of 5%. Significant results (p value < 0.05) are highlighted in bold.

Species	Stationary Models		Transect Models	
	Trend Estimate $\pm$ 95%CI	P-value	Trend Estimate $\pm$ 95%CI	P-value
Pallid Bat				
Townsend's Big-eared Bat	0.11 $\pm$ 0.05	<0.001		
Big Brown Bat	0.07 $\pm$ 0.05	0.003	0.06 $\pm$ 0.09	0.150
Spotted Bat	0.12 $\pm$ 0.15	0.114		
Eastern Red Bat				
Hoary Bat	0.08 $\pm$ 0.20	0.441	0.01 $\pm$ 0.13	0.906
Silver-haired Bat	0.04 $\pm$ 0.04	0.062	0.04 $\pm$ 0.07	0.311
California Myotis	0.01 $\pm$ 0.04	0.632	0.06 $\pm$ 0.12	0.331
Western Small-footed Myotis			0.22 $\pm$ 0.26	0.103
Long-eared Myotis	0.09 $\pm$ 0.04	<0.001	0.04 $\pm$ 0.12	0.488
Little Brown Myotis	0.01 $\pm$ 0.04	0.544	0.16 $\pm$ 0.08	
Northern Myotis				
Fringed Myotis	-0.00 $\pm$ 0.05	0.948		
Long-legged Myotis	-0.00 $\pm$ 0.04	0.935	0.12 $\pm$ 0.14	0.094
Yuma Myotis	0.08 $\pm$ 0.05	0.001	0.04 $\pm$ 0.15	0.582

Regional trend analyses revealed significant results ($p < 0.05$) across all regions with the most found in the South Coastal region (Table 6). Most significant results were positive, but significant negative trends were found in:

- Hoary Bat in the South coastal Region (**-0.60 $\pm$ 0.58**)
- Yuma Myotis in the Northern Region (**-0.22 $\pm$ 0.22**).

Table 6: Regional trend estimates from 2017-2024 NABat monitoring in BC adjusted for covariates. Trends estimated on a logarithmic scale, thus a trend estimate of 0.05 corresponds to an annual change of 5%. Significant results (p value < 0.05) are highlighted in bold.

Species	Kootenay Region		Northern Region	
	Estimate $\pm$ 95%CI	P-value	Estimate $\pm$ 95%CI	P-value
Pallid Bat				
Townsend's	0.08 $\pm$ 0.10	0.118		
Big-eared Bat				
Big Brown Bat	0.02 $\pm$ 0.12	0.707	0.02 $\pm$ 0.24	0.881
Spotted Bat				
Eastern Red Bat				
Hoary Bat	0.54 $\pm$ 0.59	0.073	1.05 $\pm$ 1.19	0.084
Silver-haired Bat	0.17 $\pm$ 0.08	<0.001	0.01 $\pm$ 0.14	0.882
California Myotis	0.03 $\pm$ 0.07	0.342	0.01 $\pm$ 0.11	0.829
Western				
Small-footed				
Myotis				
Long-eared Myotis	-0.02 $\pm$ 0.10	0.734	0.01 $\pm$ 0.15	0.934
Little Brown	0.11 $\pm$ 0.06	0.001	-0.05 $\pm$ 0.09	0.308
Myotis				
Northern Myotis				
Fringed Myotis	-0.02 $\pm$ 0.14	0.828		
Long-legged Myotis	0.06 $\pm$ 0.06	0.088	-0.03 $\pm$ 0.11	0.617
Yuma Myotis	-0.09 $\pm$ 0.15	0.222	-0.22 $\pm$ 0.22	0.043

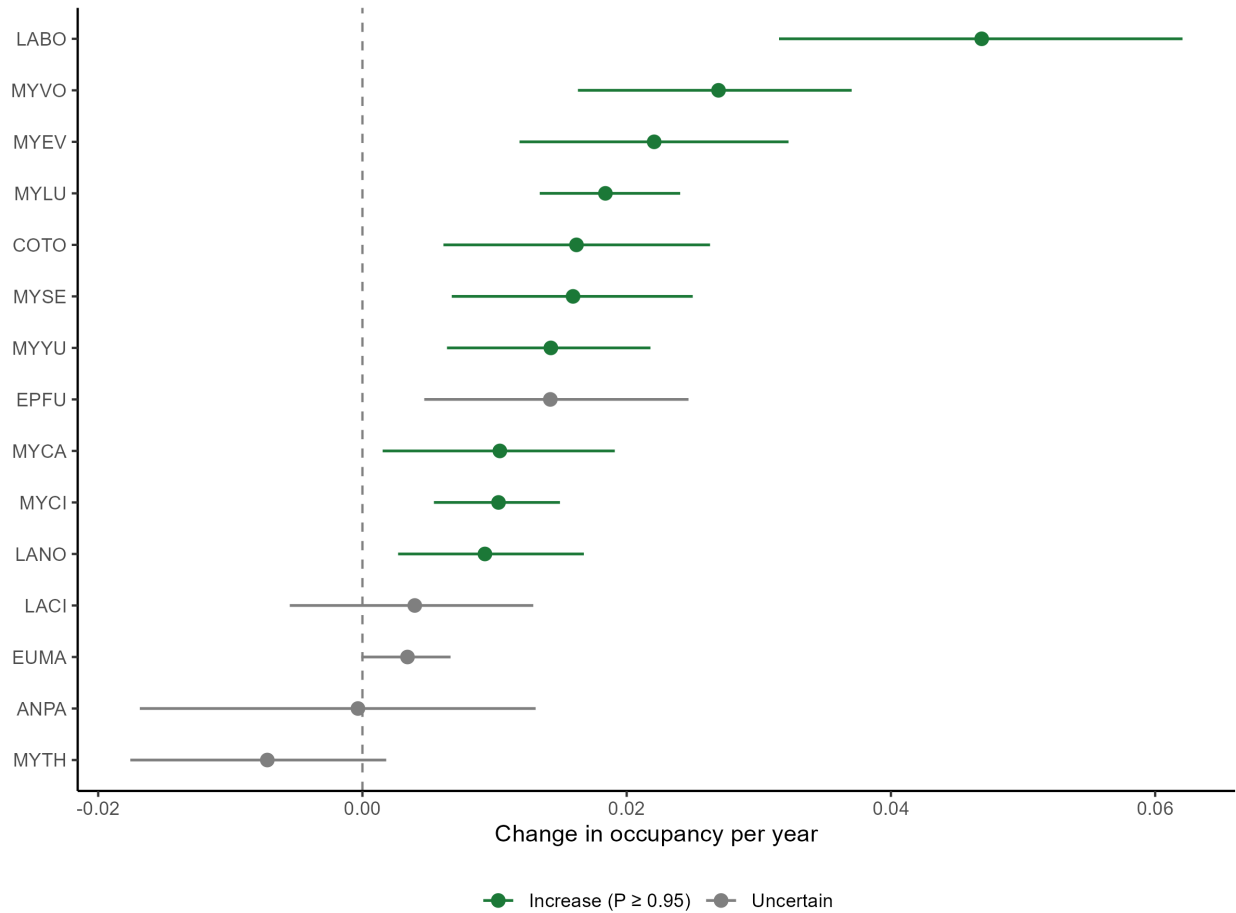
5.3 Dynamic Occupancy Models

Models converged for all 15 bat species in British Columbia.

ANPA and LABO needs to be explored more carefully since there was a big increase in the detection probability of 2025. Need to explore the data more carefully to see the reason for it.

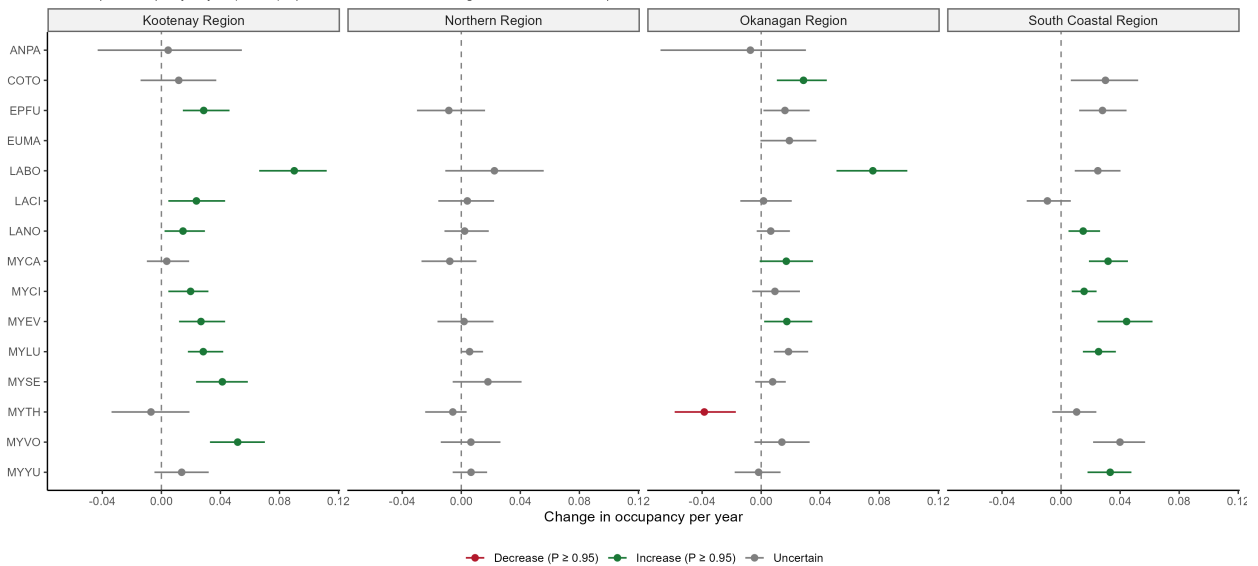
BC-wide occupancy trend by species

Linear slope of occupancy on year (95% CI), all sites. Colour: posterior probability that occupancy in the last year exceeds the first.



Regional occupancy trends by species

Linear slope of occupancy on year (95% CI). Species with no detections in a region are omitted from that panel.



6 Discussion

6.1 Species Presence

Species ranges across the project period (2016–2025) continue to appear broadly stable, with no grid cells showing conspicuous or sustained absences for previously detected species.

6.2 Trend analyses

6.2.1 Activity Trends

—we still have positive trends even though we are using auto ID. How do these compare to Dynamic occupancy?

Overall, results suggest that bat populations remain broadly stable across British Columbia, with no species showing widespread negative trends.

Several species showed statistically significant positive trends in both province-wide and South Coastal regional models. However, these results should be interpreted with caution. Caution is warranted when evaluating the magnitude of the trend estimates. Several models exhibited underdispersion, which, while not unexpected given the structural characteristics of acoustic bat activity data, means that confidence intervals and effect sizes should be interpreted carefully rather than taken at face value.

The significant negative trend detected for Hoary Bat in the South Coastal Region warrants attention and should be monitored closely in coming years to determine whether this represents a genuine localized decline or reflects sampling variability.

More broadly, as threats to bat populations continue to grow, it may be worth exploring complementary analytical approaches alongside activity-based trend models. Acoustic activity metrics carry inherent noise, and detecting meaningful population signals can be challenging, particularly for species with low detection rates or patchy distributions. We recommend exploring occupancy modelling as a complementary approach in future analyses, as it may provide a clearer population signal by reducing the noise inherent in activity-based trend estimates from stationary acoustic data.

6.2.2 Dynamic Occupancy

The dynamic occupancy estimates suggest that the bat community in BC appears to be generally stable to increasing over 2017–2025, with no species showing evidence of decline. Species showing strongest increases are *Myotis* species and LABO. Two species can't be reliably assessed (PAHE, TABR) due to very low detection rates. Distance to road has an effect on colonization and persistence dynamics for several species, with sites closer to roads tending to have higher turnover.

6.3 Challenges and Lessons Learned

6.3.0.1 Timing of Data Retrieval and Submission

Field crews across British Columbia continue to reliably collect recordings for their grid cells. However, organizing and submitting these recordings, along with associated metadata, remains a challenge. Delays in data submission from grid leaders extend processing and analysis times, making it difficult to complete reports by the end of the fiscal year. The transition in project management this year further complicated data submission. We anticipate these challenges will be resolved in future years as teams adjust to the new process. To improve efficiency, we plan to increase reminders to grid leaders and provide additional support to address ongoing challenges in data organization and submission.

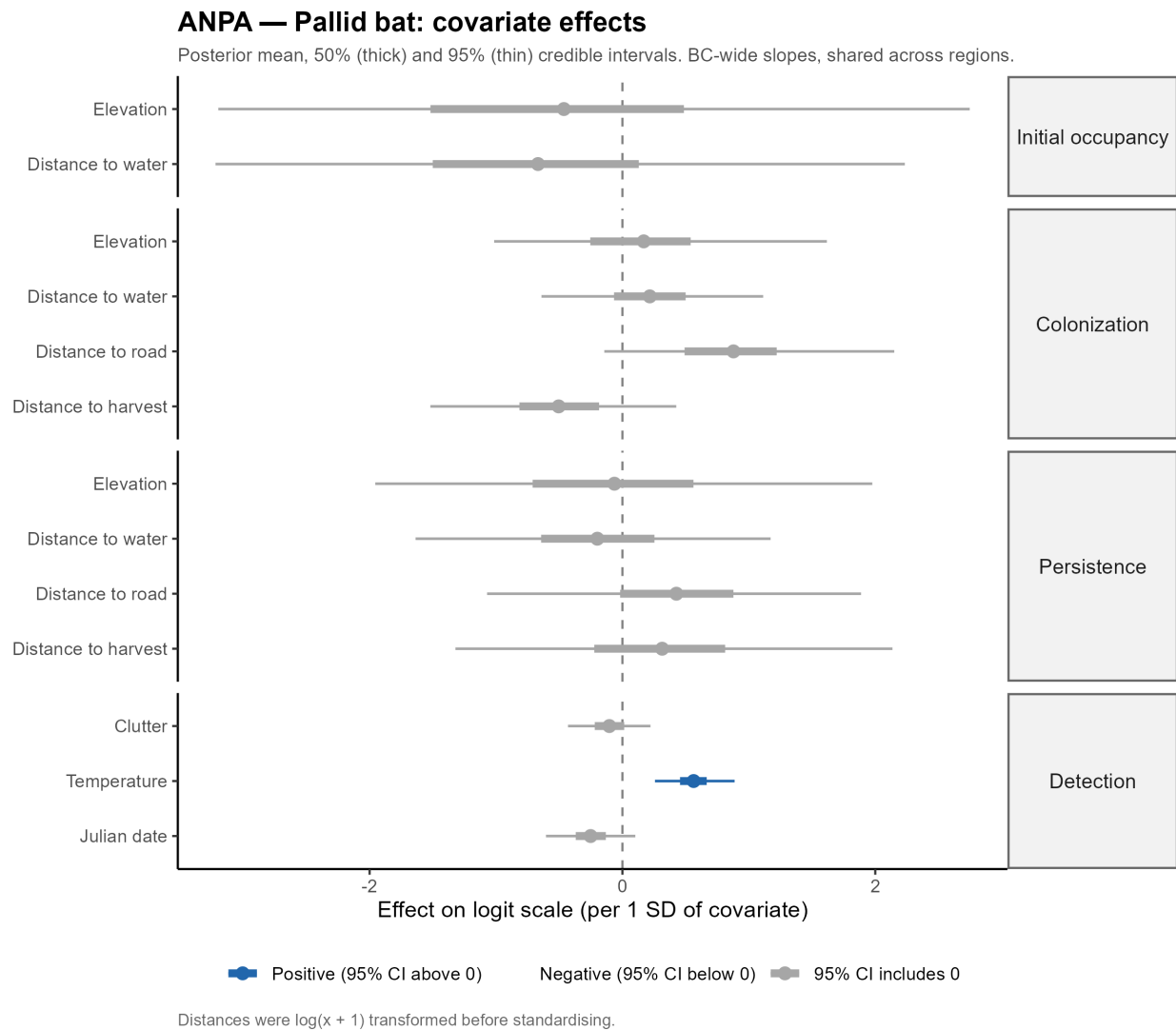
6.4 Looking Forward

The BC NABat program has yielded a large and robust acoustic dataset that provides solid baseline information on distribution and relative abundance of 12 of 15 resident BC bat species. The data that has been collected so far are an invaluable reference point for BC's bat distributions and activity that improve with each additional year of monitoring. Key considerations for the future of the program include:

- **Manual Verification Efficiency.** This was the first year that the BC monitoring program reduced the proportion of manual vetting carried out in the recordings to ~50% based on the findings from Stratton et al. (2022). Careful consideration of these results compared to previous years will be considered as a guide for future models and analyses.
- **Transect Sampling.** Transects are a relatively high labour/effort source of data than stationary detectors that require participants to work late into the night. They also often do not collect many recordings of bats in BC relative to the stationary detectors. However, transects are still the only source of reliable estimates of relative abundance. While continuing to support current transect sampling efforts, the NNW Bat Hub will work along side other partners to also explore other methods of creating reliable abundance estimates that are less labour intensive.
- **Adding new grid cells.** Efforts will continue to be made to monitor as many of the same grid cells as possible, as the power of the dataset improves with each additional year. Moreover, looking for new opportunities to add new grid cells especially in under sampled regions of the province can help fill knowledge gaps for species distributions. New cells could also be added within ranges of species that are not widespread but easily monitored using acoustics. For example, Fringed Myotis and Spotted Bats are highly suited to acoustic surveillance, but only occur within a small portion of the grid cells being monitored currently

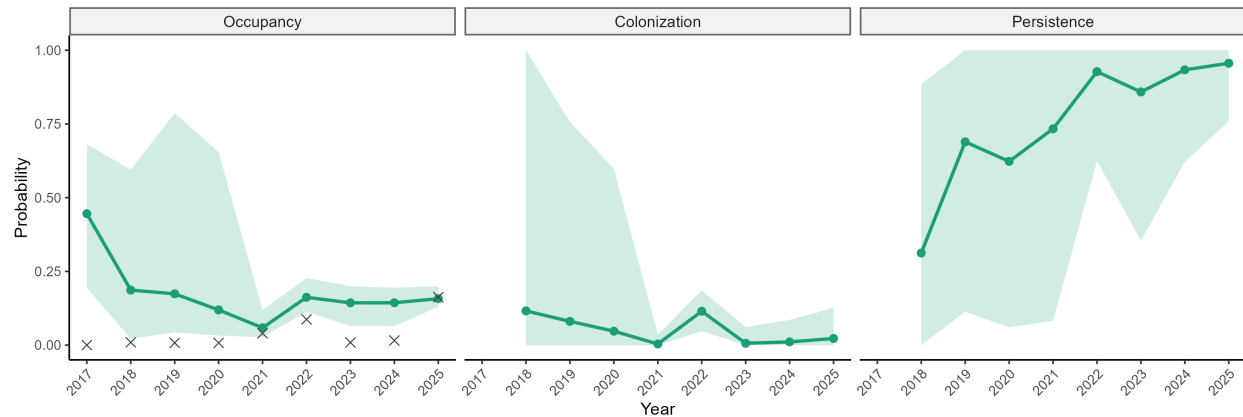
7 Appendix A

7.1 Pallid bat (*Antrozous pallidus*)



ANPA — Pallid bat: BC-wide dynamics

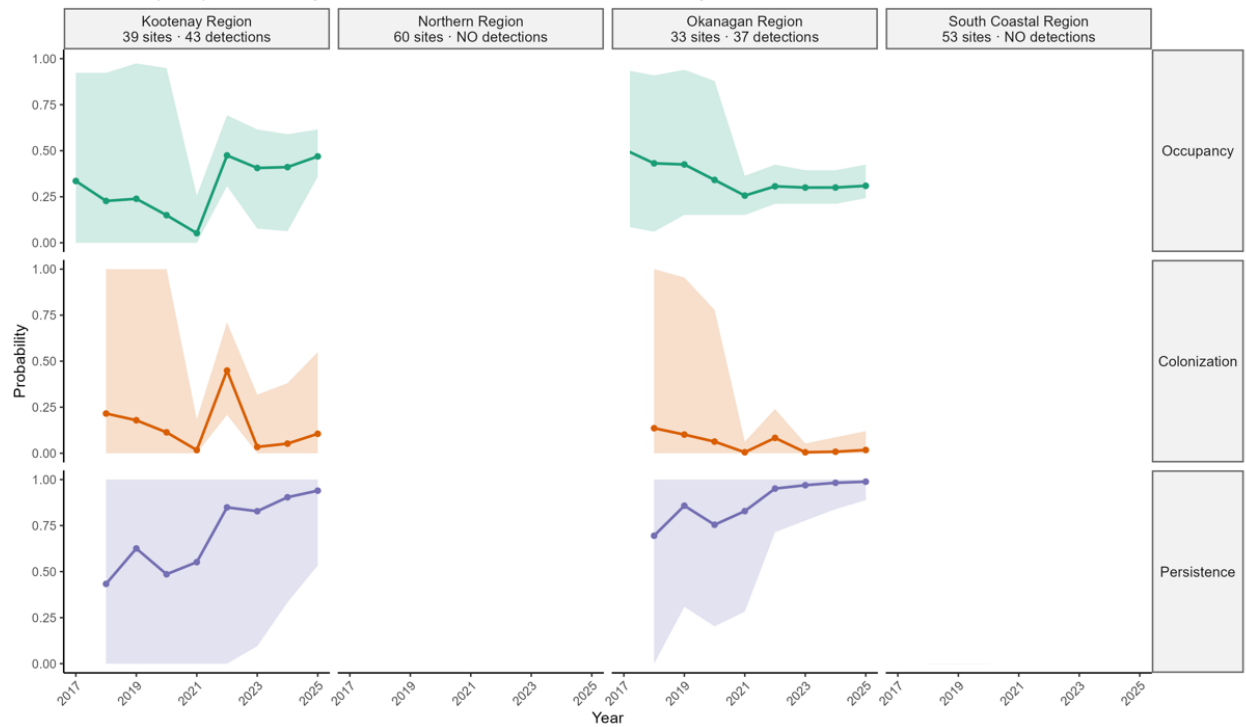
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

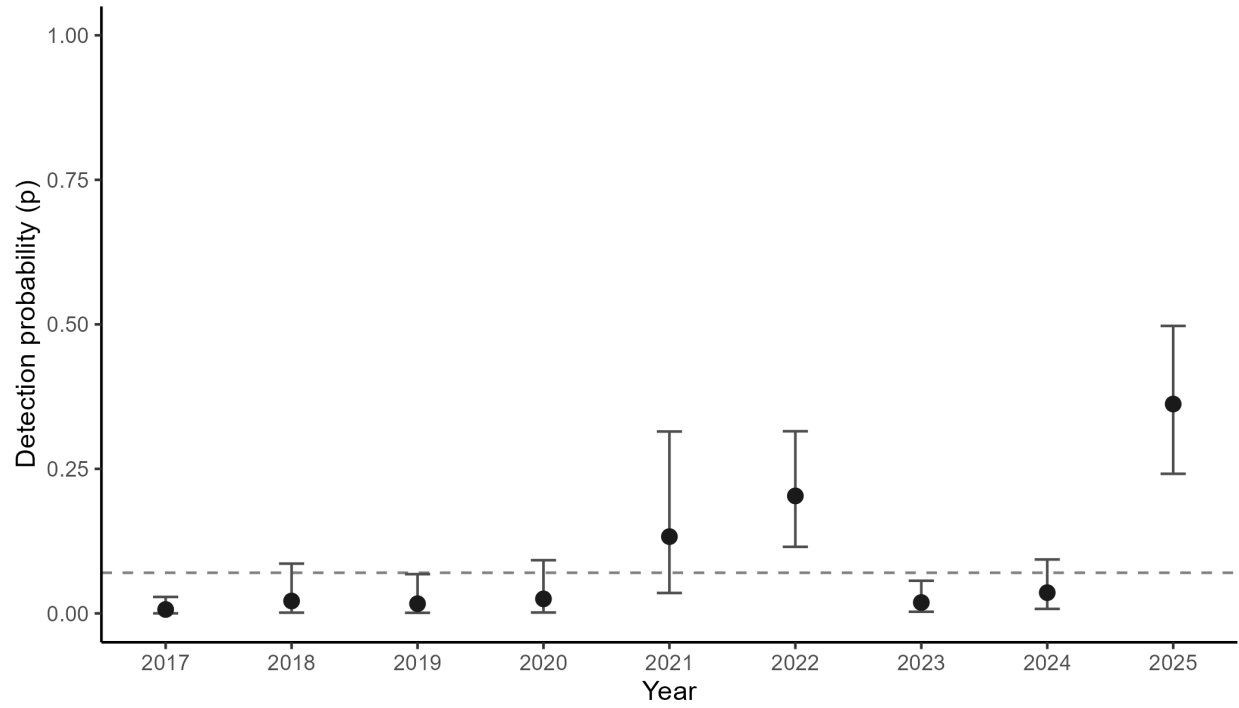
ANPA — Pallid bat: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



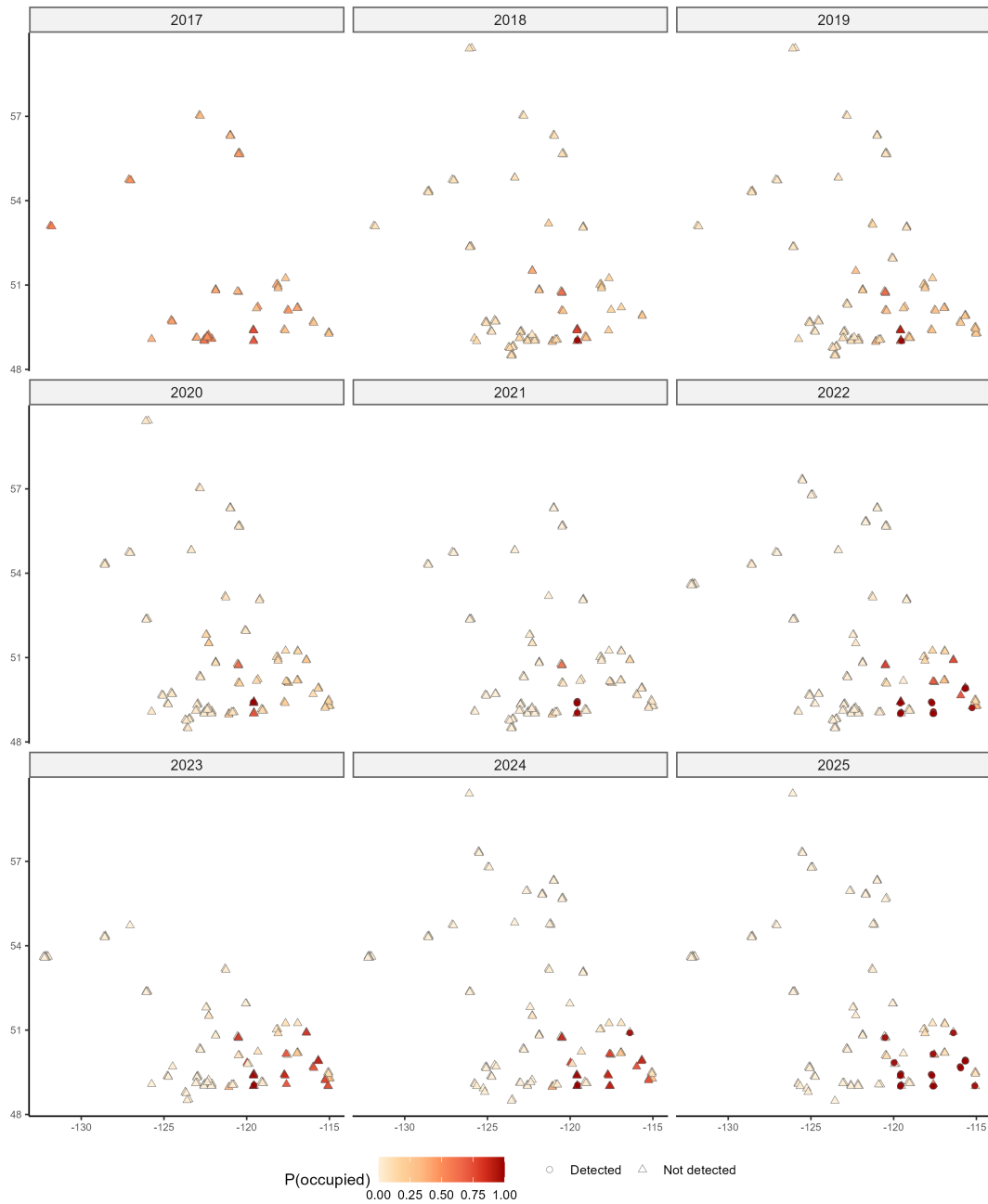
ANPA — Pallid bat: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



ANPA — Pallid bat: P(occupancy) by site and year

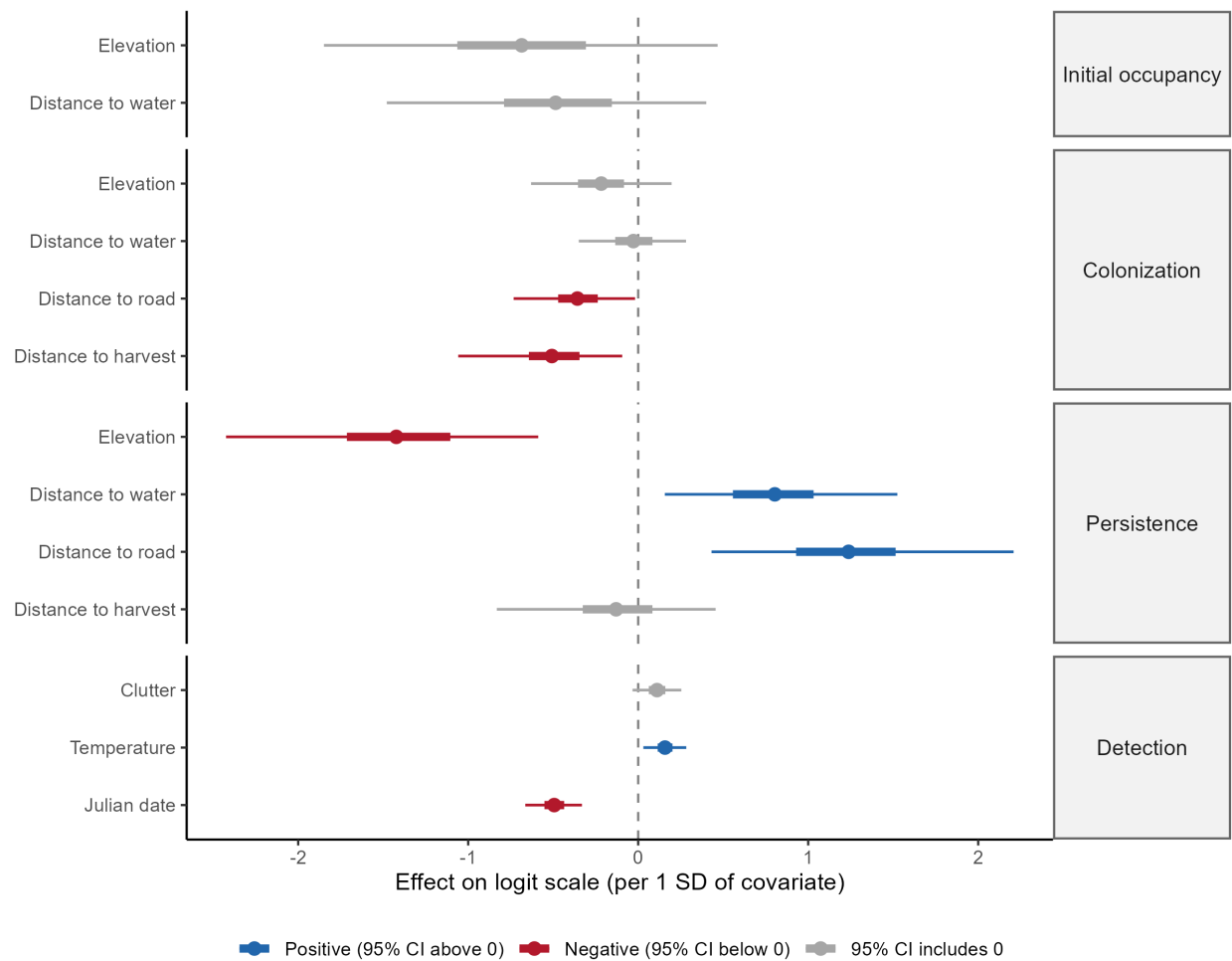
Surveyed site-years only. Triangles = not detected that year; their colour is model inference.



7.2 Townsend’s big-eared bat (*Corynorhinus townsendii*)

COTO — Townsend's big-eared bat: covariate effects

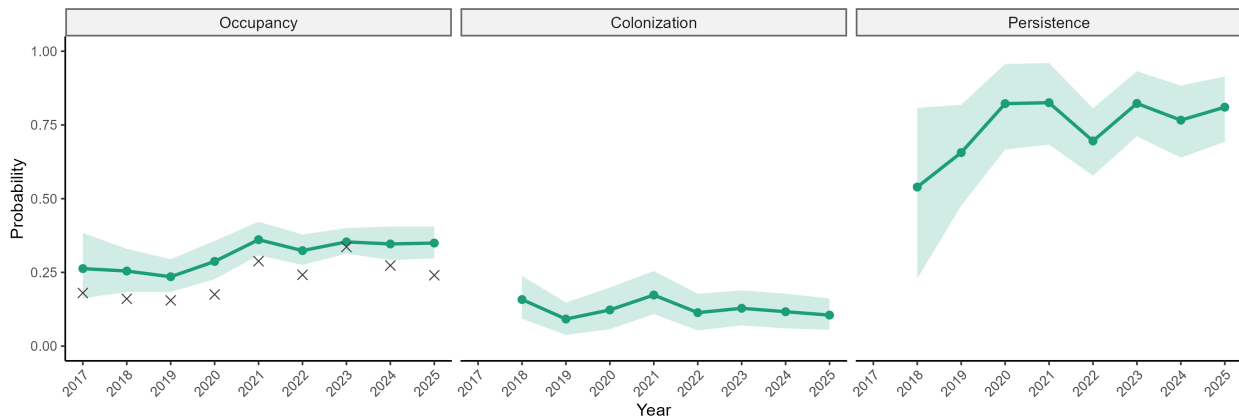
Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



Distances were log(x + 1) transformed before standardising.

COTO — Townsend's big-eared bat: BC-wide dynamics

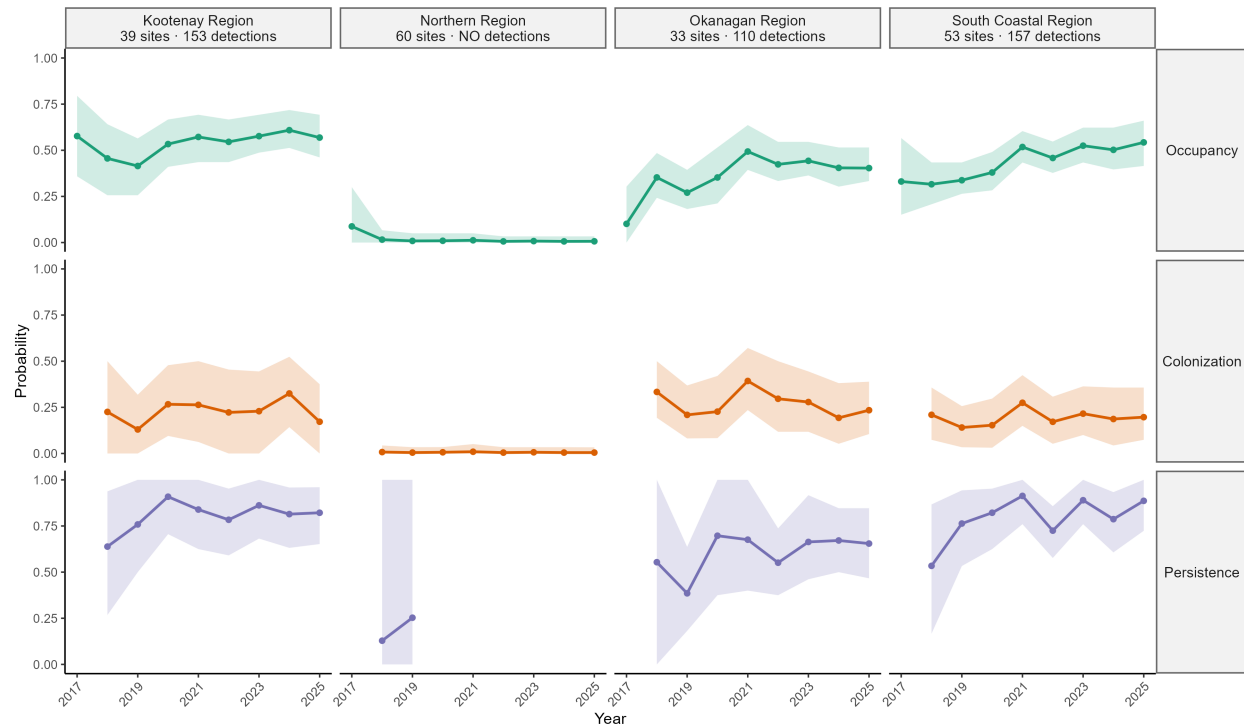
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

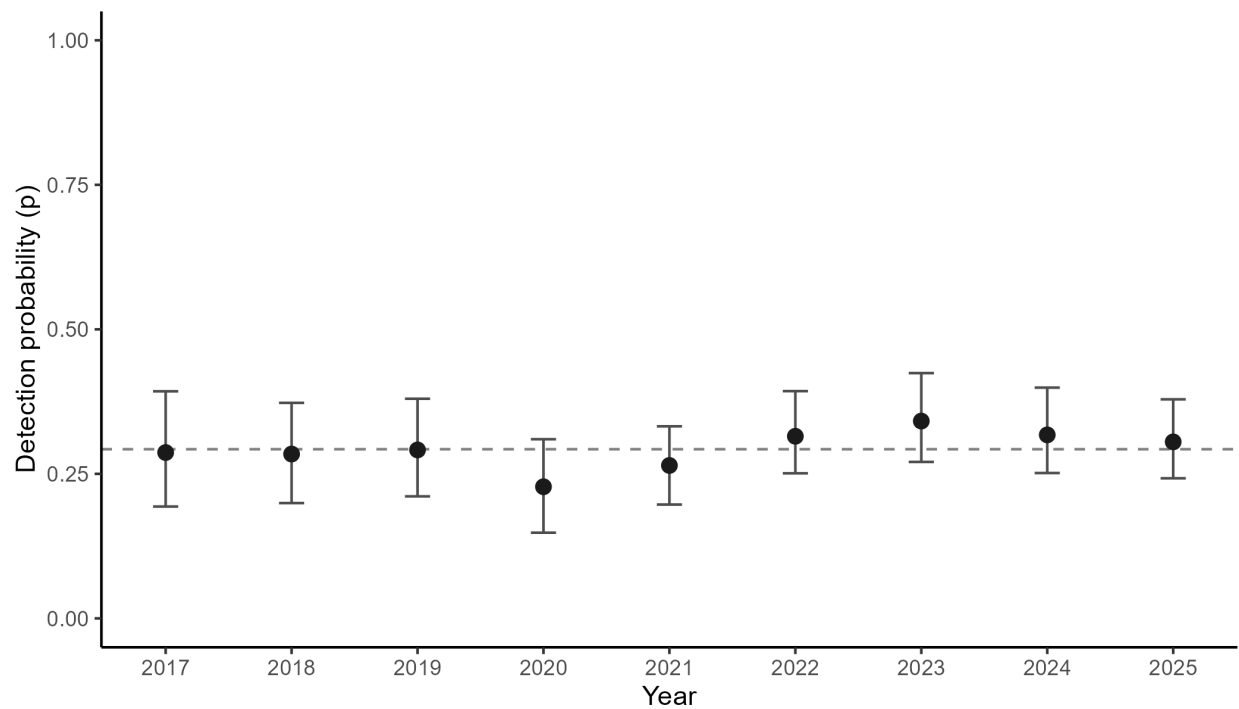
COTO — Townsend's big-eared bat: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



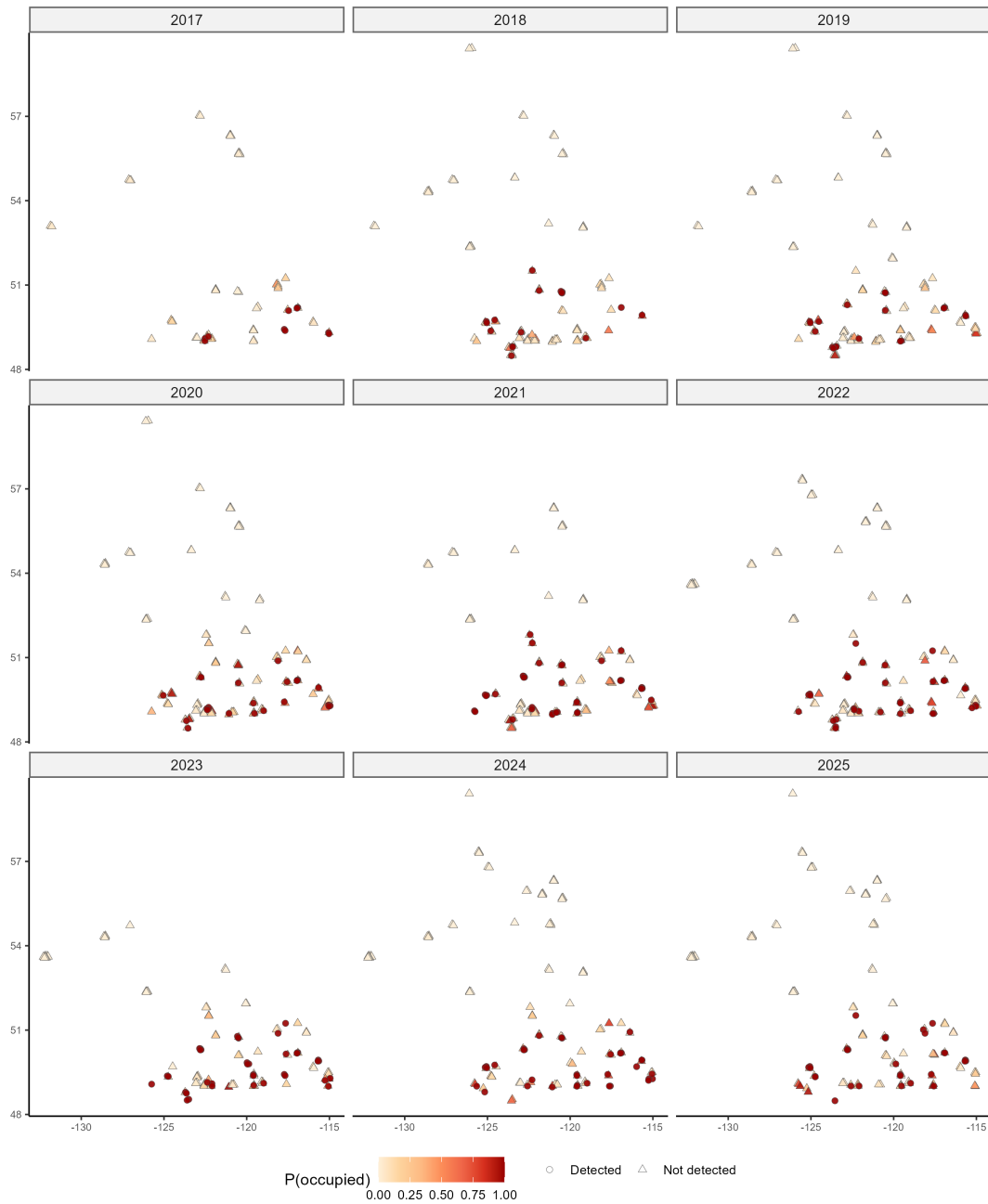
COTO — Townsend's big-eared bat: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

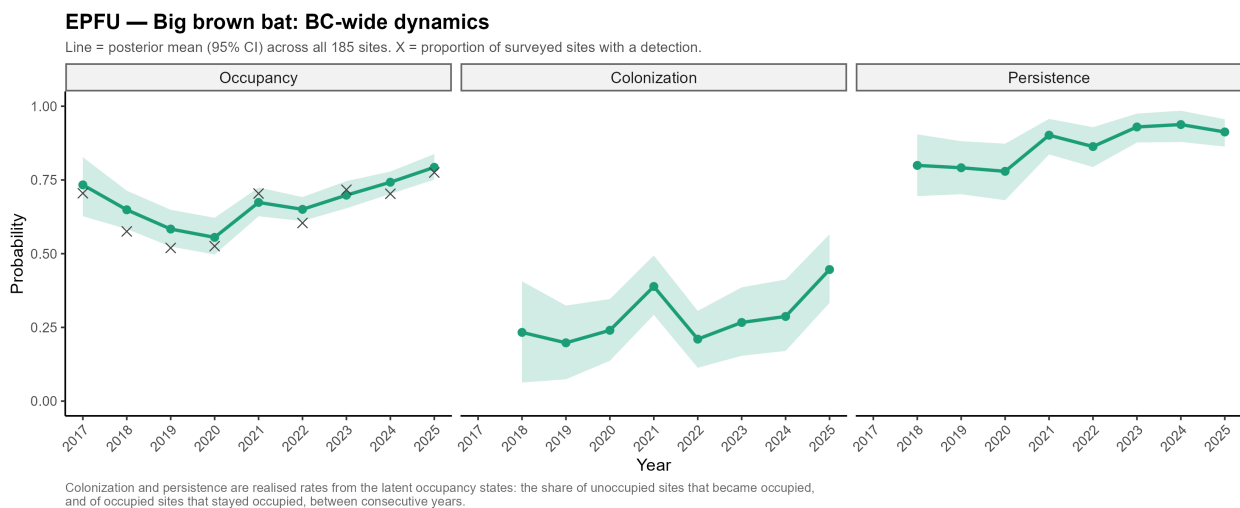
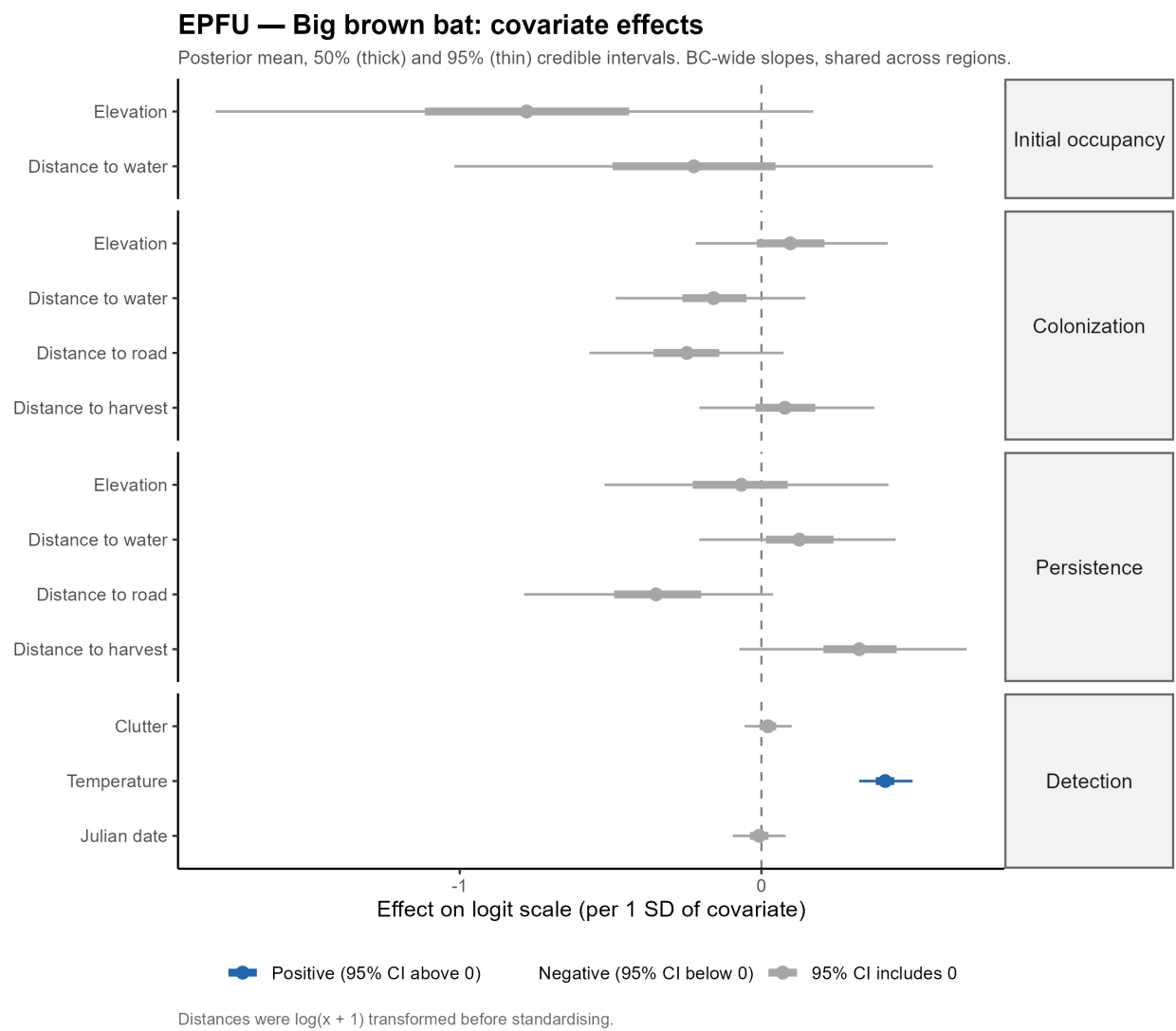


COTO — Townsend's big-eared bat: P(occupancy) by site and year

Surveyed site-years only. Triangles = not detected that year; their colour is model inference.

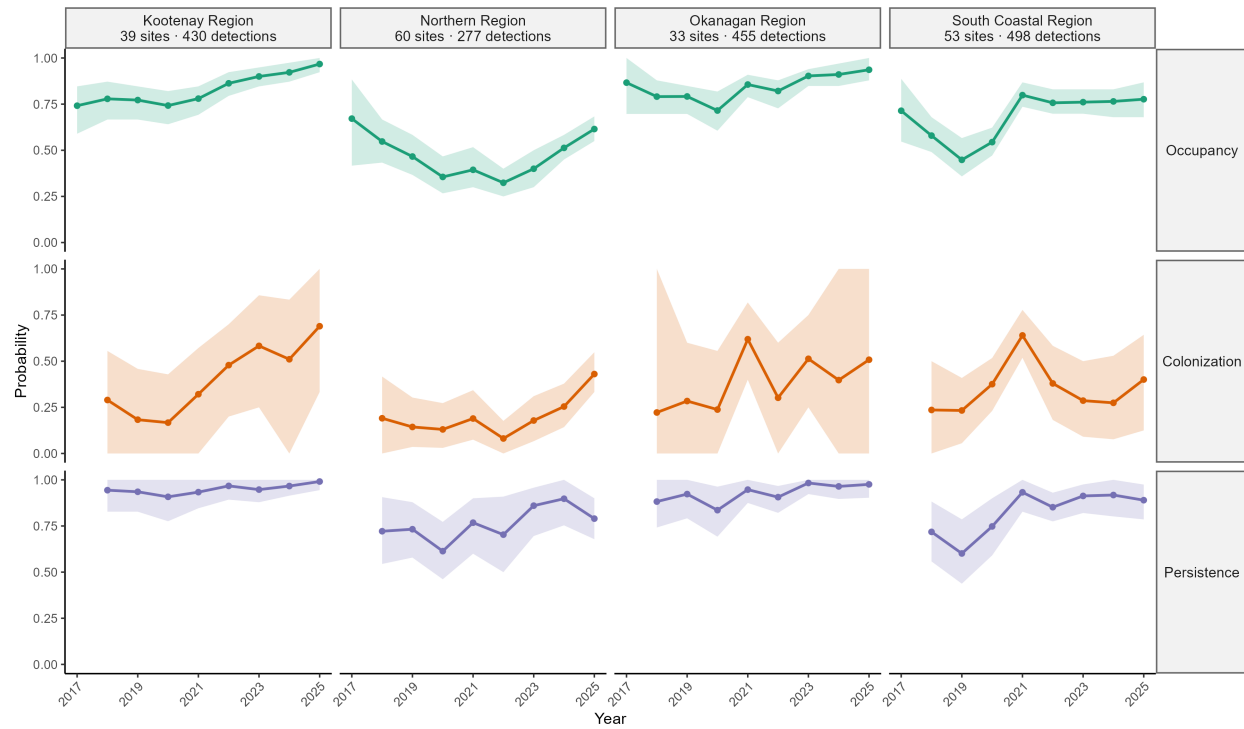


7.3 Big brown bat (*Eptesicus fuscus*)



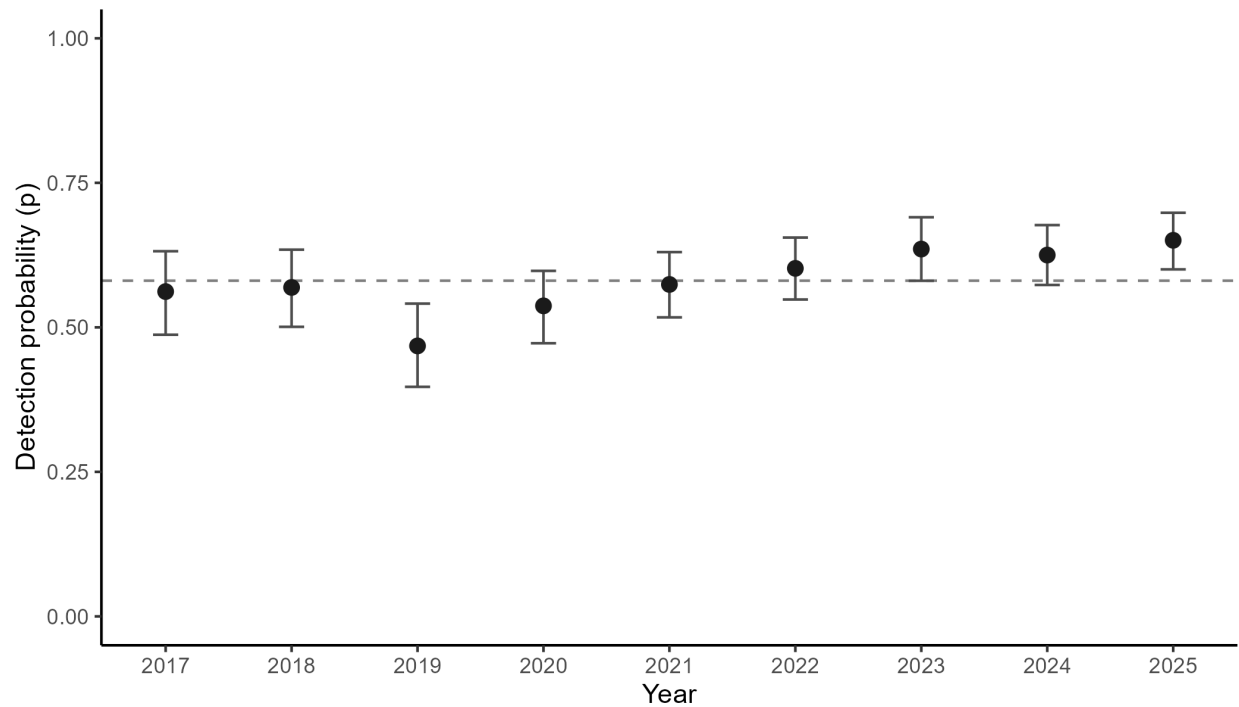
EPFU — Big brown bat: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



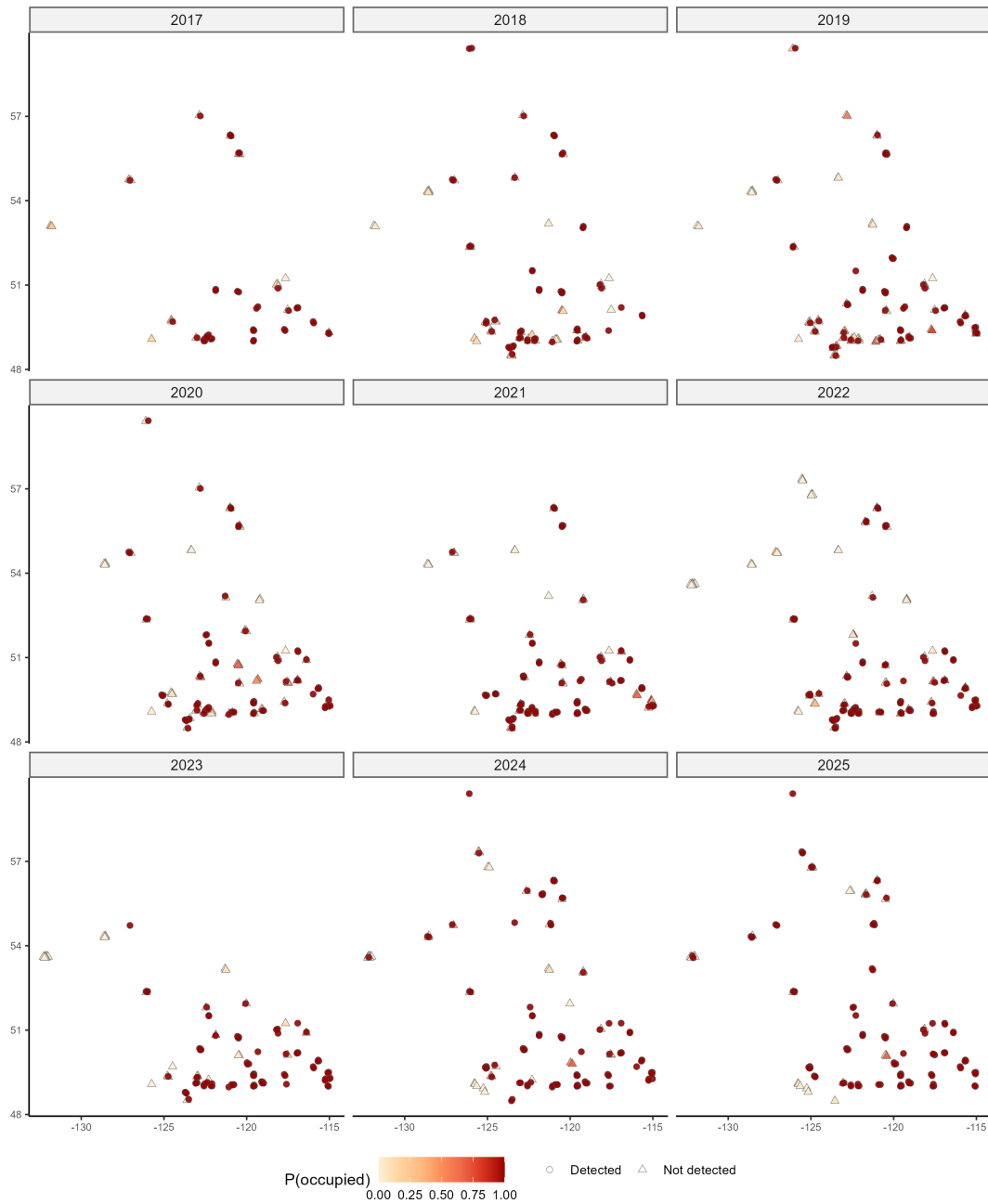
EPFU — Big brown bat: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



EPFU — Big brown bat: P(occupancy) by site and year

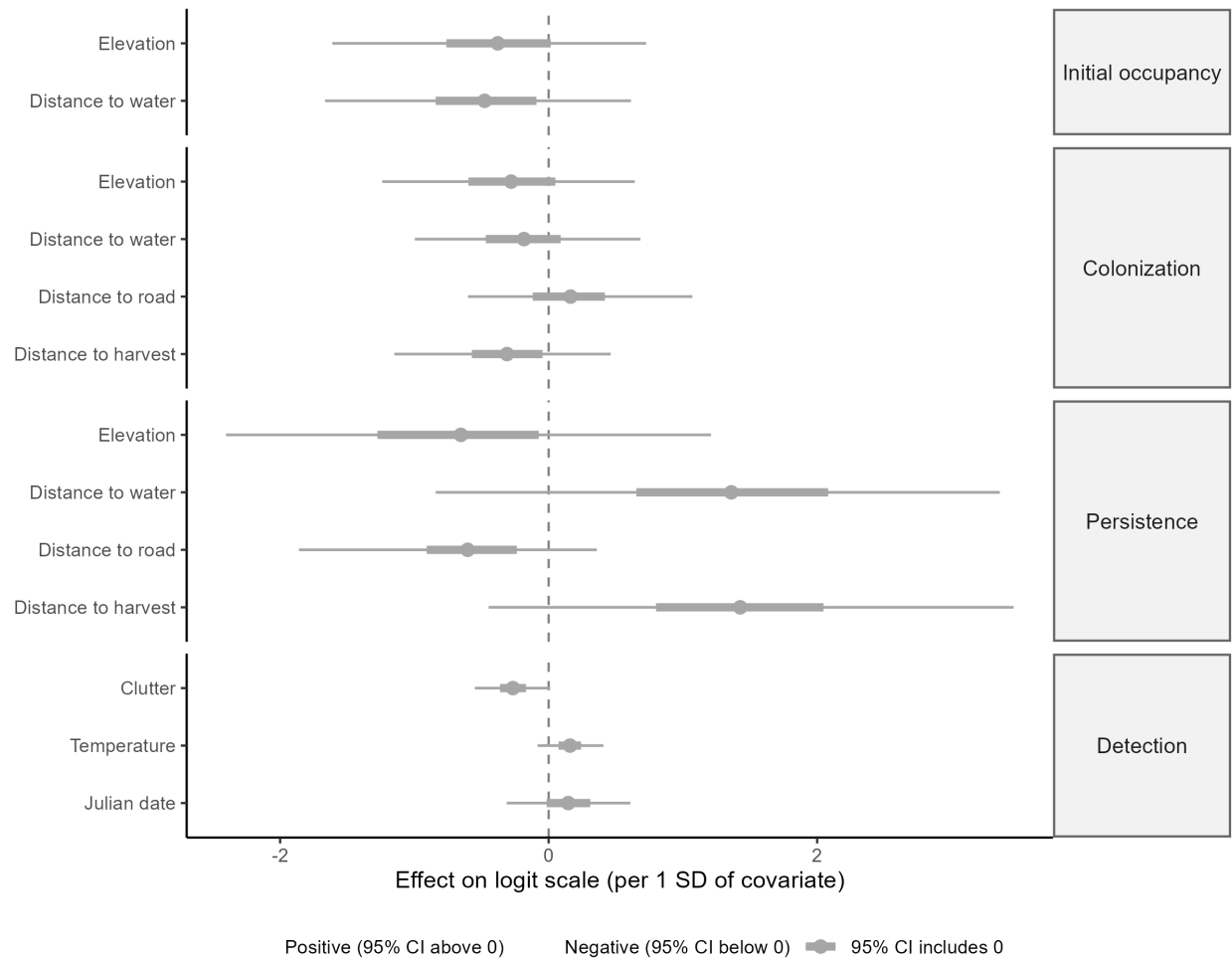
Surveyed site-years only. Triangles = not detected that year; their colour is model inference.



7.4 Spotted bat (*Euderma maculatum*)

EUMA — Spotted bat: covariate effects

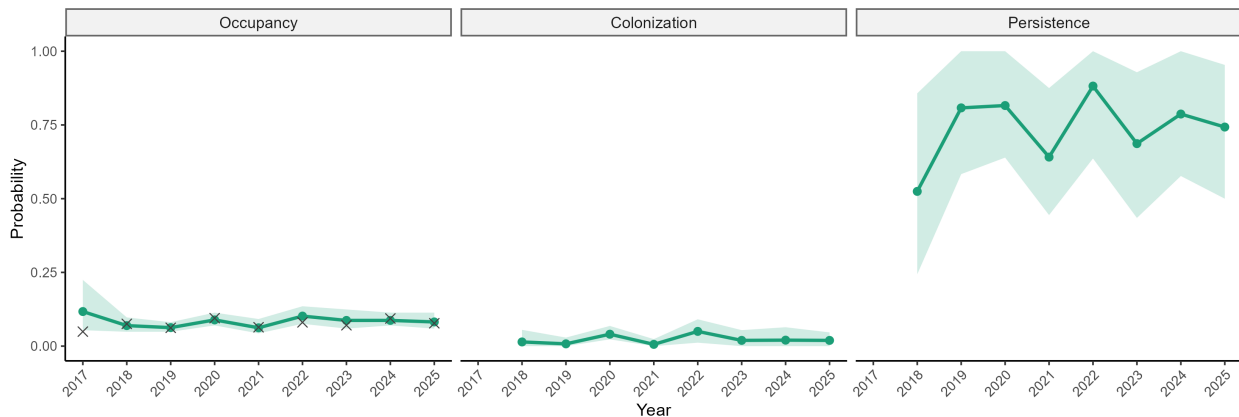
Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



Distances were log(x + 1) transformed before standardising.

EUMA — Spotted bat: BC-wide dynamics

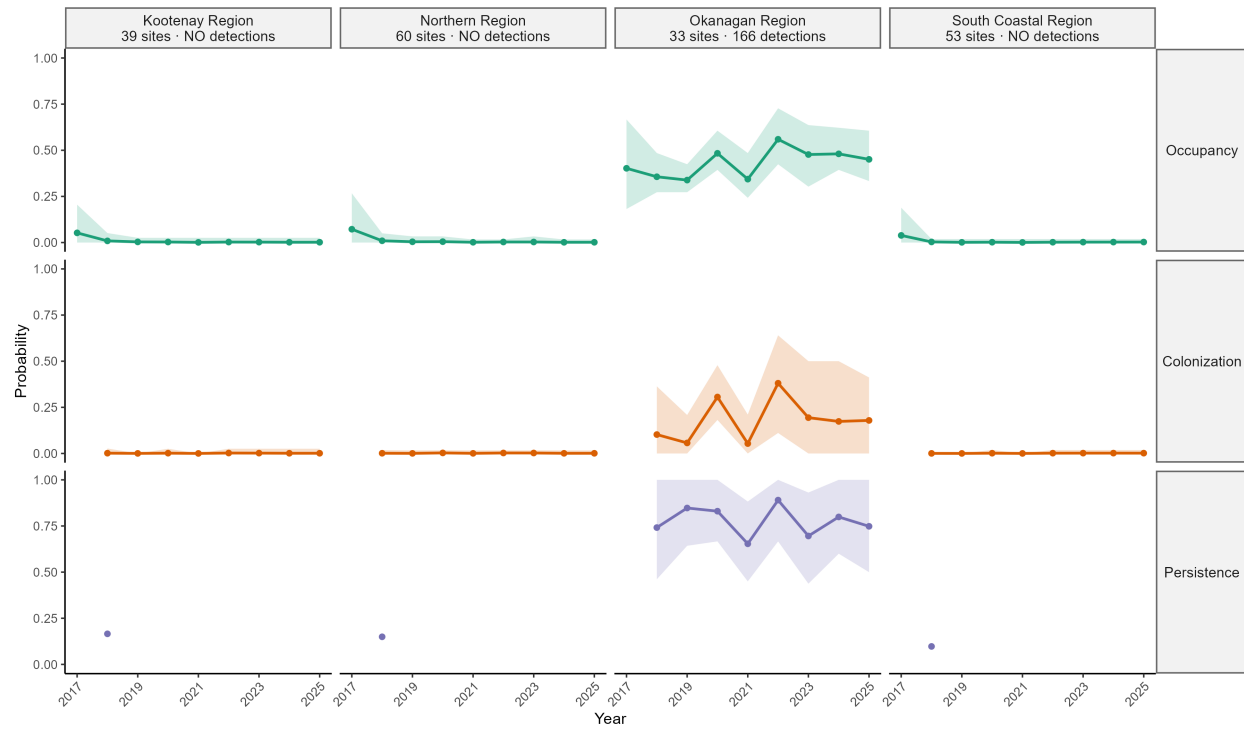
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

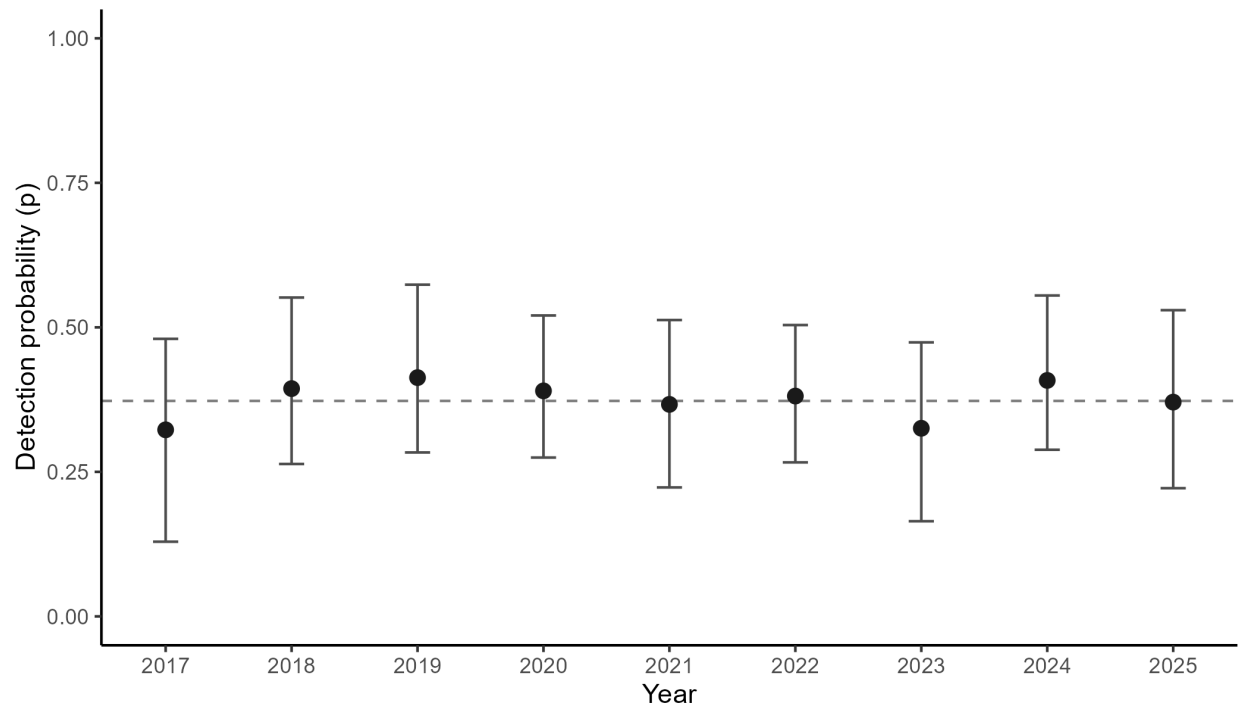
EUMA — Spotted bat: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



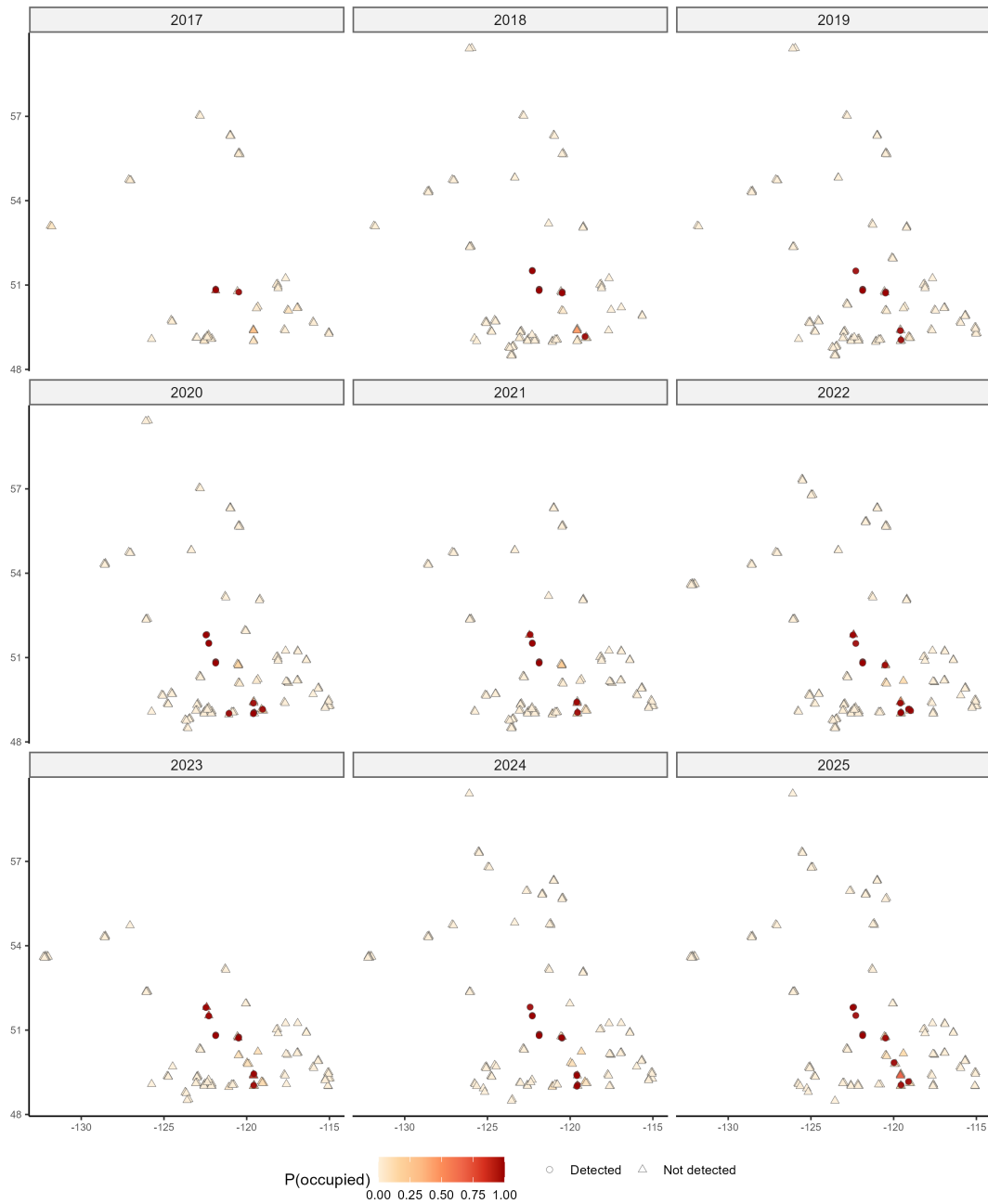
EUMA — Spotted bat: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



EUMA — Spotted bat: P(occupancy) by site and year

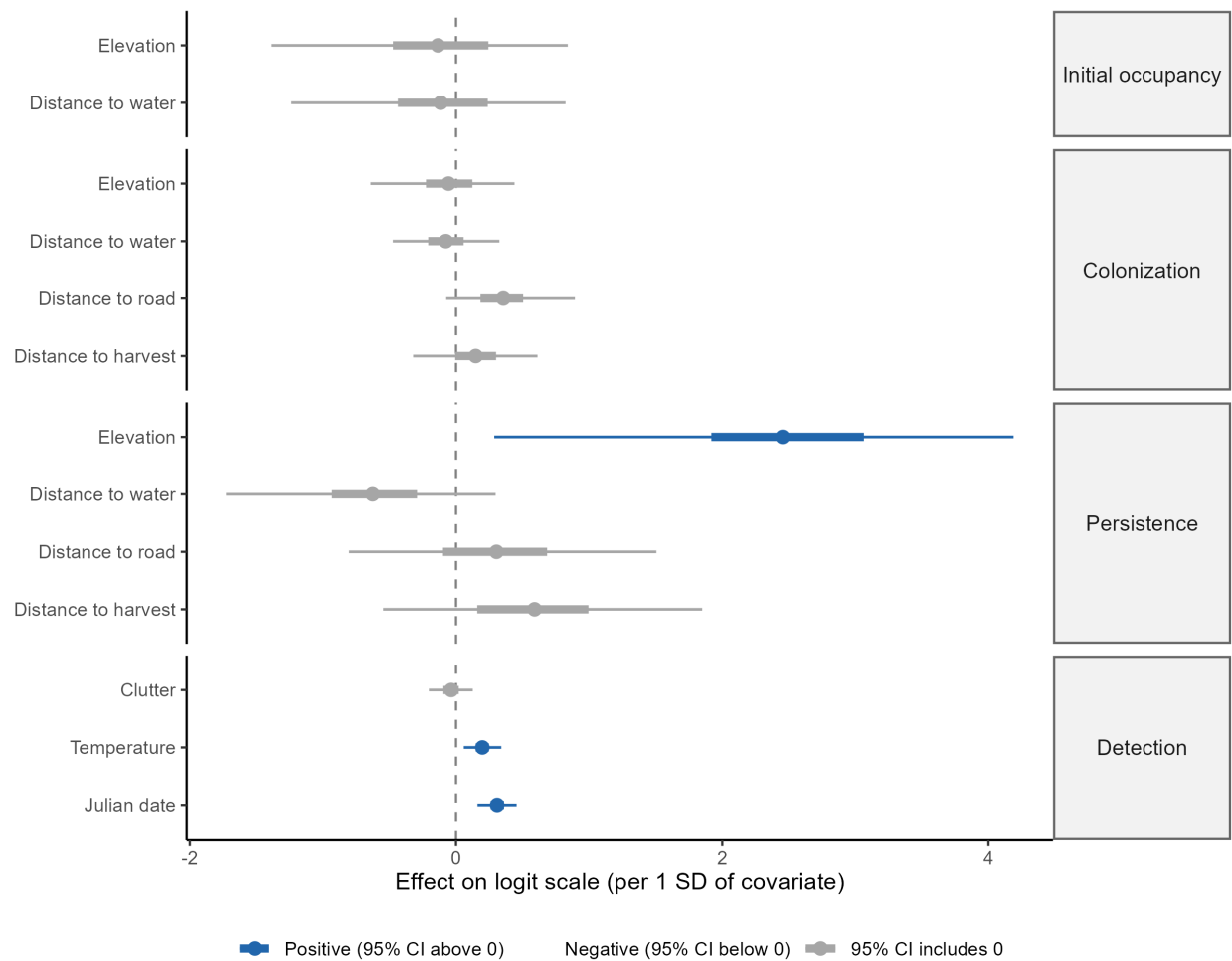
Surveyed site-years only. Triangles = not detected that year; their colour is model inference.



7.5 Eastern-red bat (*Lasiurus borealis*)

LABO — Eastern red bat: covariate effects

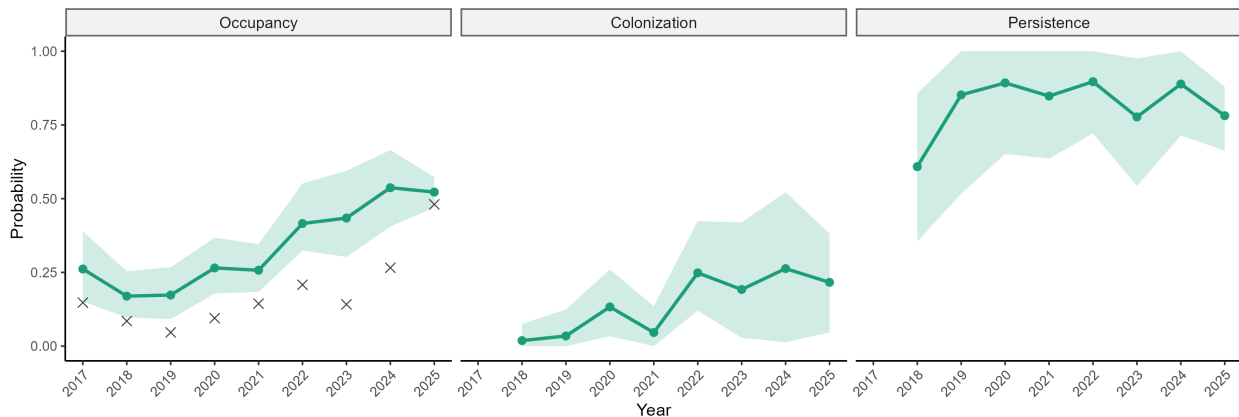
Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



Distances were $\log(x + 1)$ transformed before standardising.

LABO — Eastern red bat: BC-wide dynamics

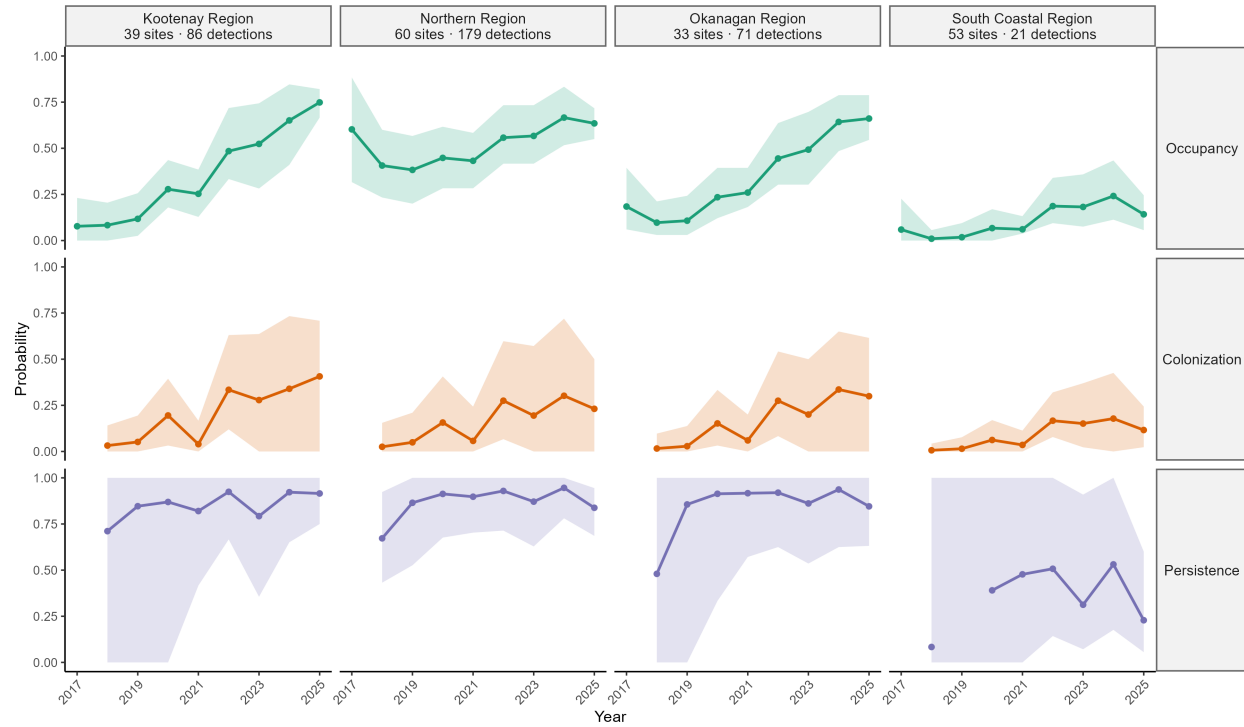
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

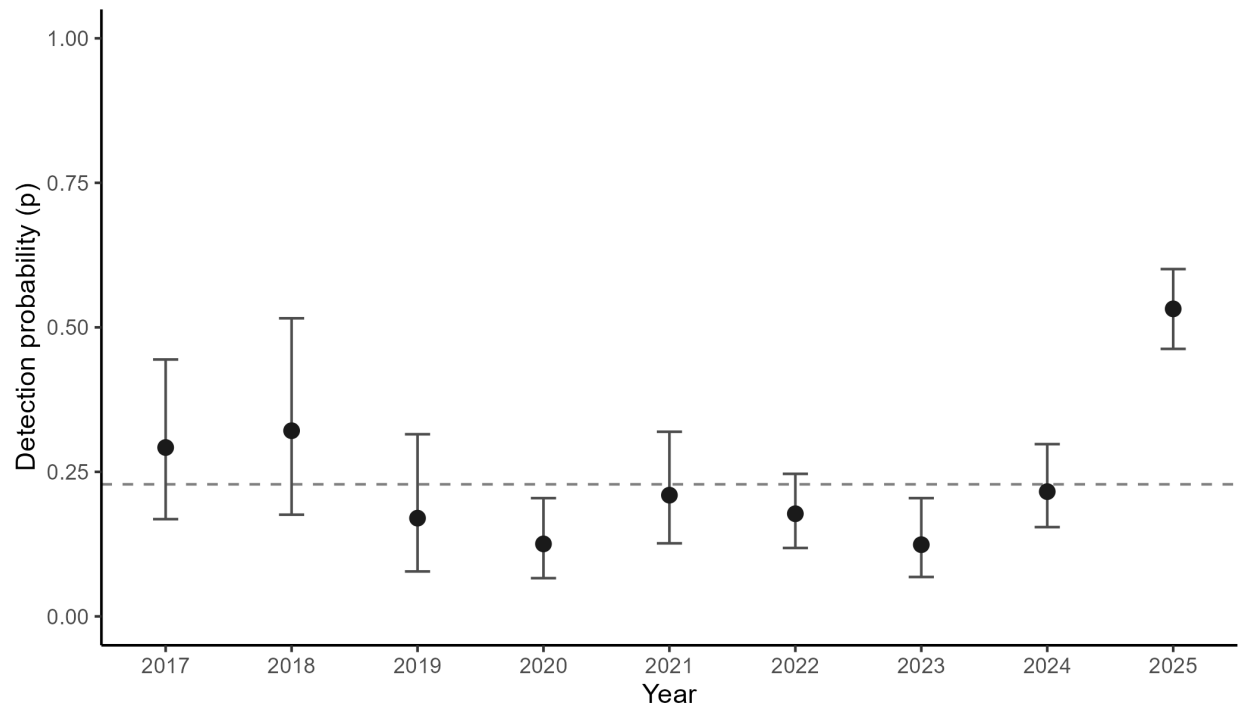
LABO — Eastern red bat: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



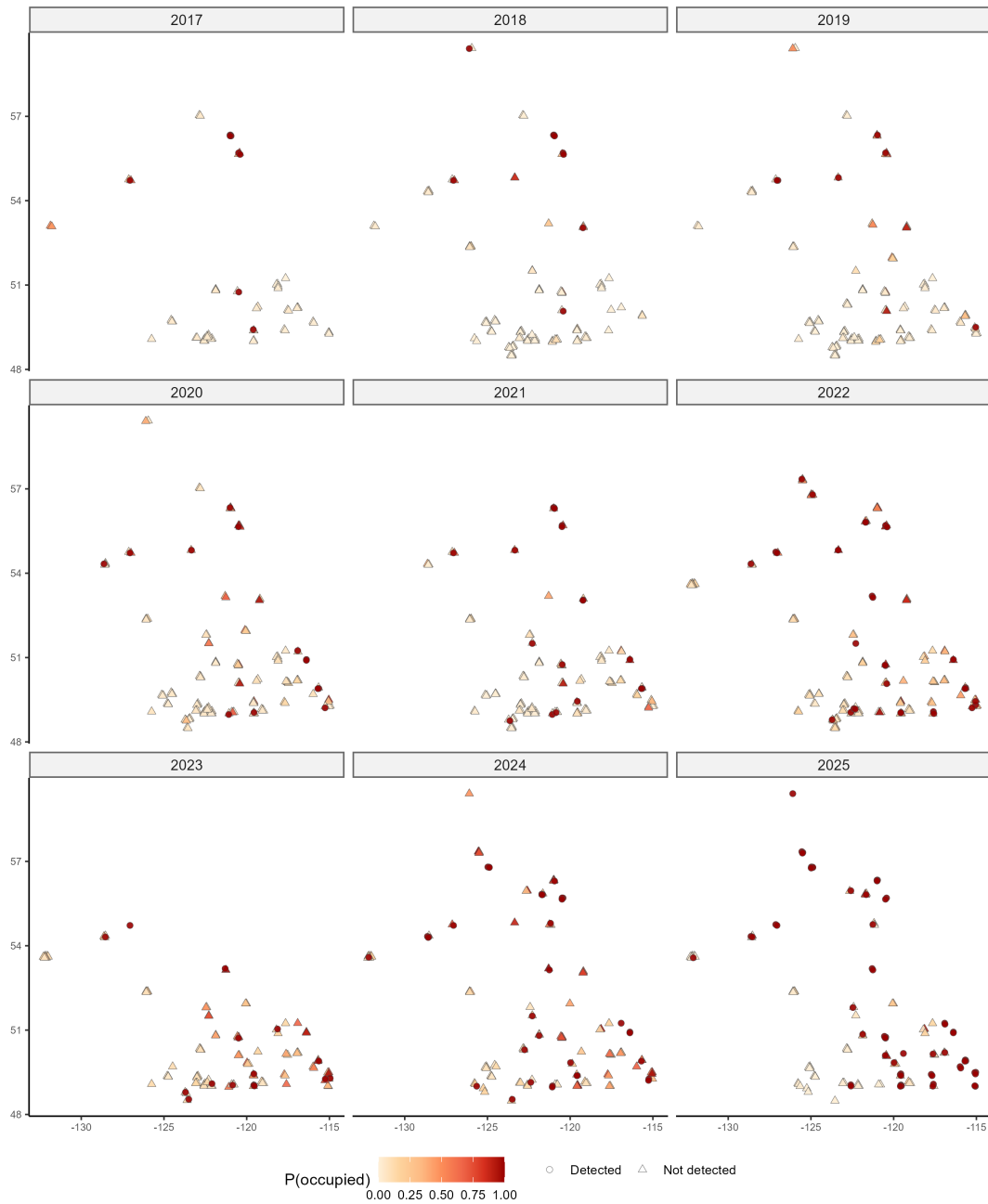
LABO — Eastern red bat: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

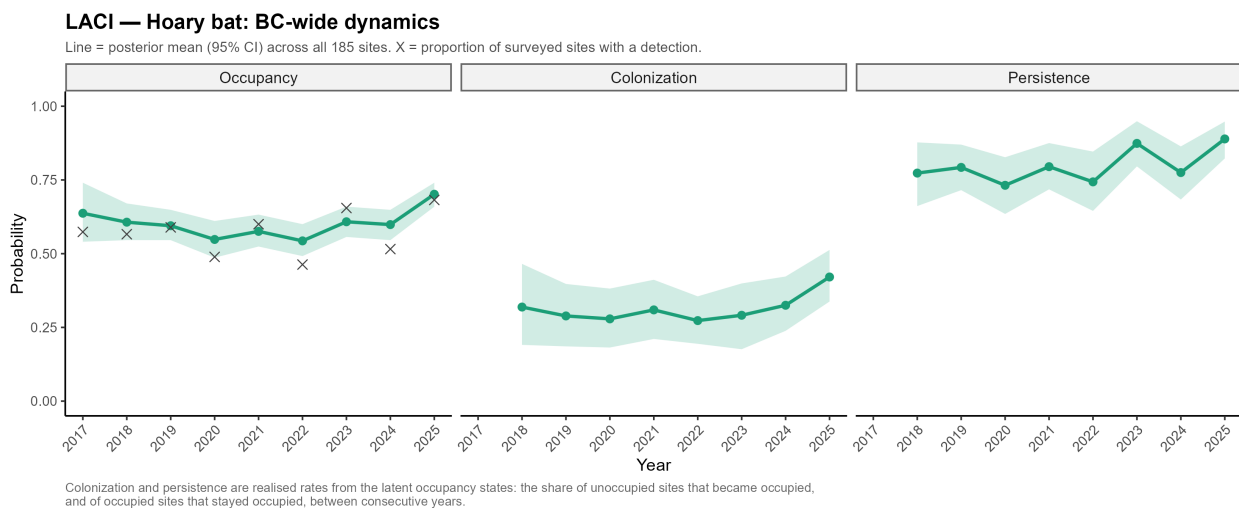
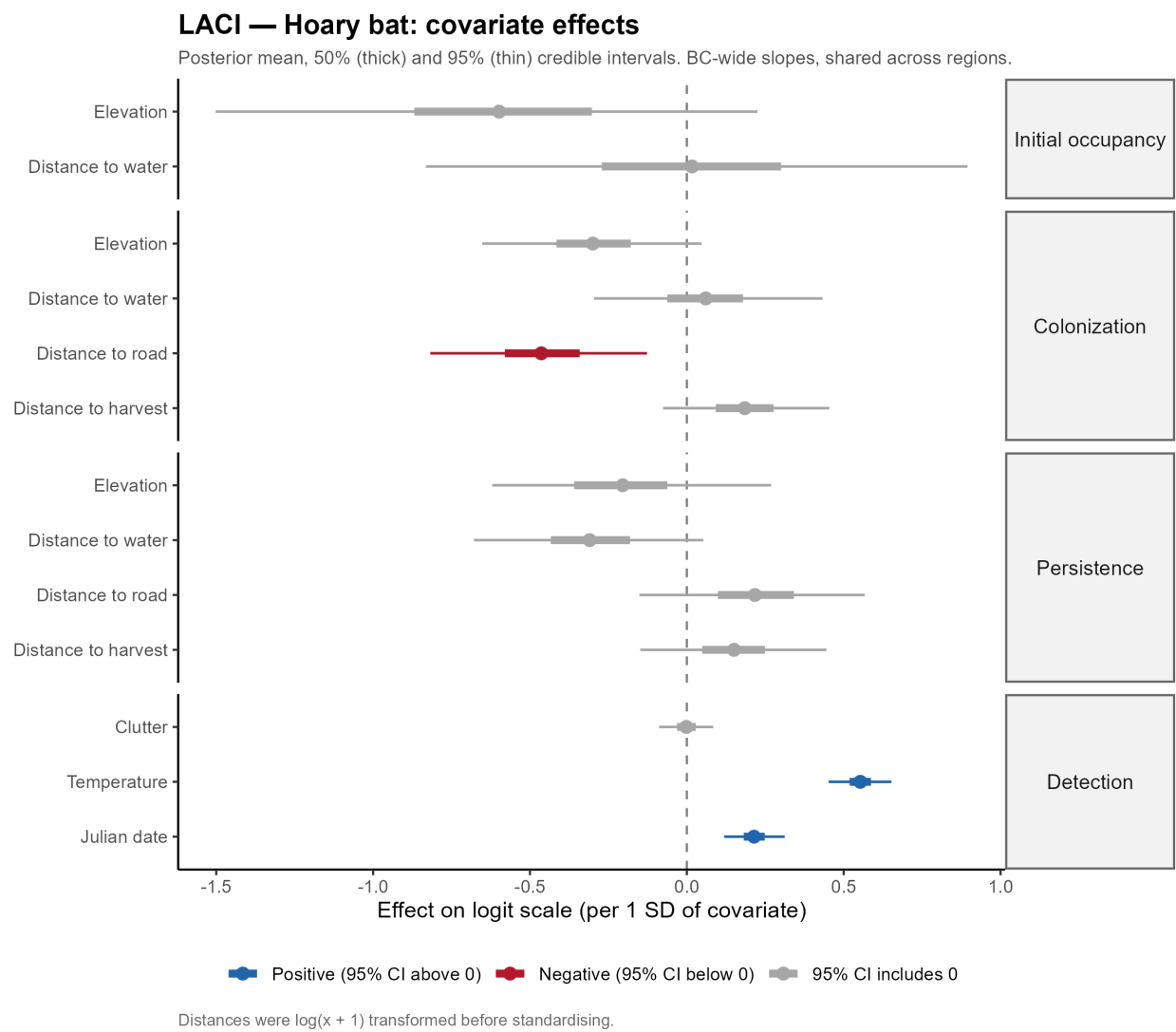


LABO — Eastern red bat: P(occupancy) by site and year

Surveyed site-years only. Triangles = not detected that year; their colour is model inference.

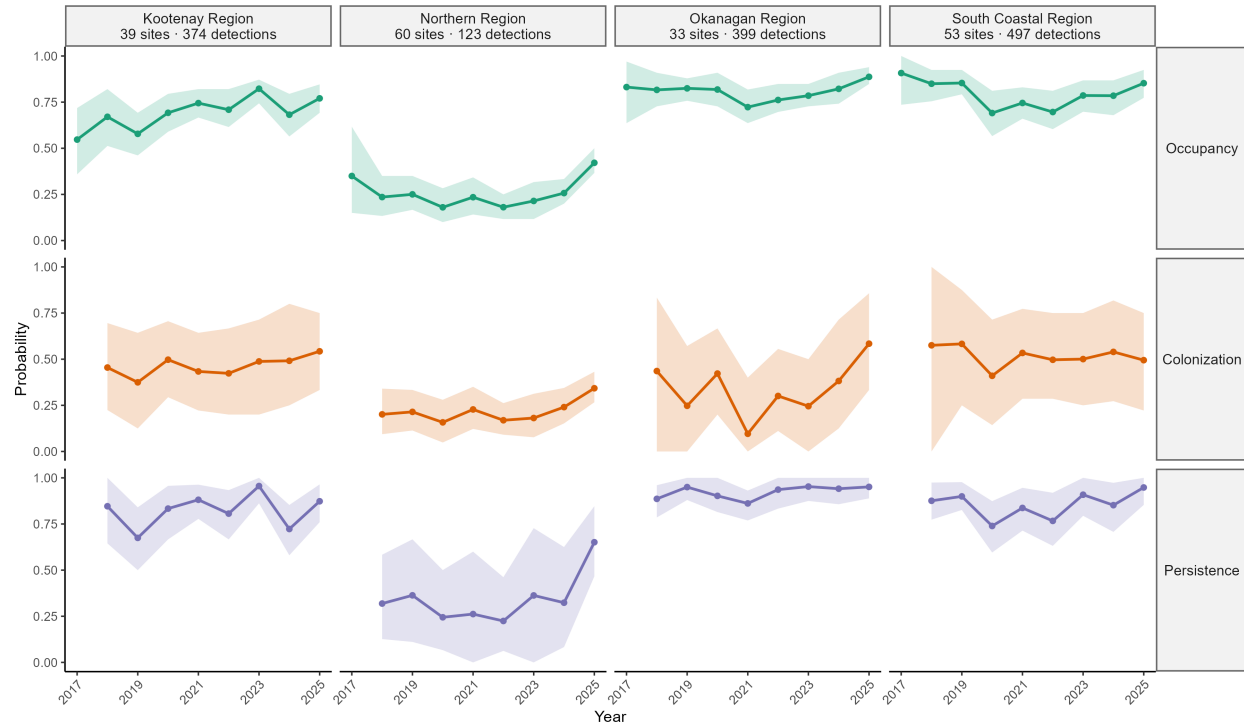


7.6 Hoary bat (*Lasiurus cinereus*)



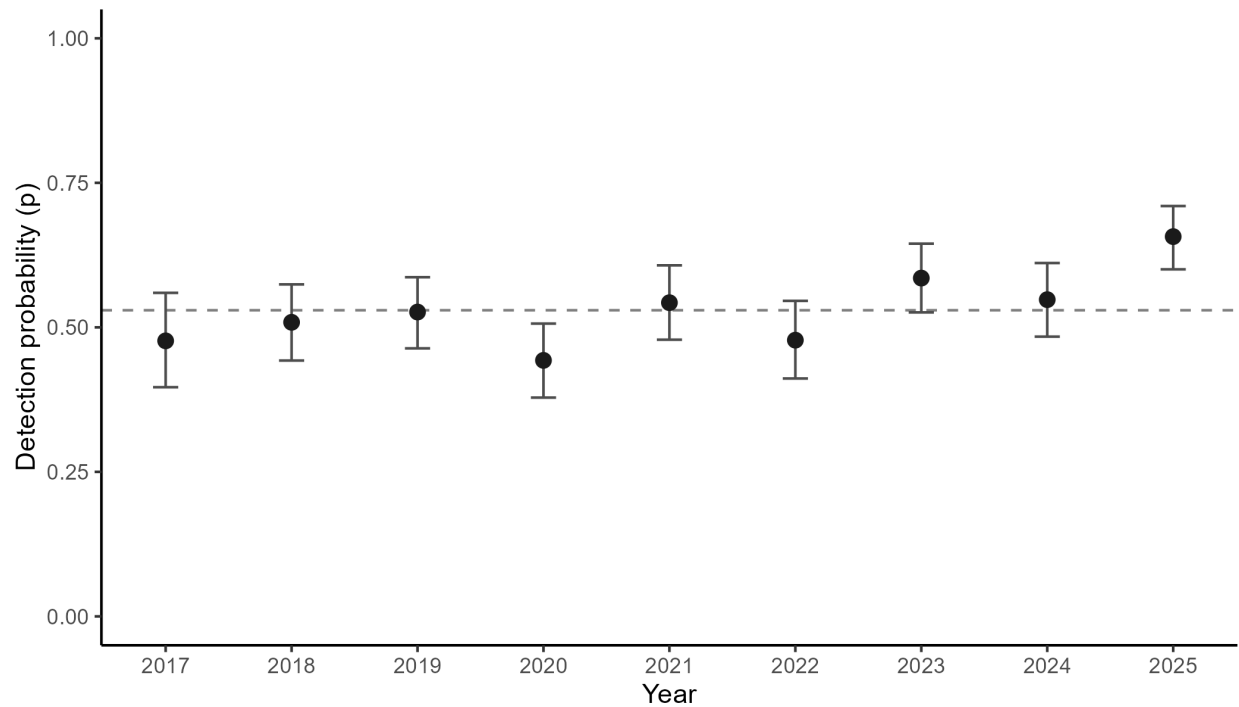
LACI — Hoary bat: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



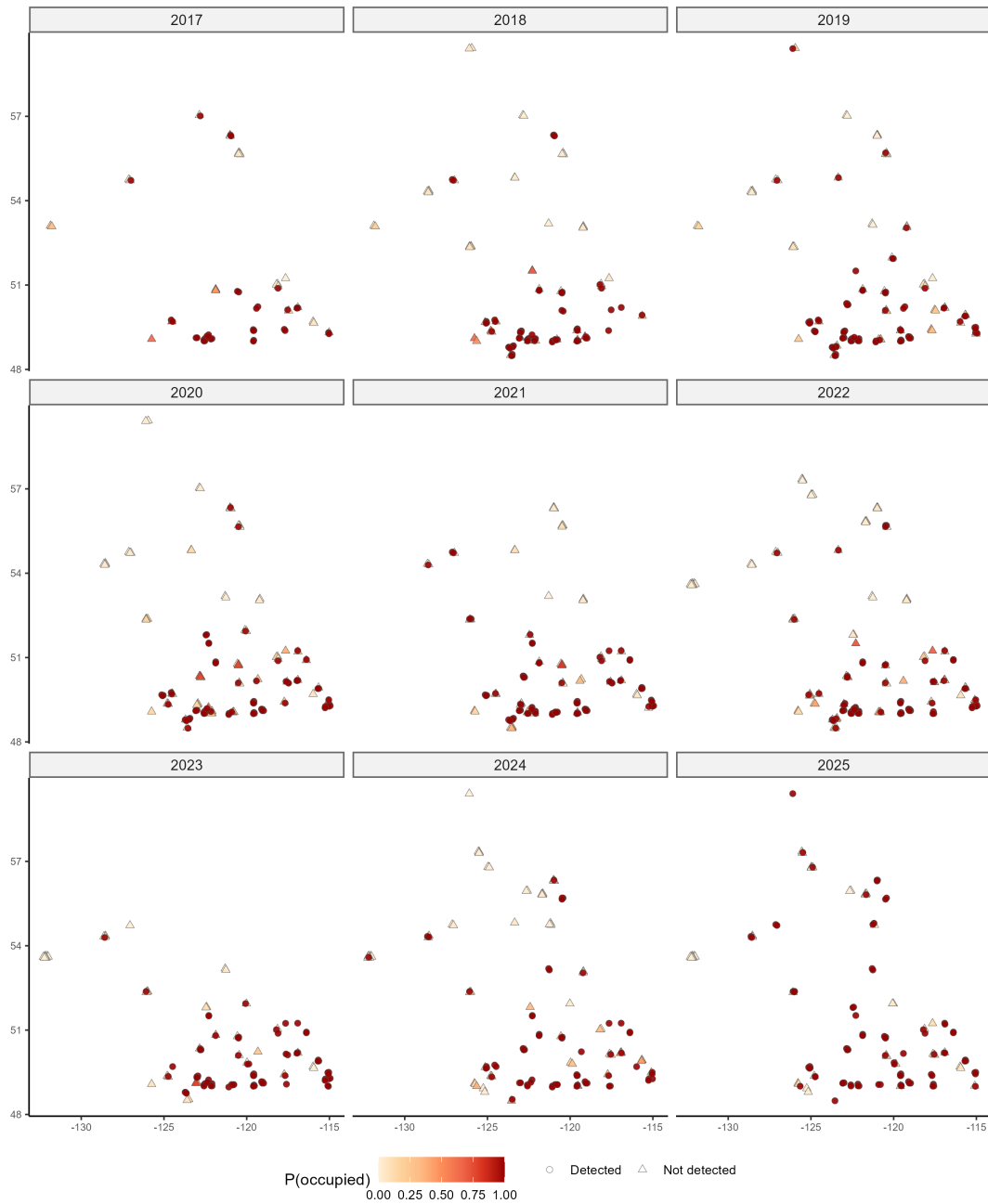
LACI — Hoary bat: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

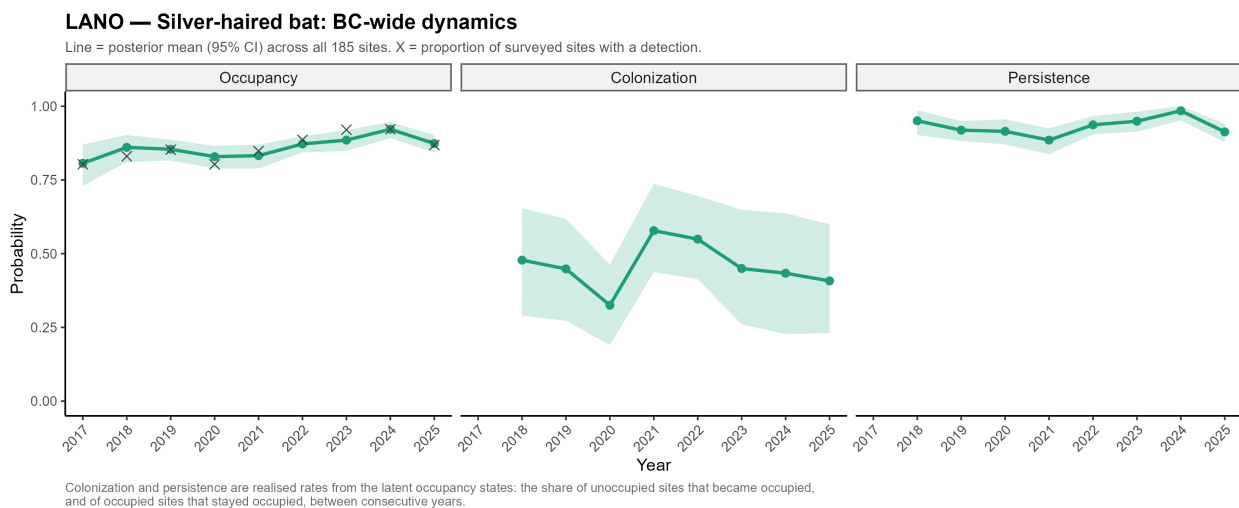
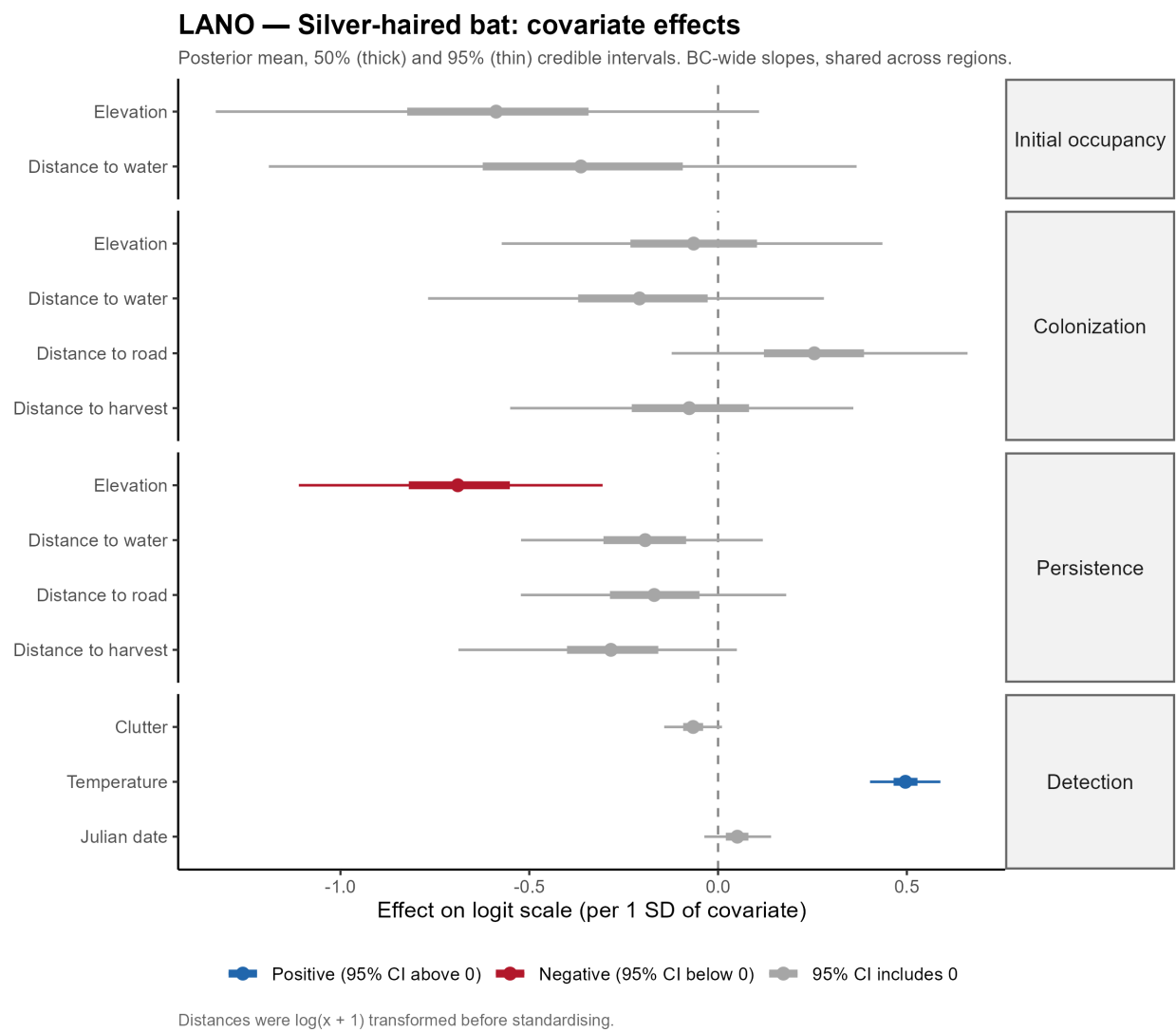


LACI — Hoary bat: P(occupancy) by site and year

Surveyed site-years only. Triangles = not detected that year; their colour is model inference.

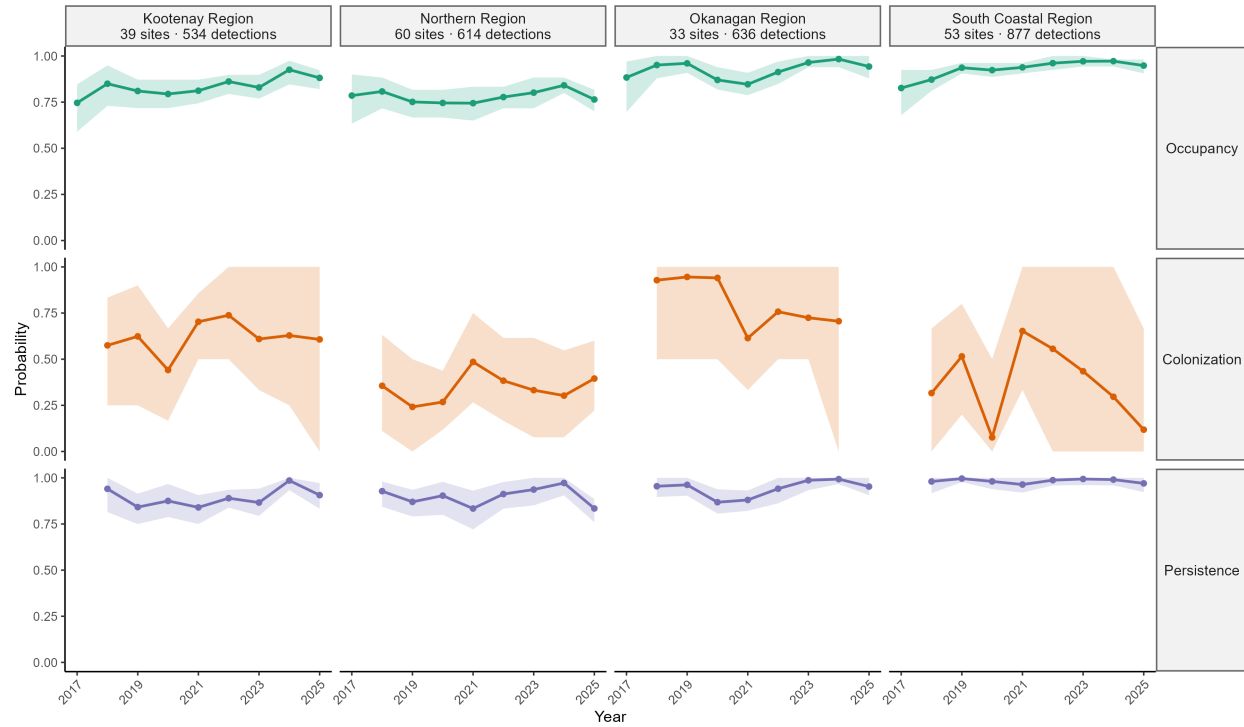


7.7 Silver-haired bat (*Lasionycteris noctivagans*)



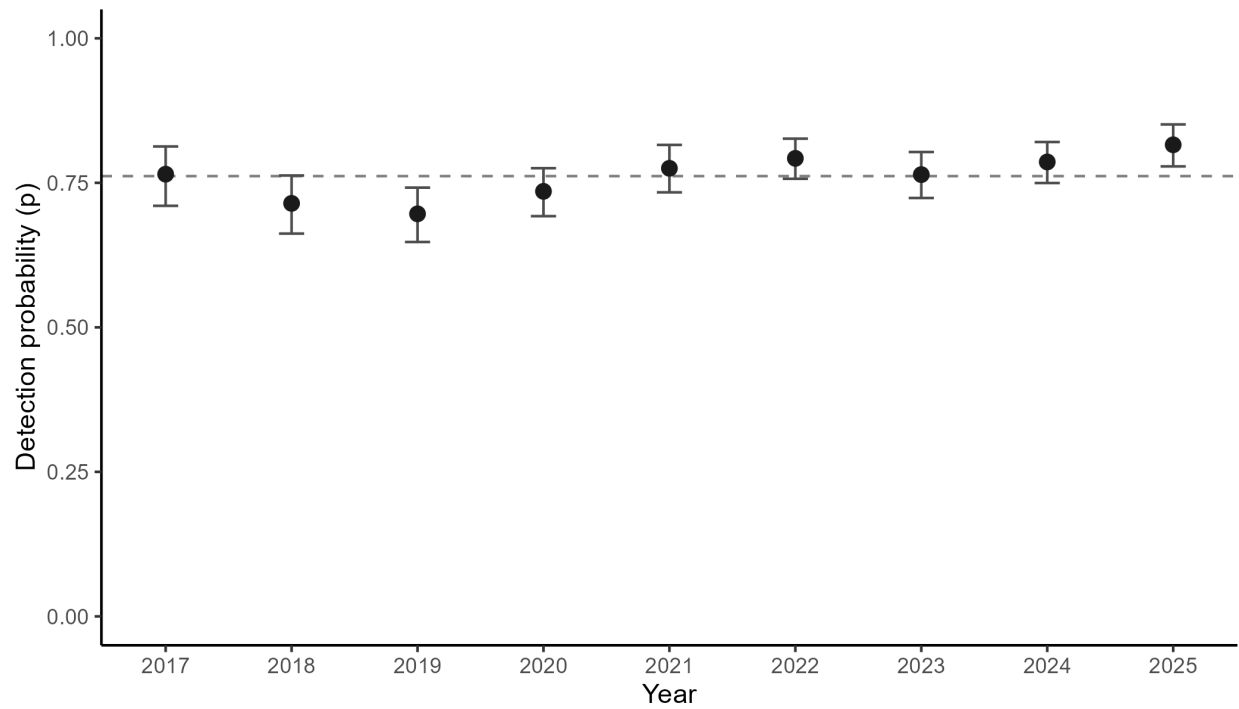
LANO — Silver-haired bat: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



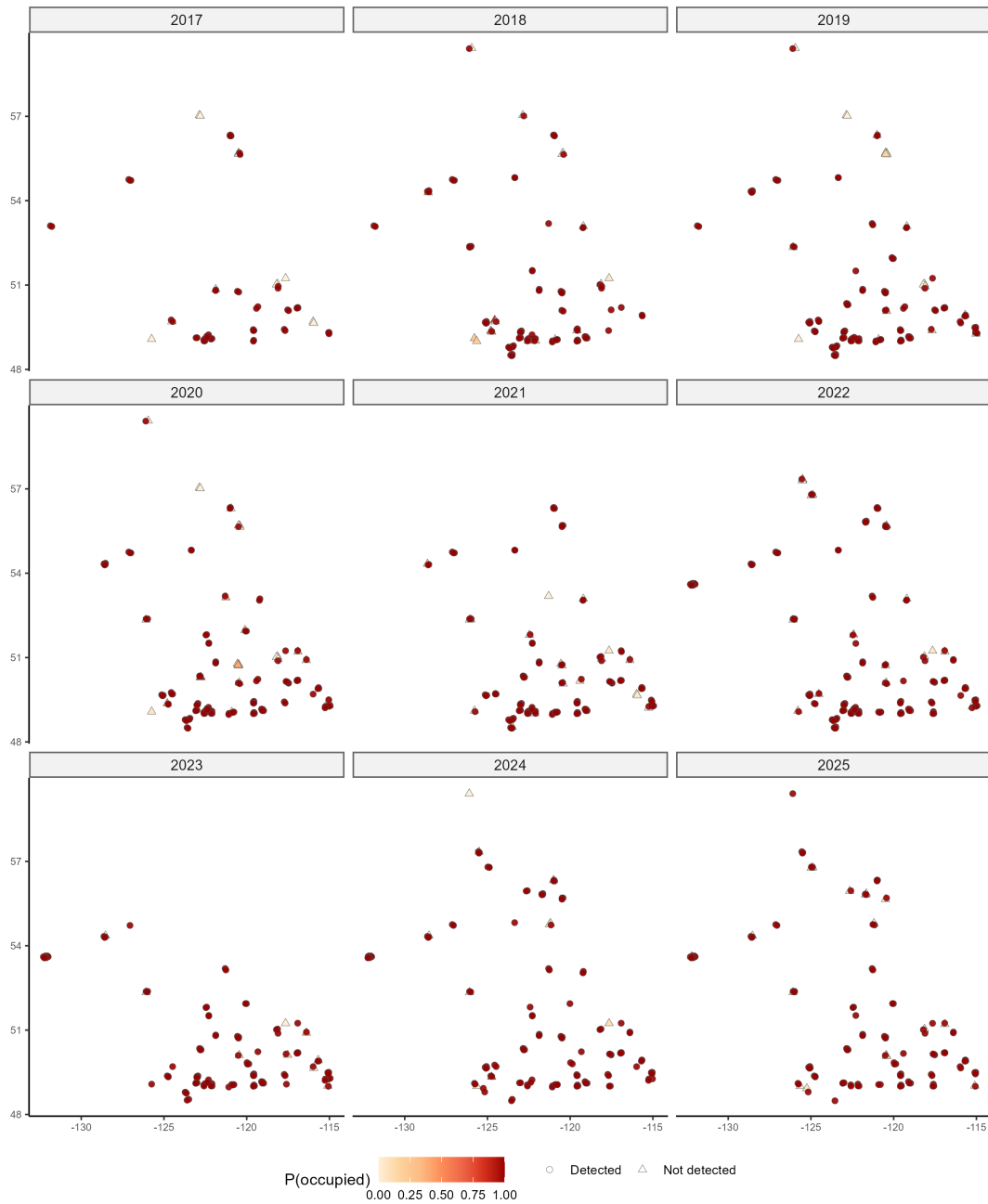
LANO — Silver-haired bat: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

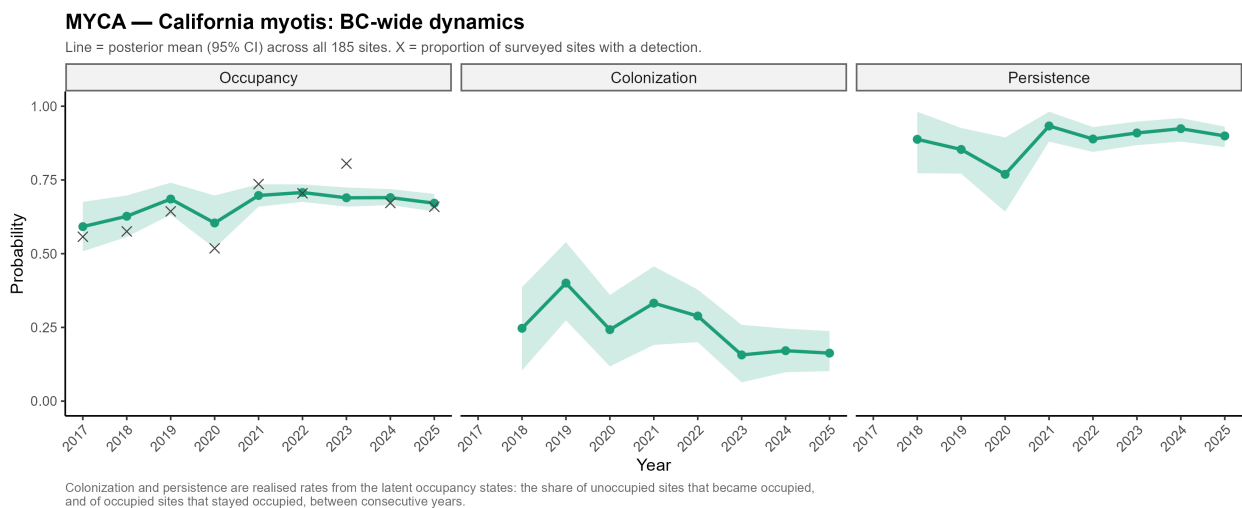
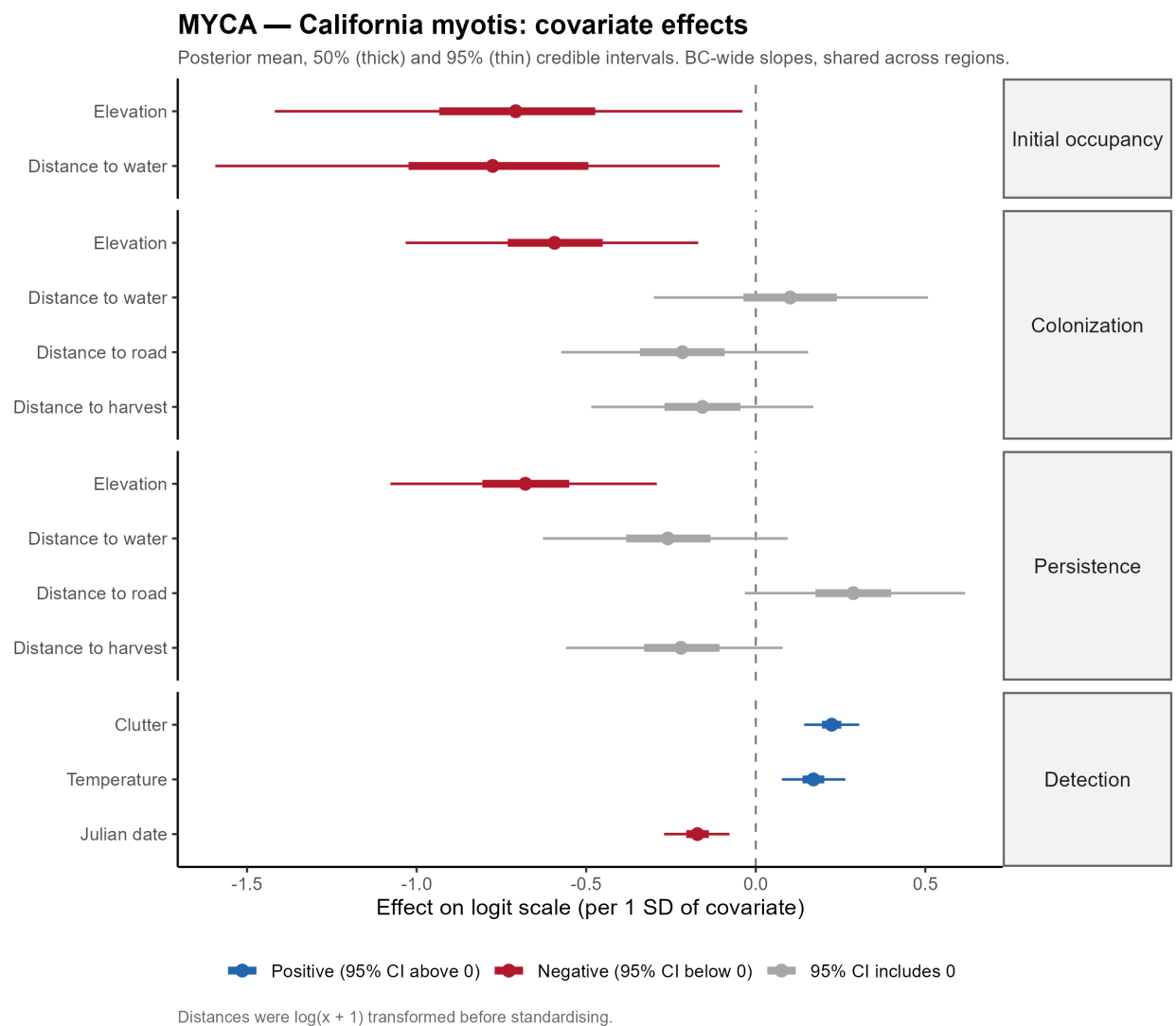


LANO — Silver-haired bat: P(occupancy) by site and year

Surveyed site-years only. Triangles = not detected that year; their colour is model inference.

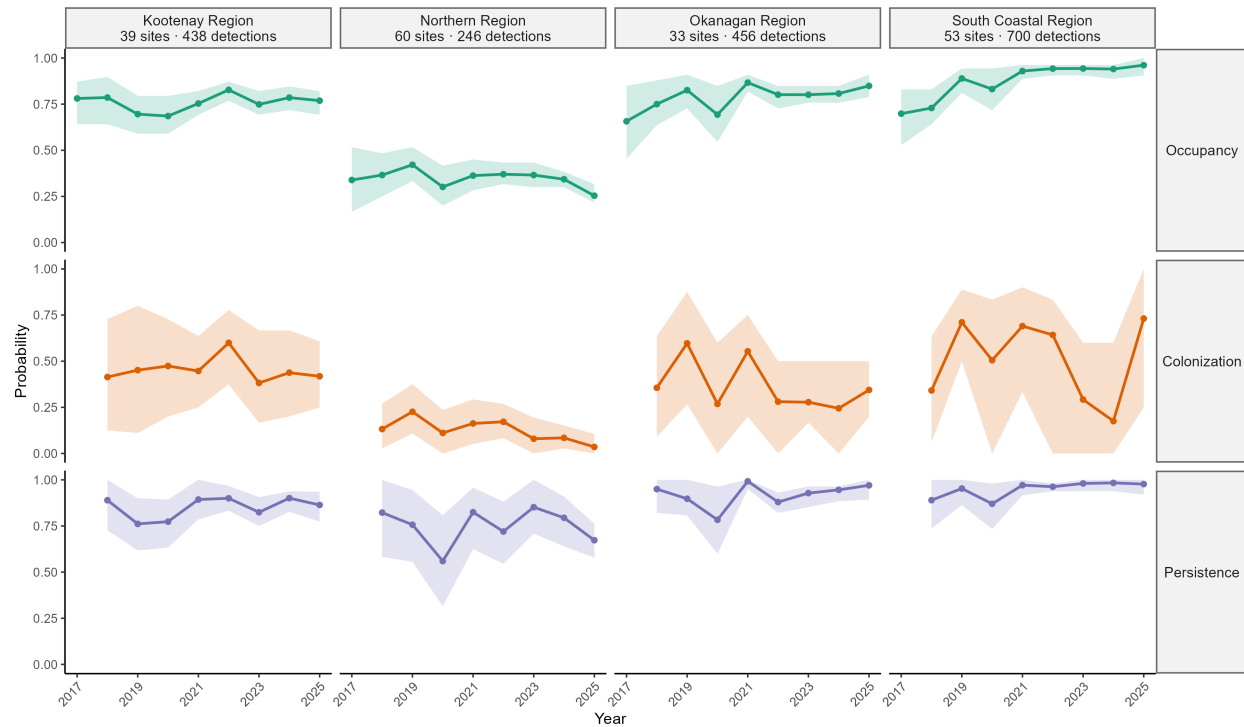


7.8 California Myotis (*Myotis californicus*)



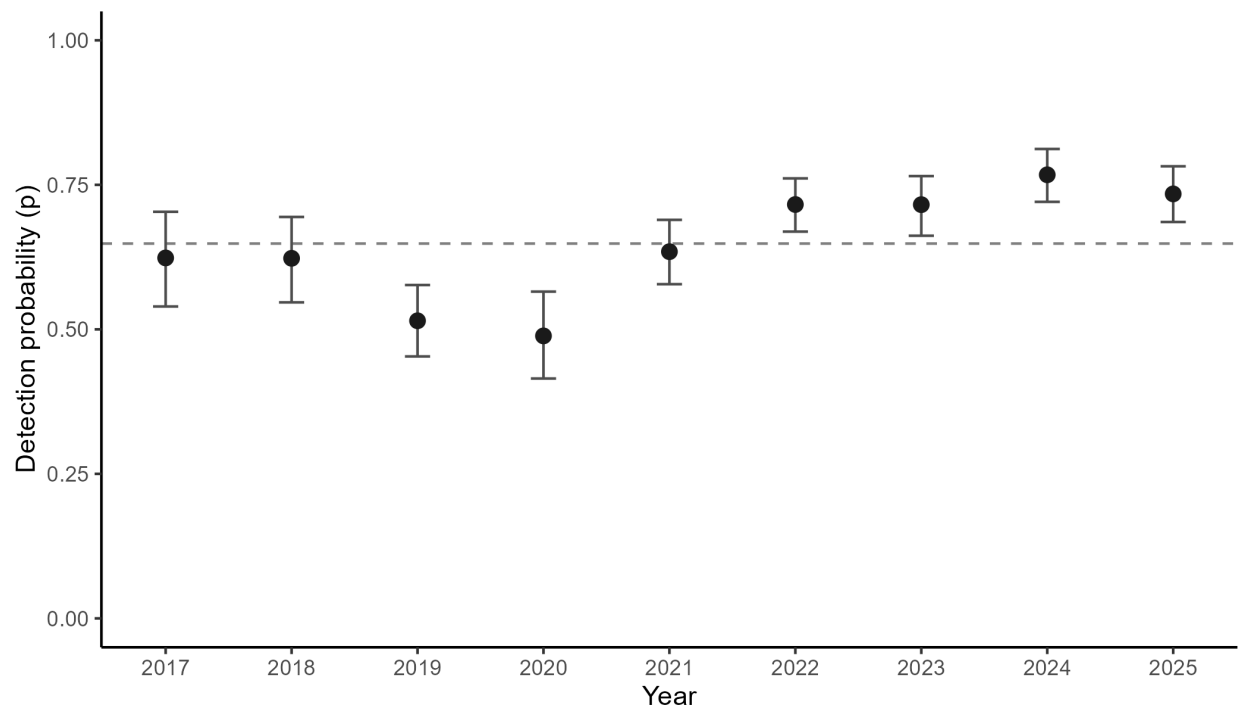
MYCA — California myotis: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



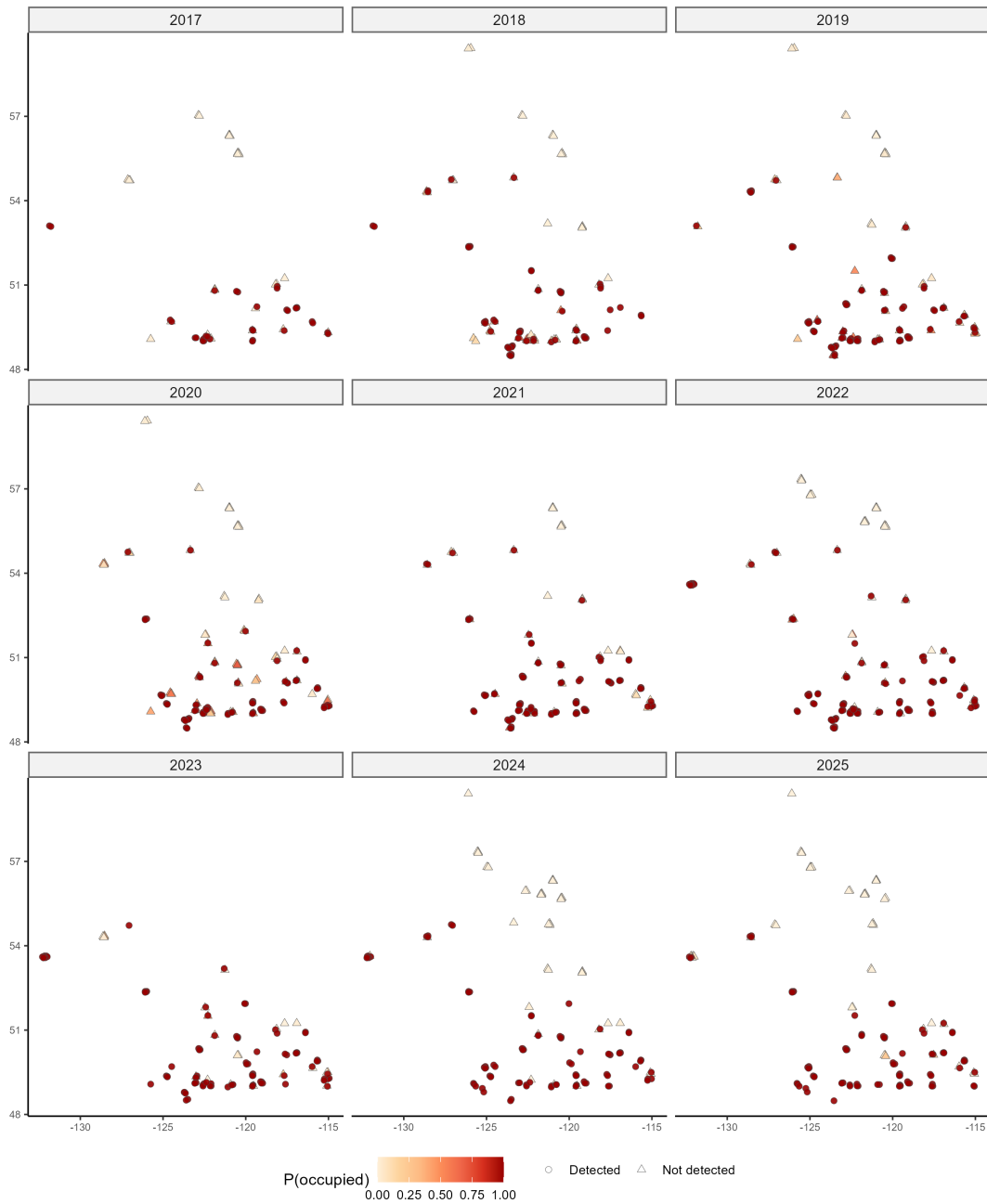
MYCA — California myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



MYCA — California myotis: P(occupancy) by site and year

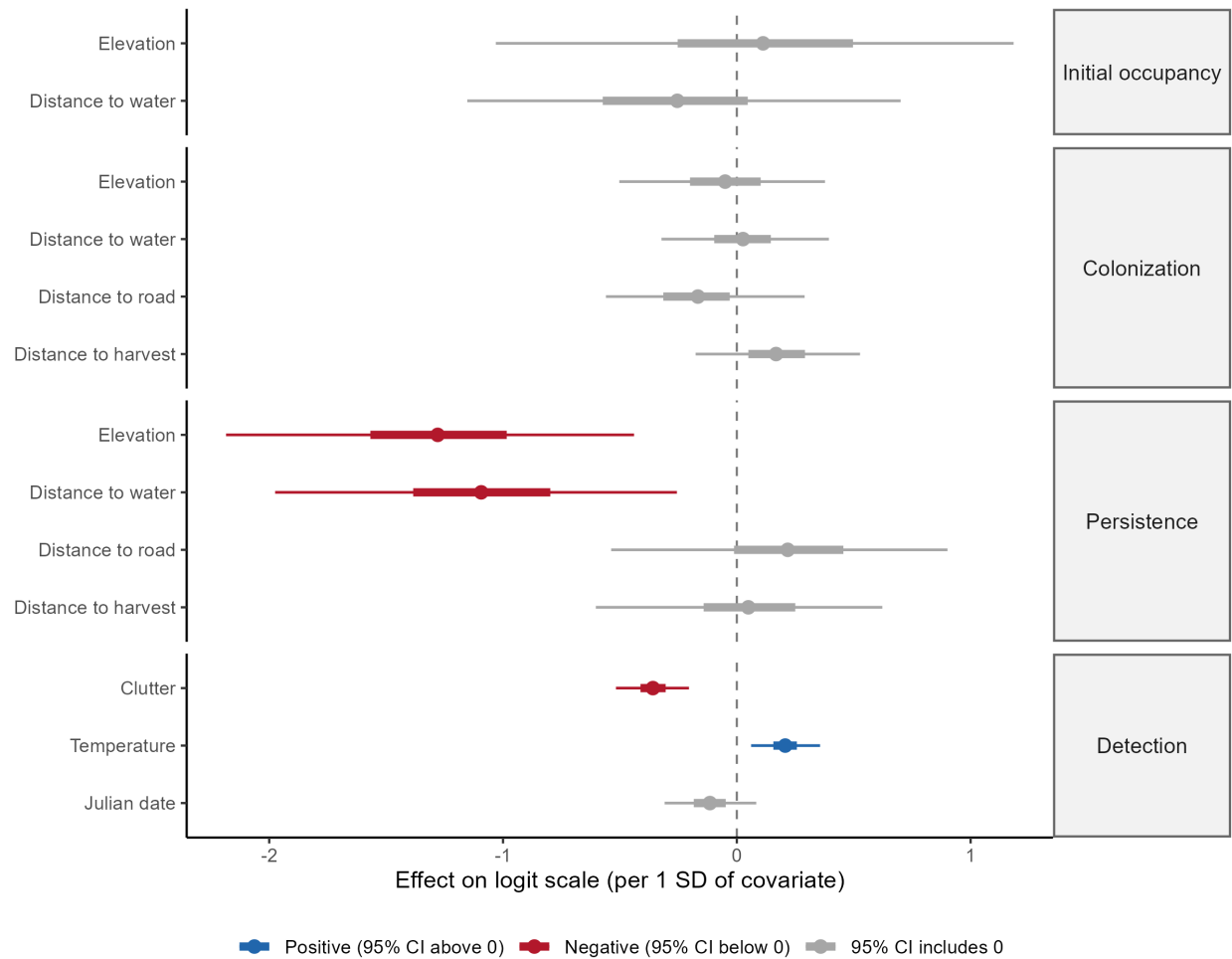
Surveyed site-years only. Triangles = not detected that year; their colour is model inference.



7.9 Western small-footed Myotis (*Myotis ciliolabrum*)

MYCI — Western small-footed myotis: covariate effects

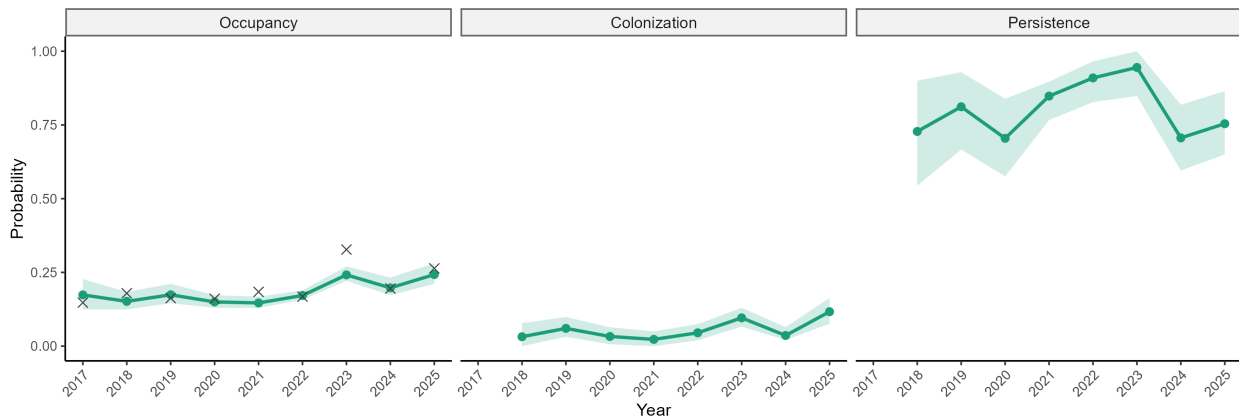
Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



Distances were log(x + 1) transformed before standardising.

MYCI — Western small-footed myotis: BC-wide dynamics

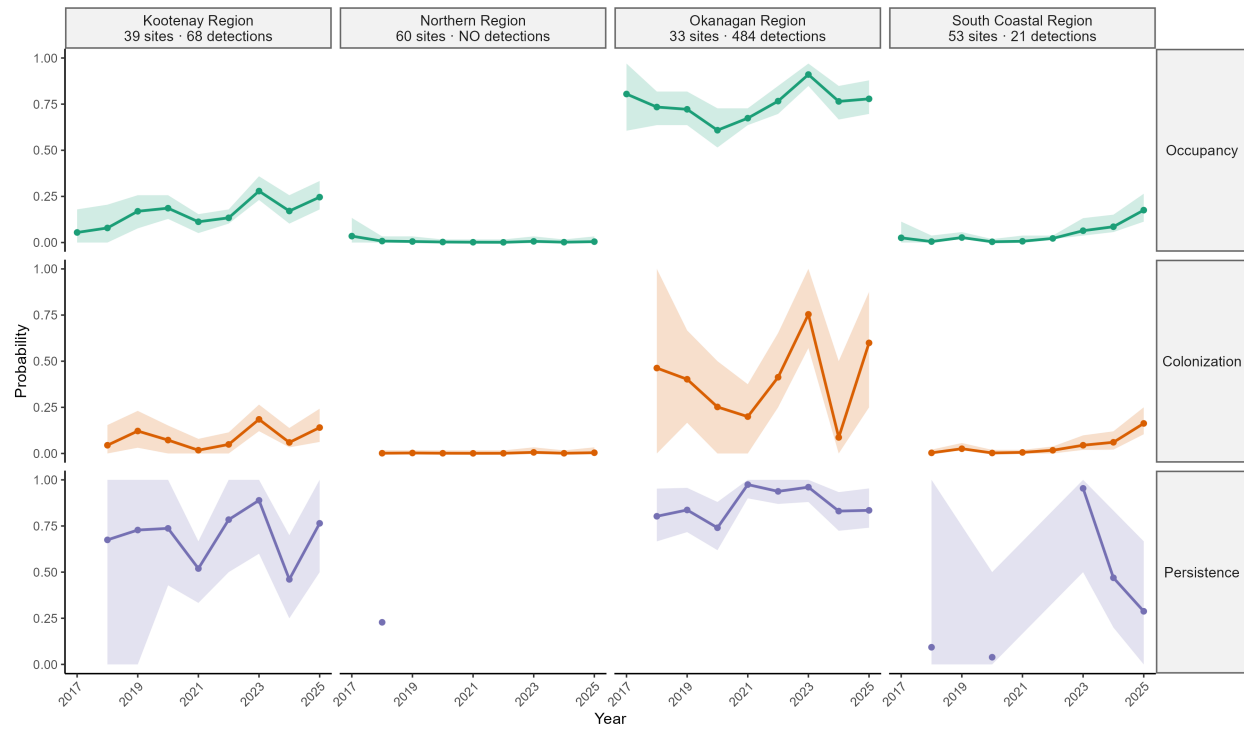
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

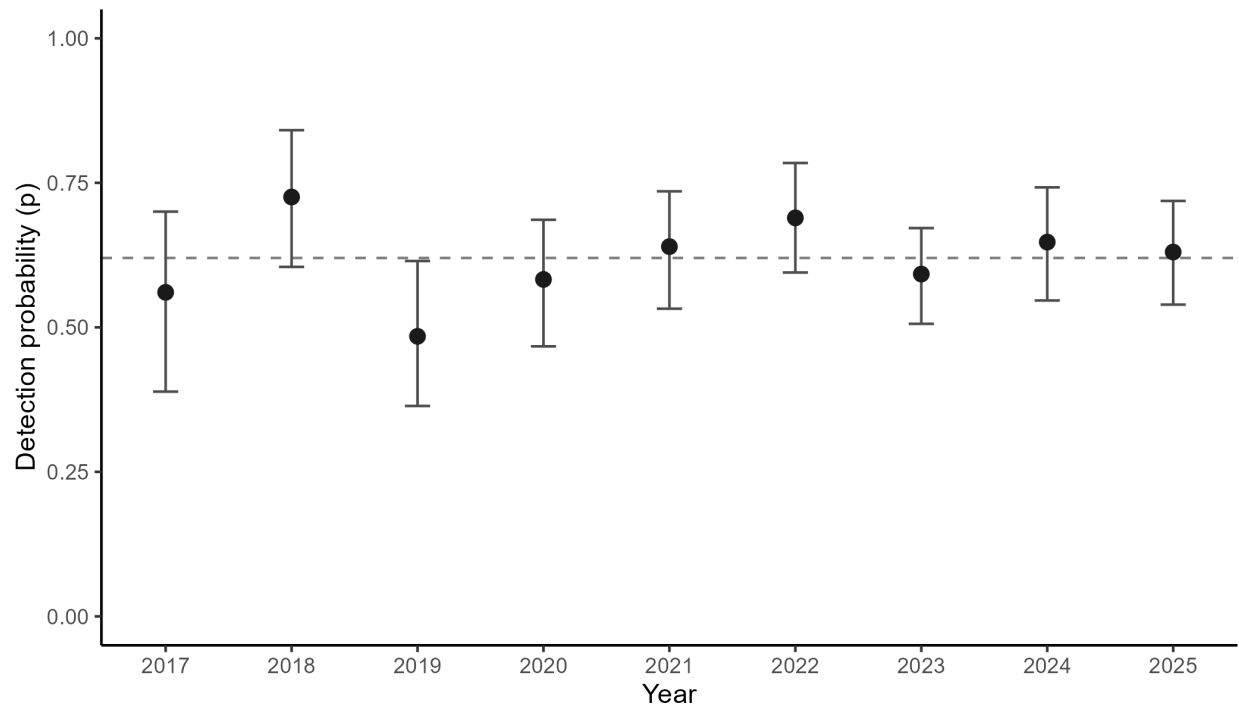
MYCI — Western small-footed myotis: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



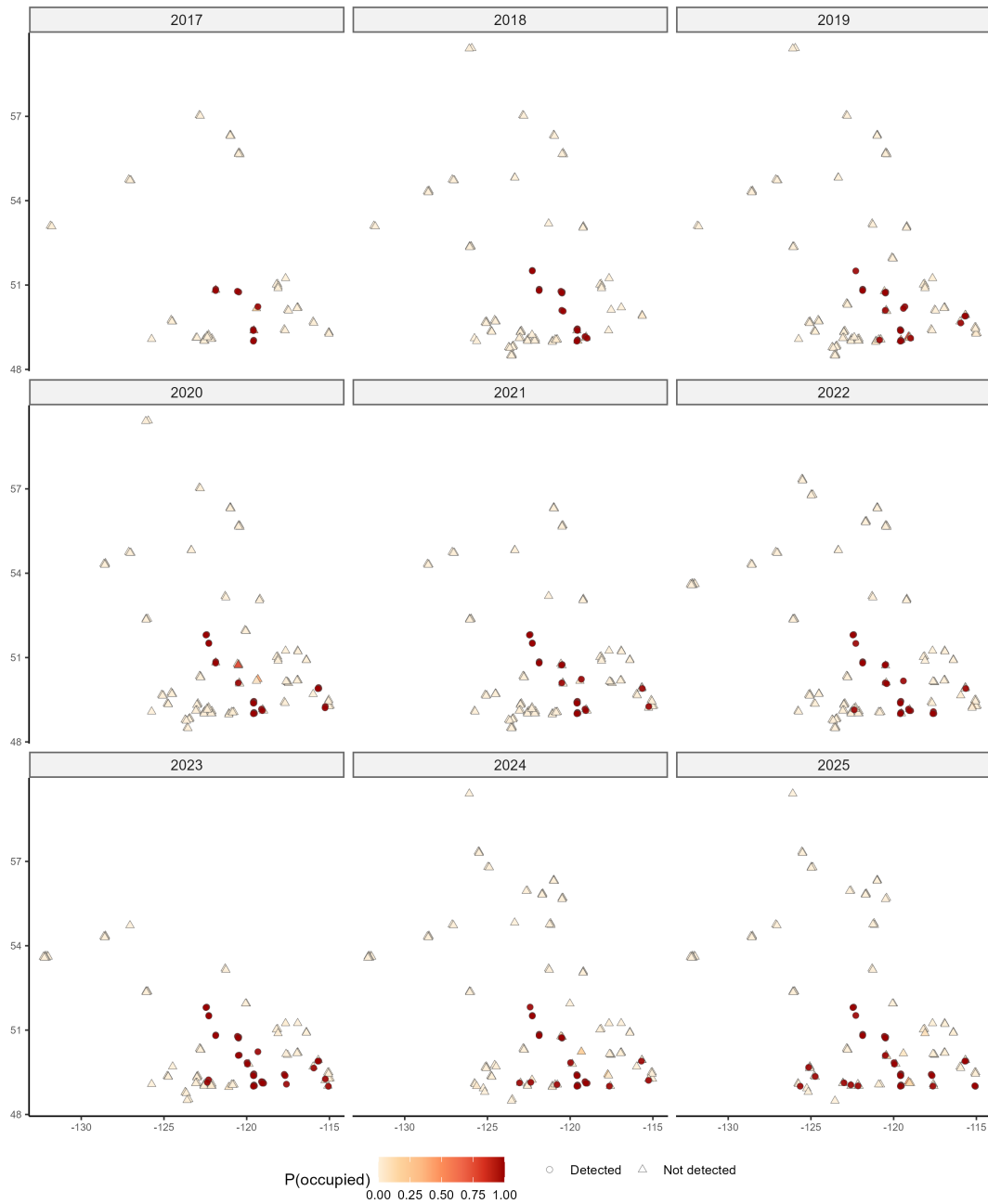
MYCI — Western small-footed myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



MYCI — Western small-footed myotis: P(occupancy) by site and year

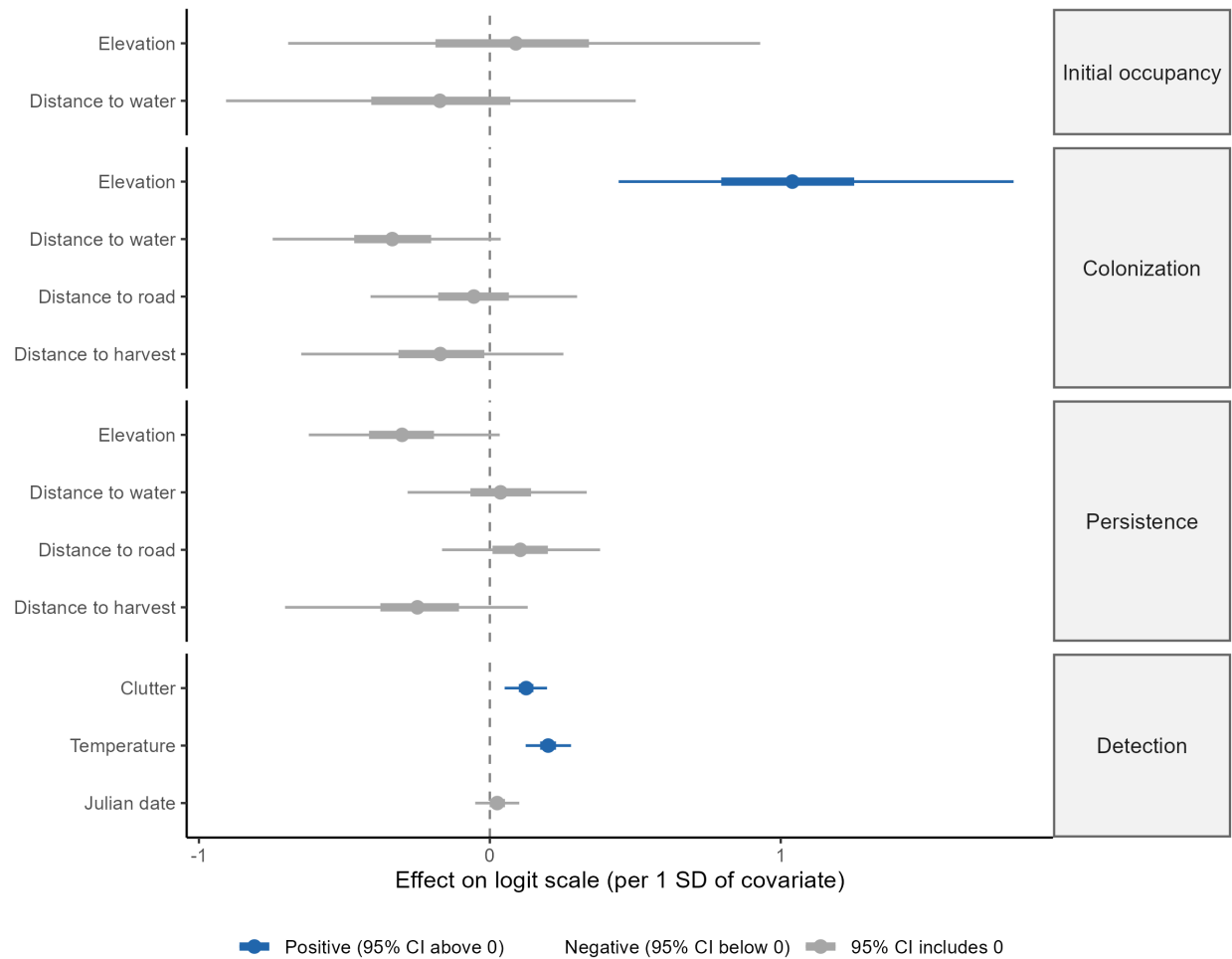
Surveyed site-years only. Triangles = not detected that year; their colour is model inference.



7.10 Long-eared Myotis (*Myotis evotis*)

MYEV — Western long-eared myotis: covariate effects

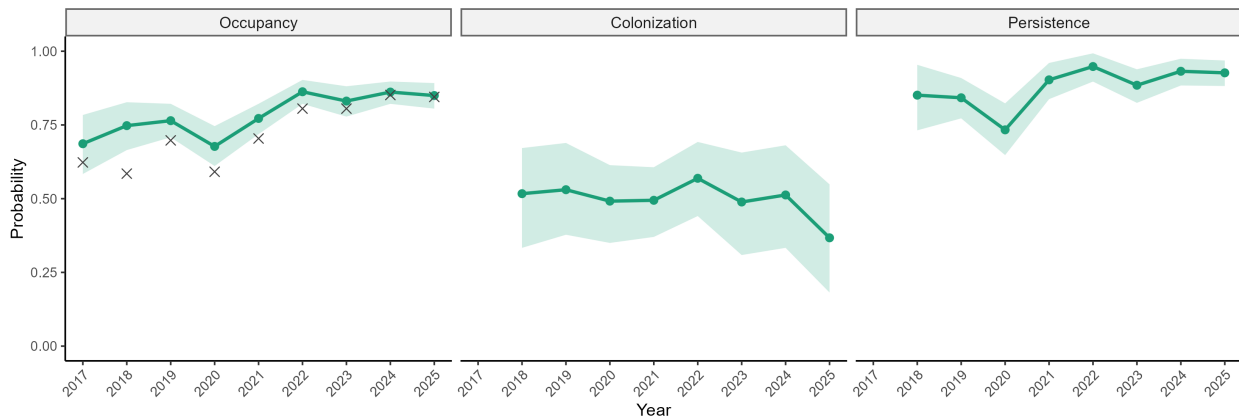
Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



Distances were log(x + 1) transformed before standardising.

MYEV — Western long-eared myotis: BC-wide dynamics

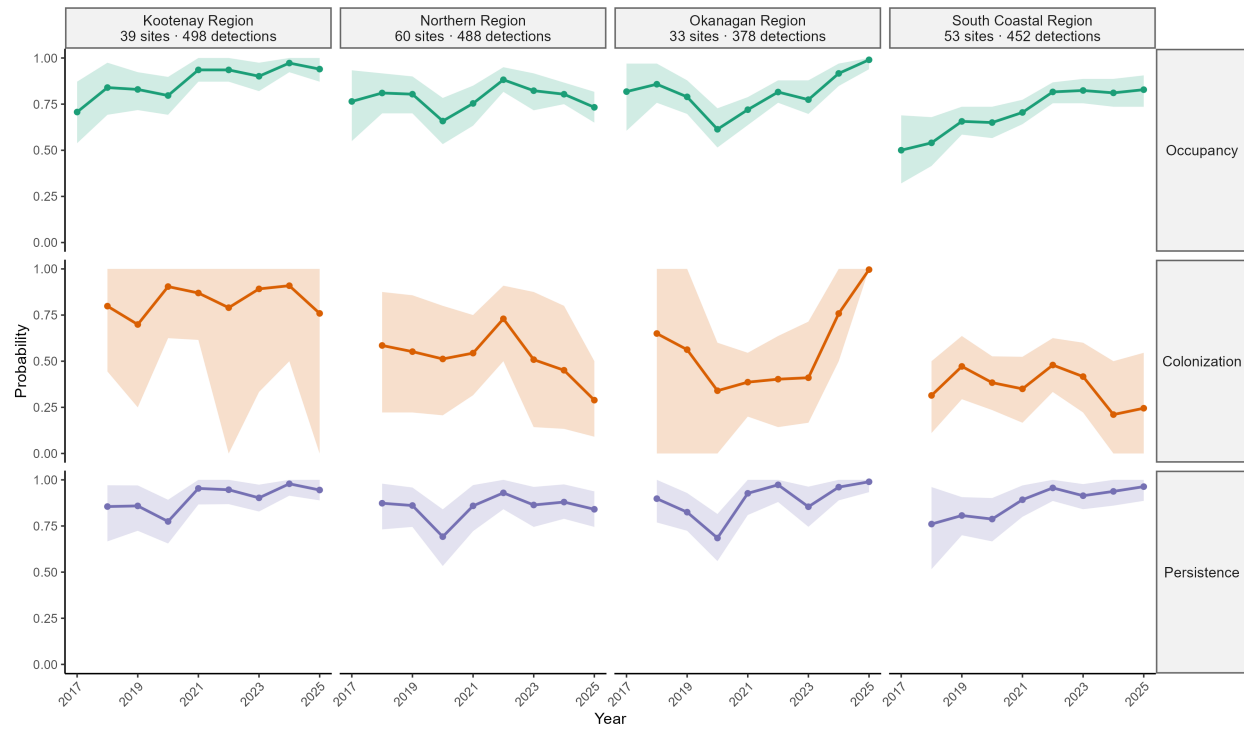
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

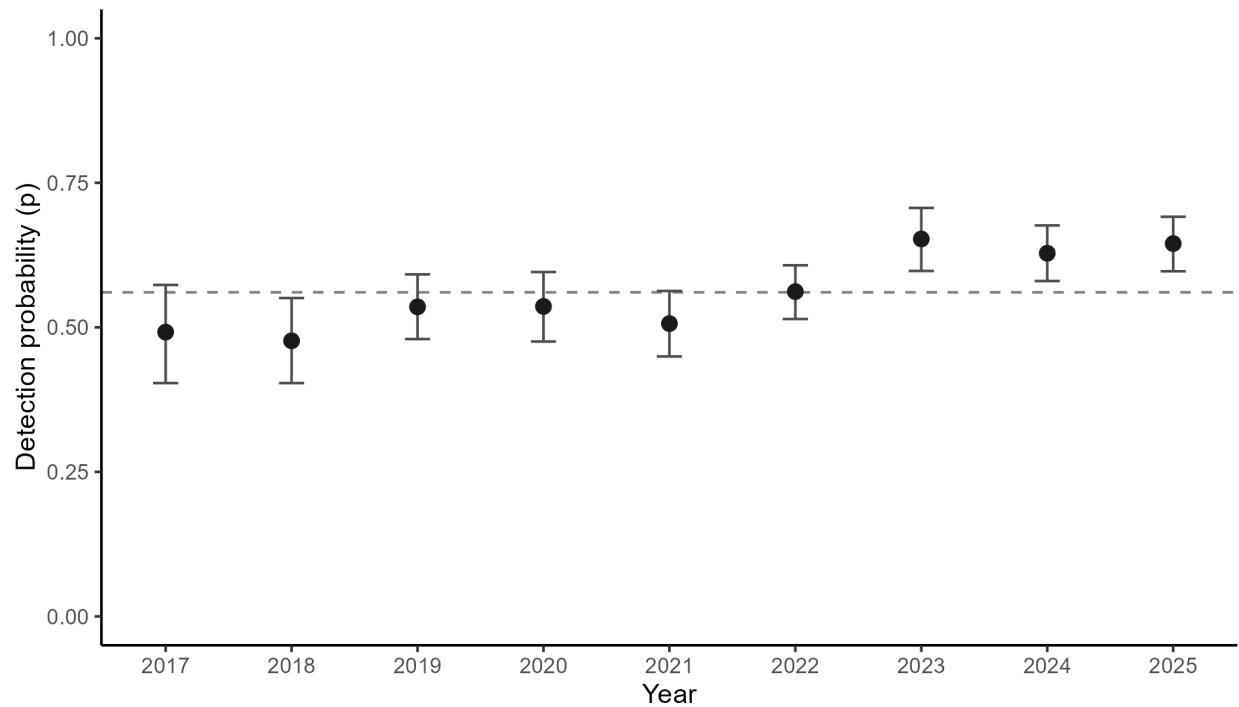
MYEV — Western long-eared myotis: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



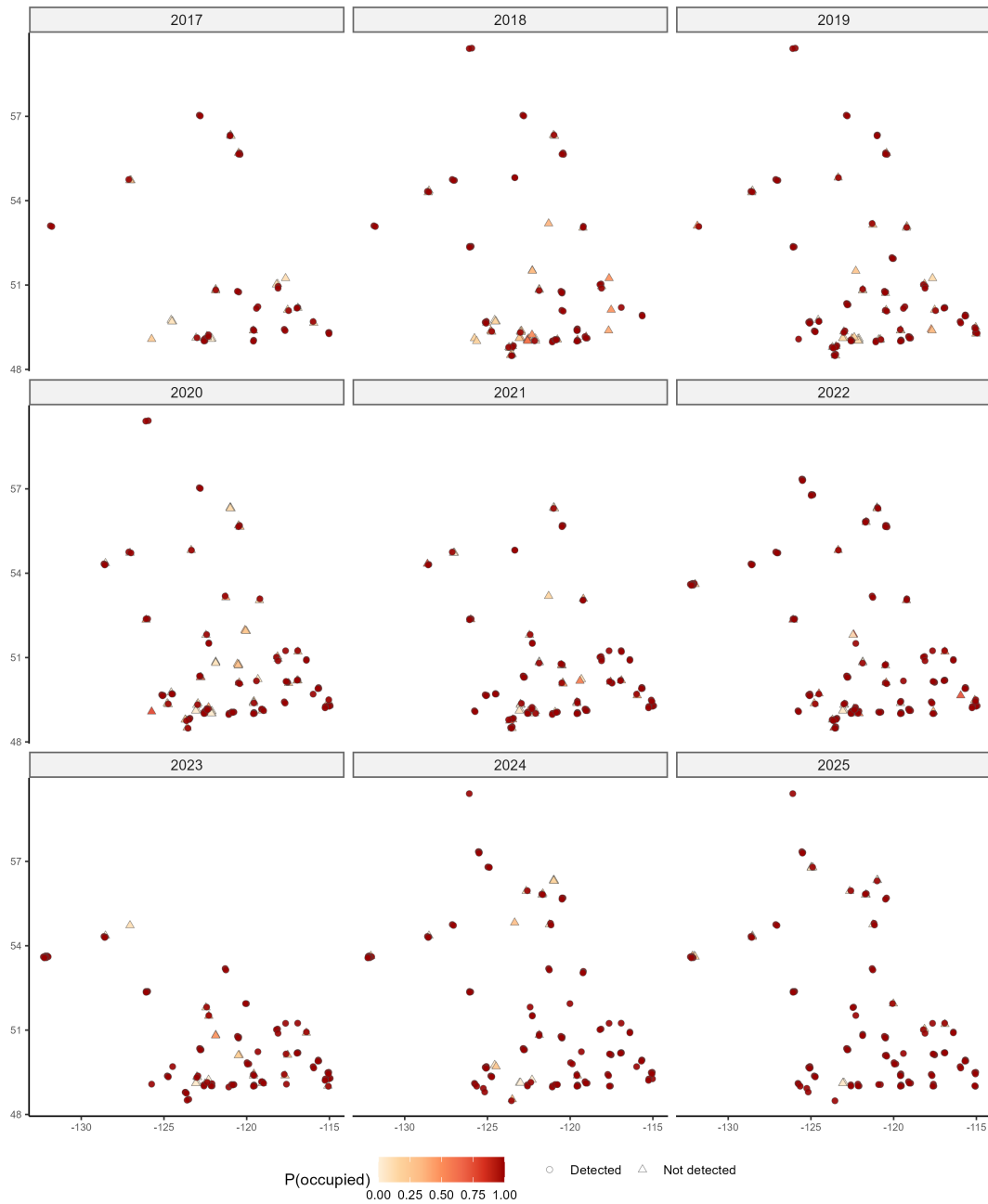
MYEV — Western long-eared myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



MYEV — Western long-eared myotis: P(occupancy) by site and year

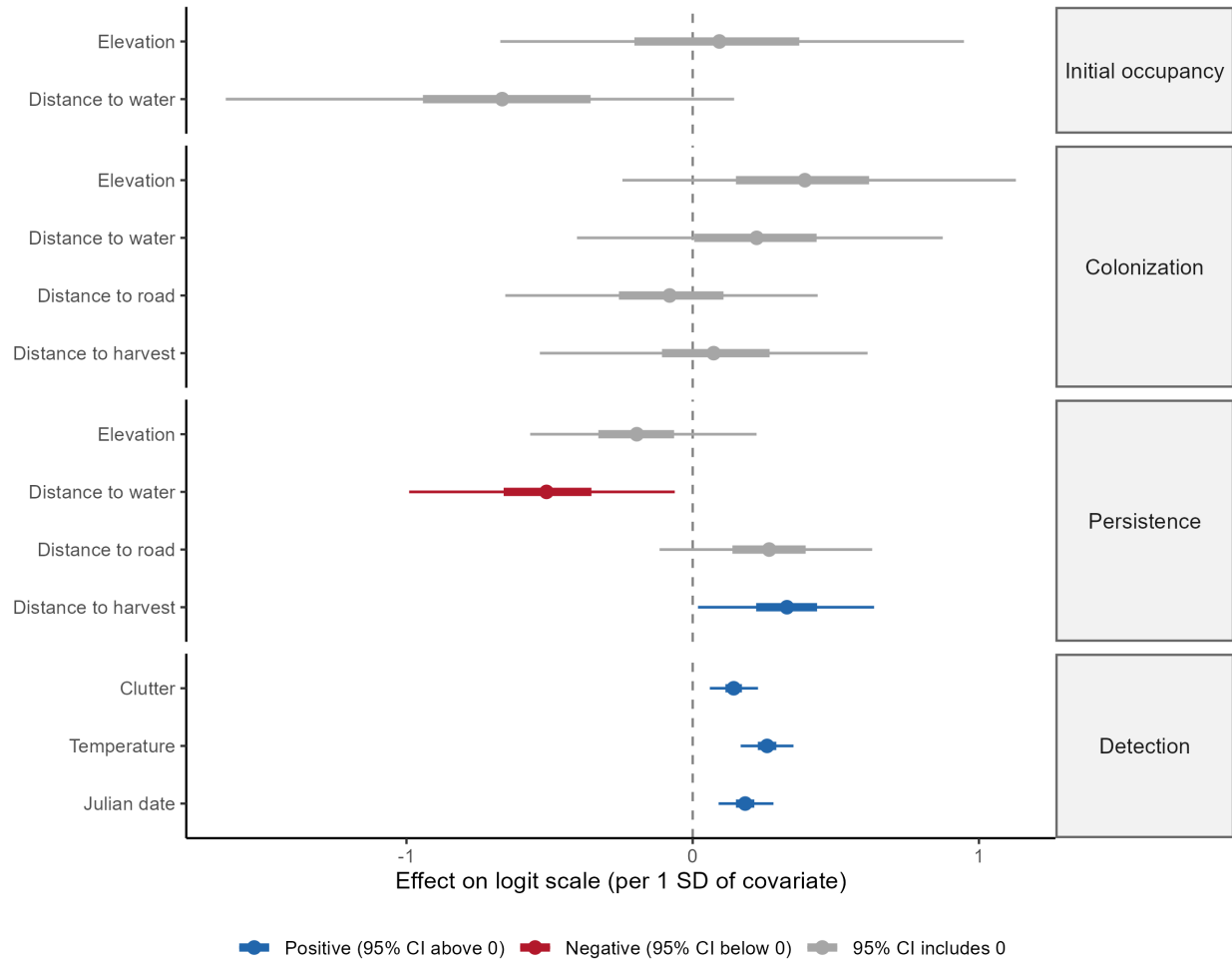
Surveyed site-years only. Triangles = not detected that year; their colour is model inference.



7.11 Little brown Myotis (*Myotis lucifugus*)

MYLU — Little brown myotis: covariate effects

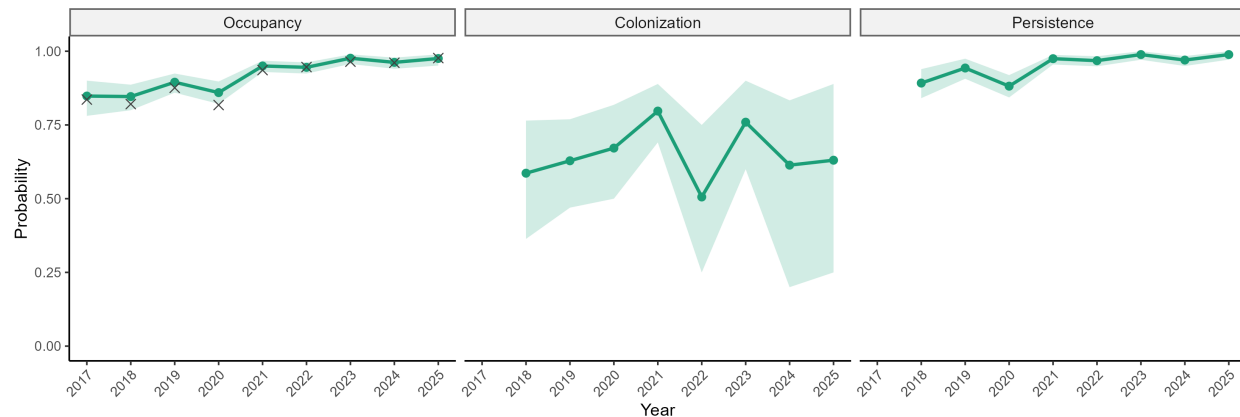
Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



Distances were $\log(x + 1)$ transformed before standardising.

MYLU — Little brown myotis: BC-wide dynamics

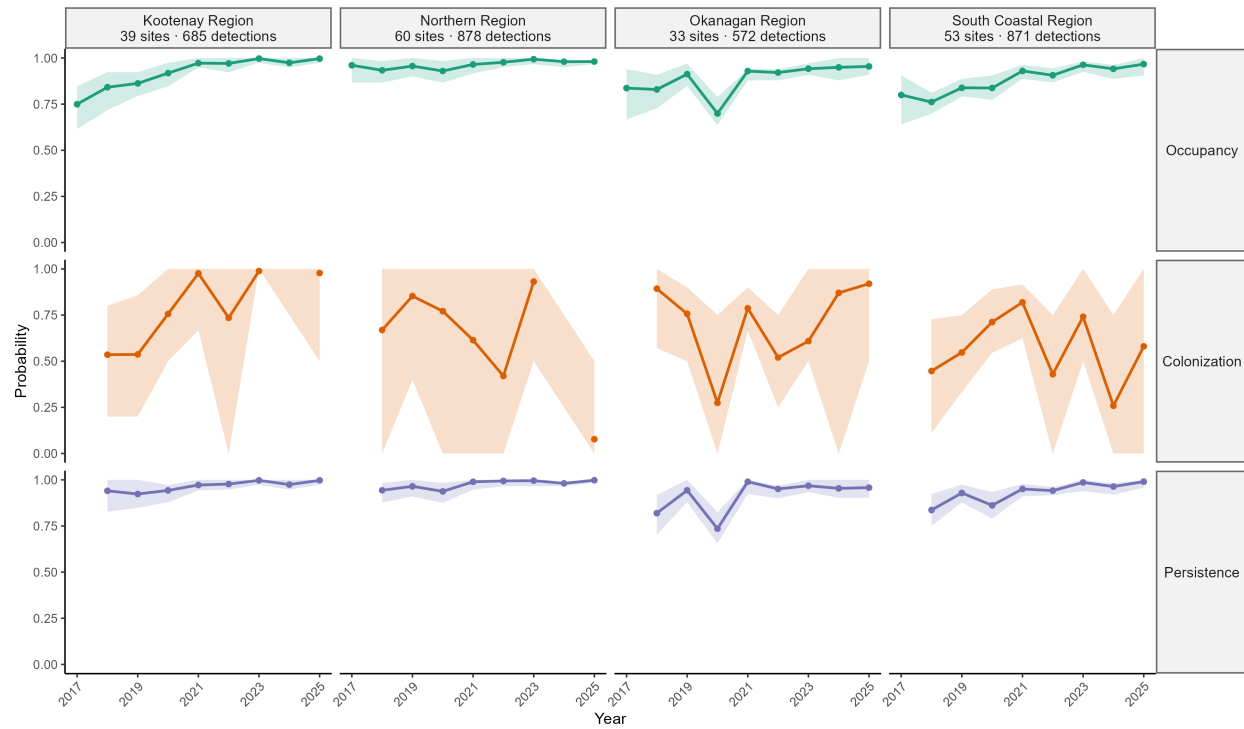
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

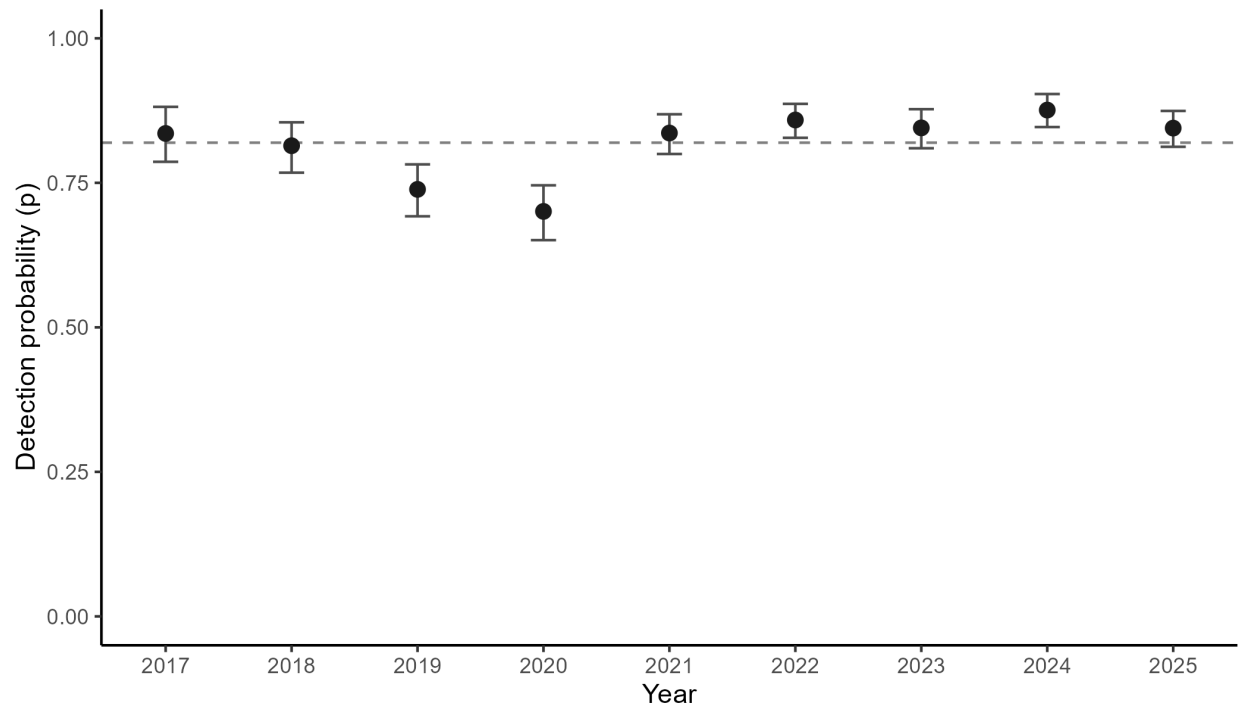
MYLU — Little brown myotis: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



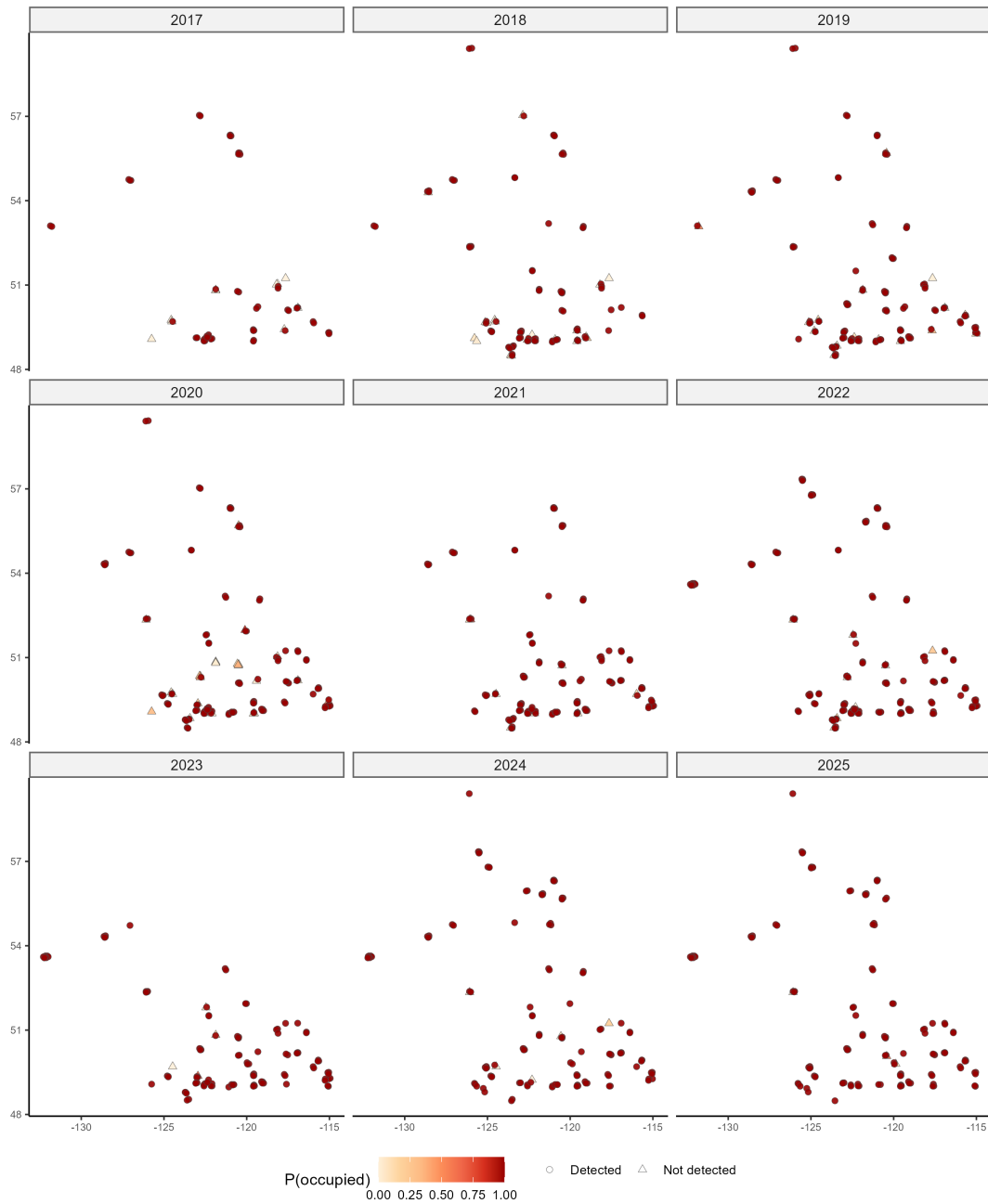
MYLU — Little brown myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

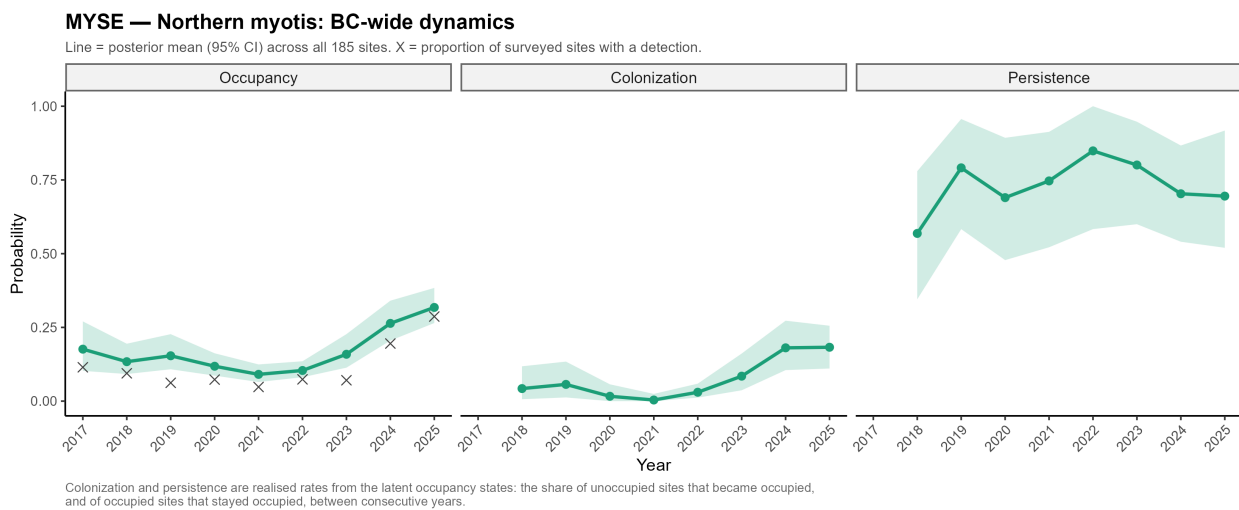
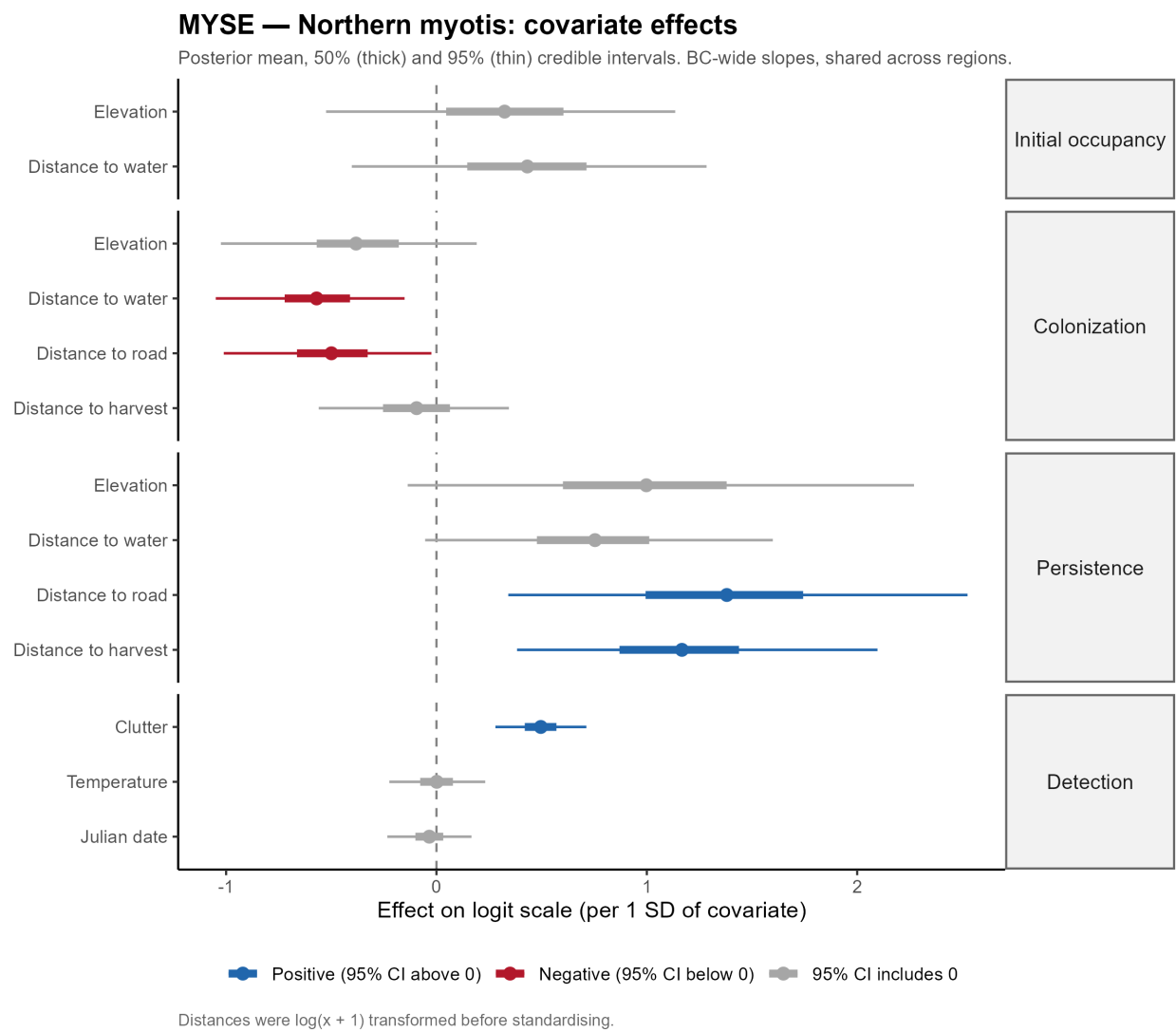


MYLU — Little brown myotis: P(occupancy) by site and year

Surveyed site-years only. Triangles = not detected that year; their colour is model inference.

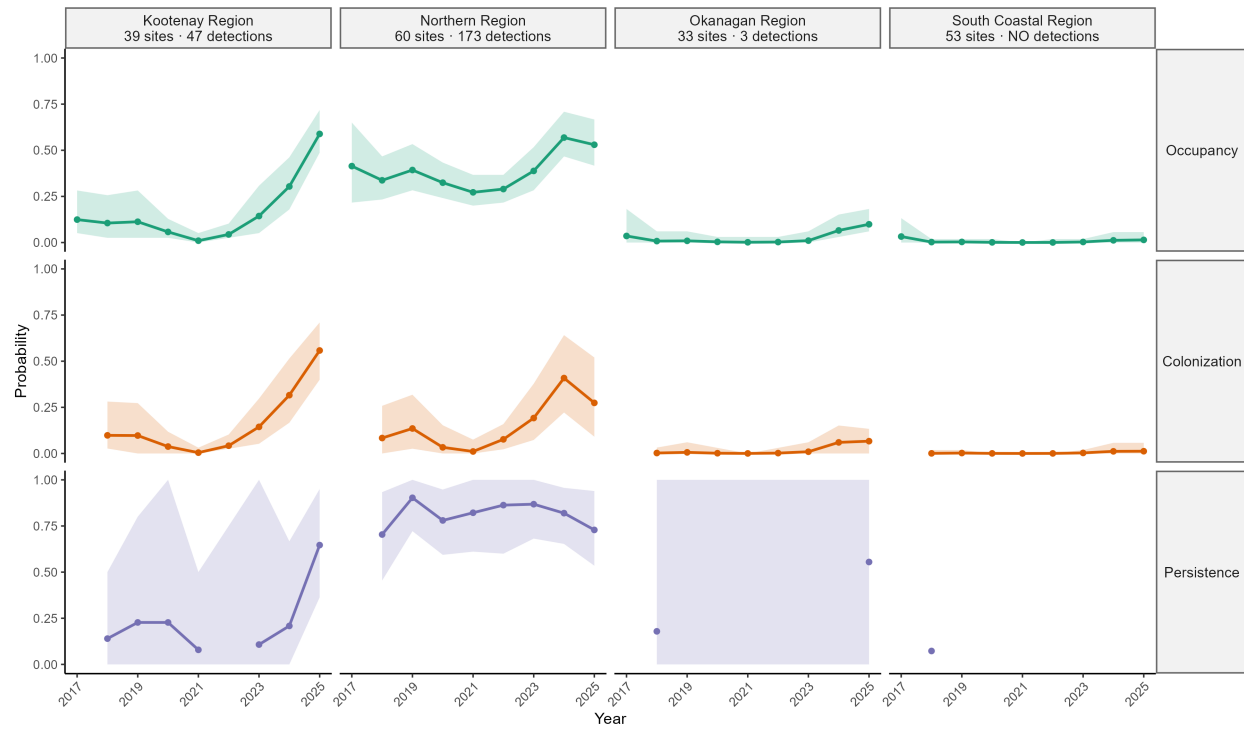


7.12 Northern Myotis (*Myotis septentrionalis*)



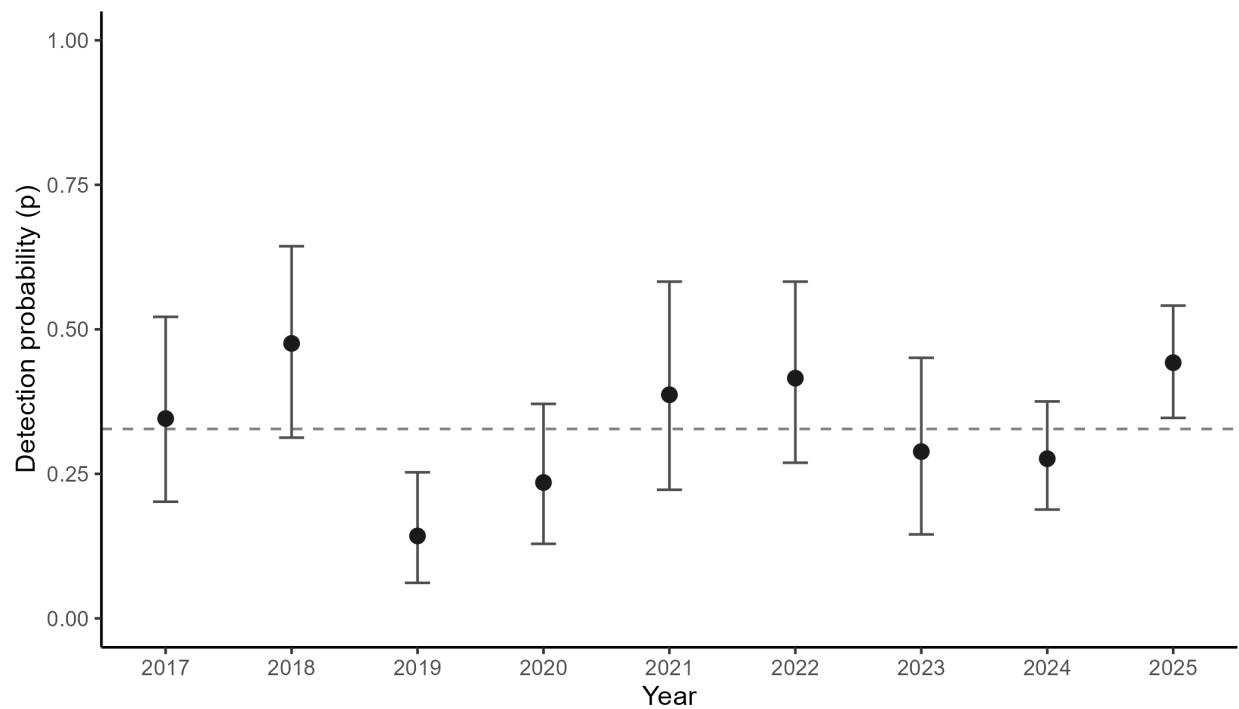
MYSE — Northern myotis: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



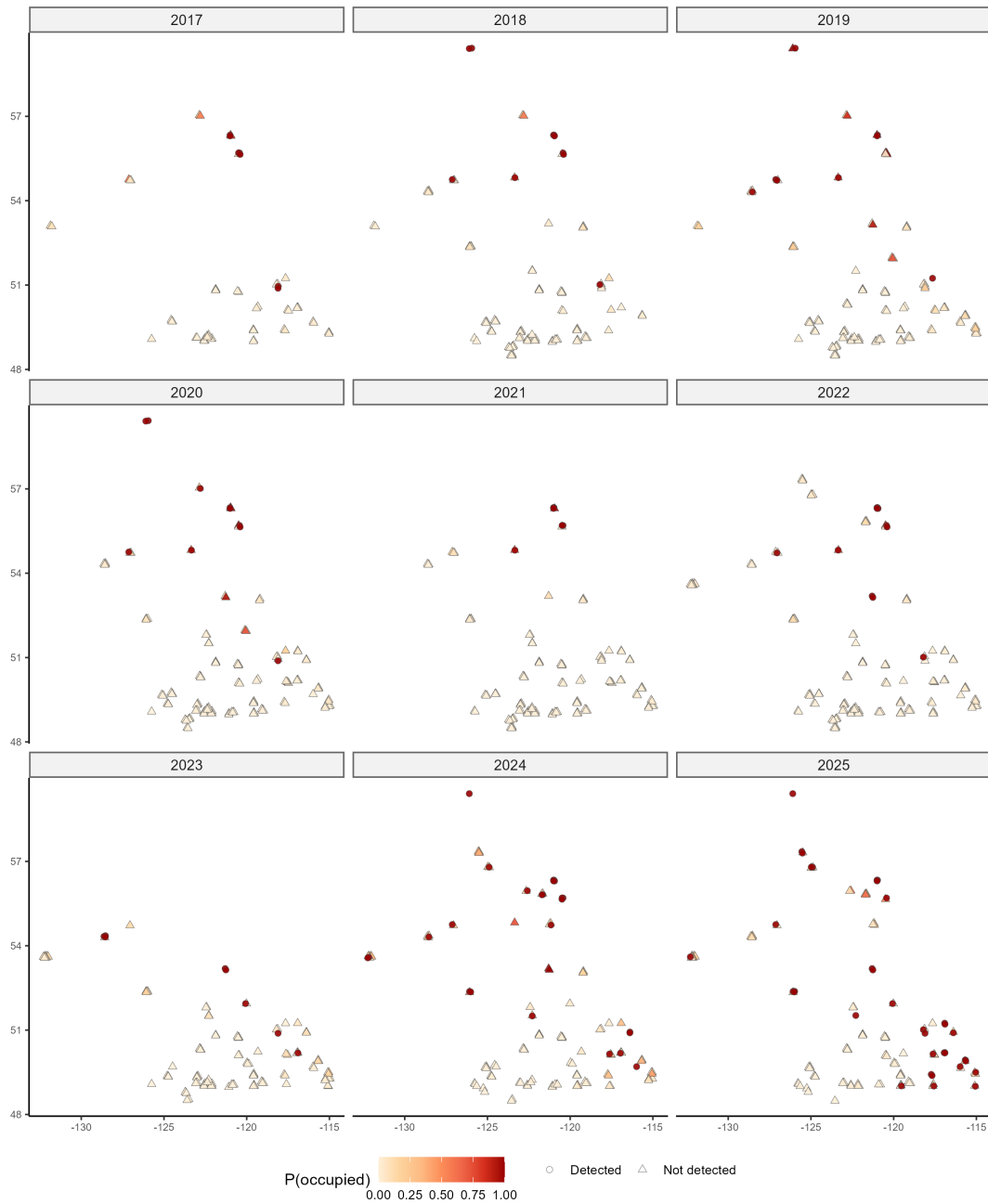
MYSE — Northern myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

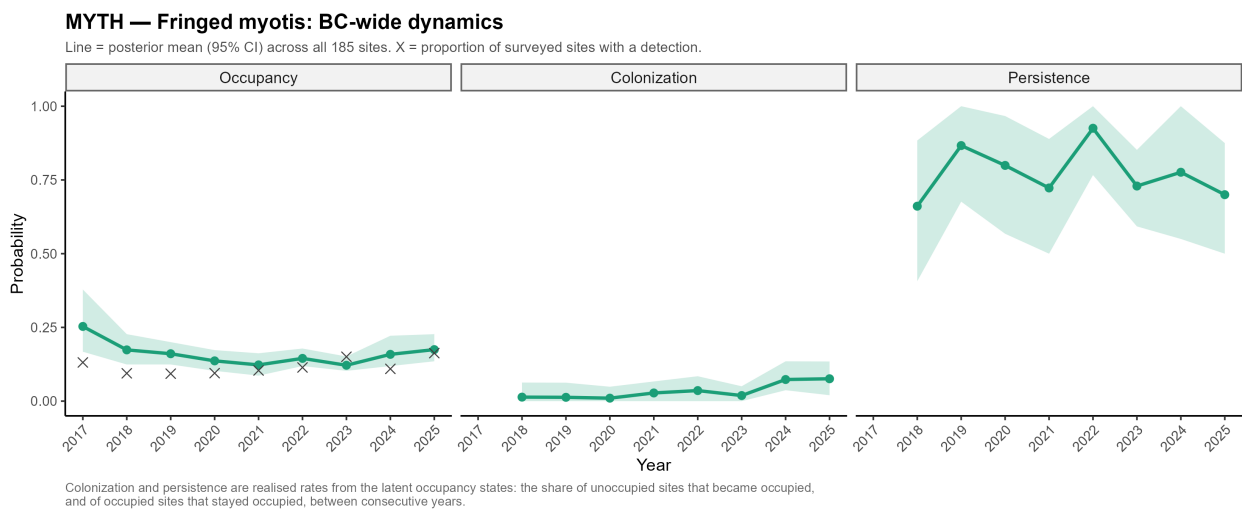
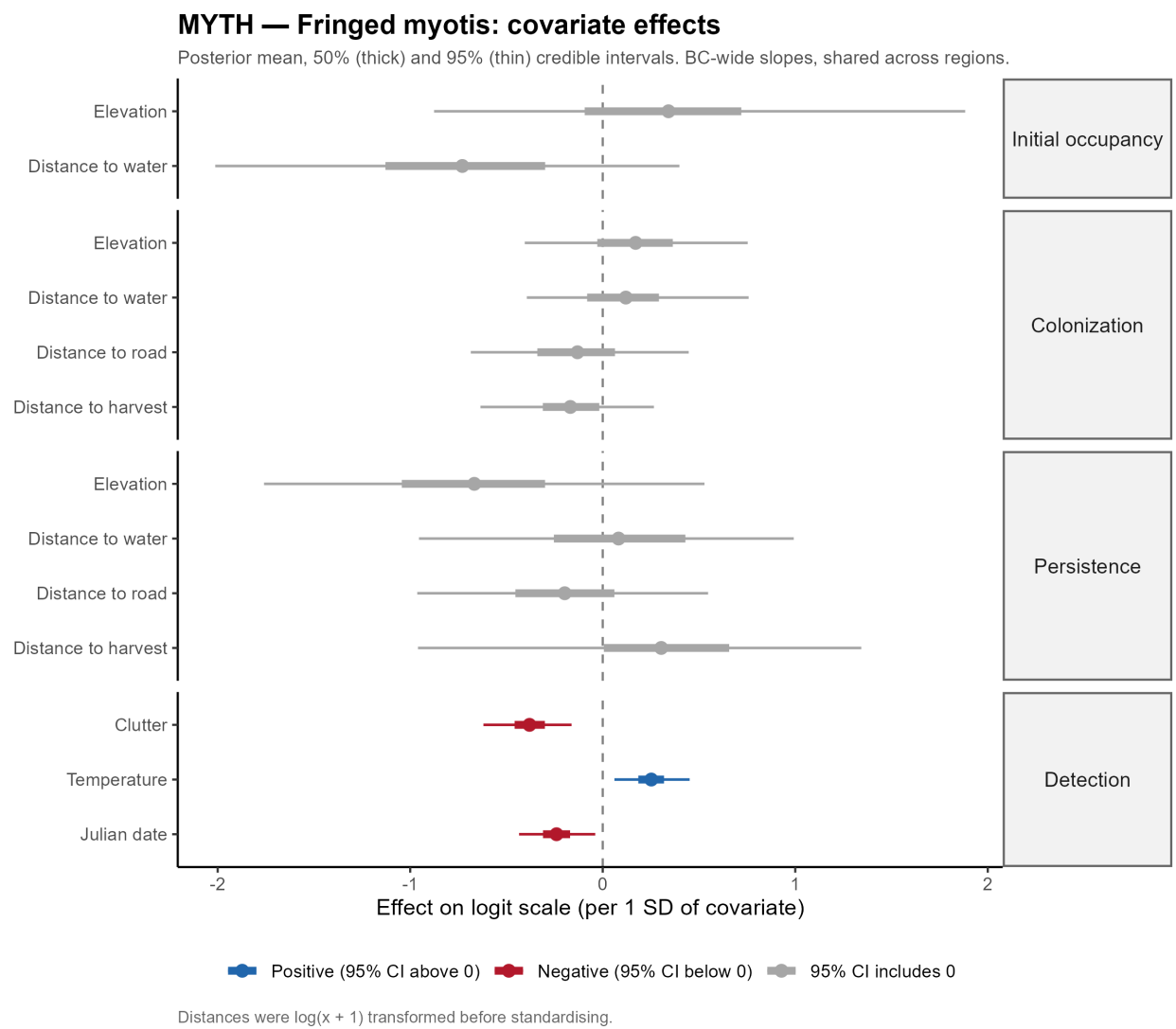


MYSE — Northern myotis: P(occupancy) by site and year

Surveyed site-years only. Triangles = not detected that year; their colour is model inference.

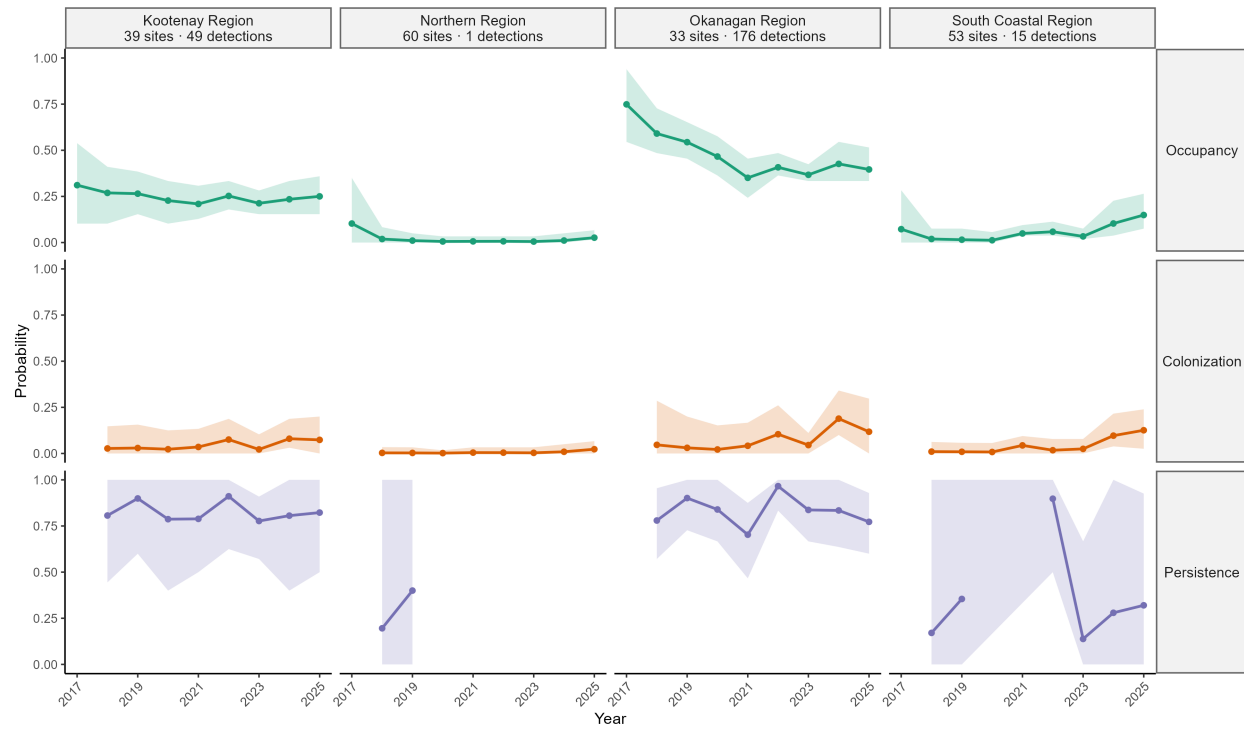


7.13 Fringed Myotis (*Myotis thysanodes*)



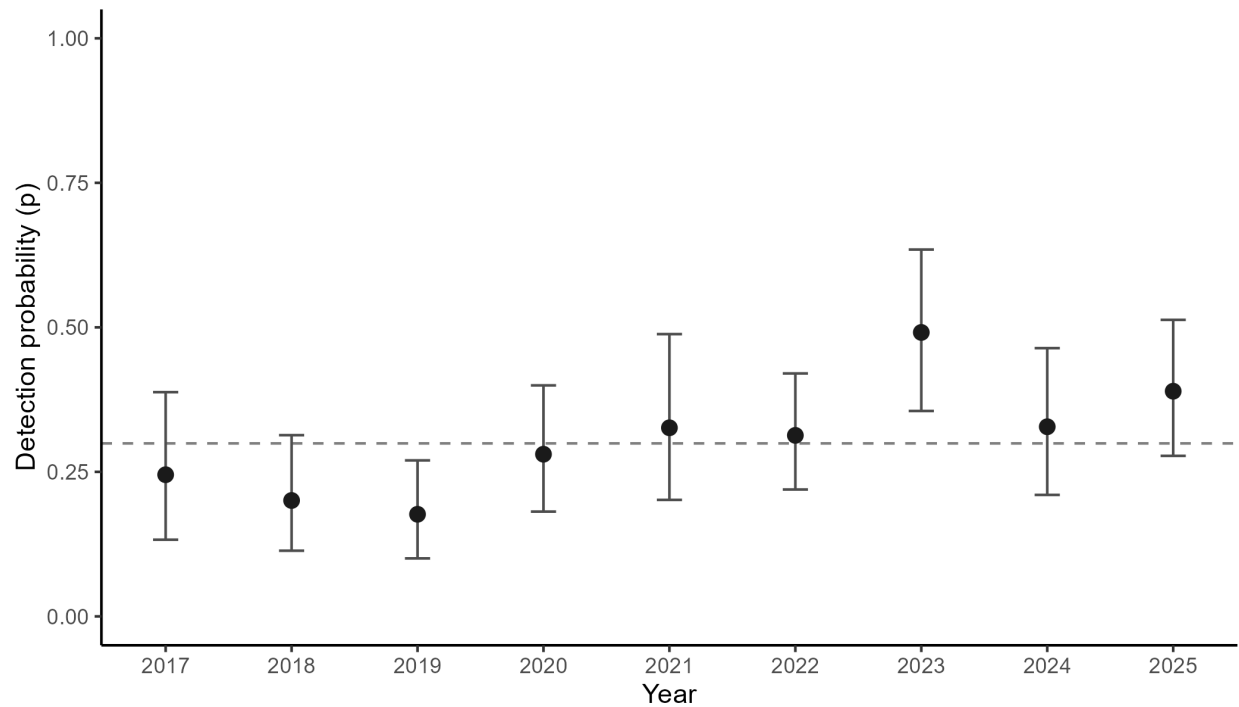
MYTH — Fringed myotis: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



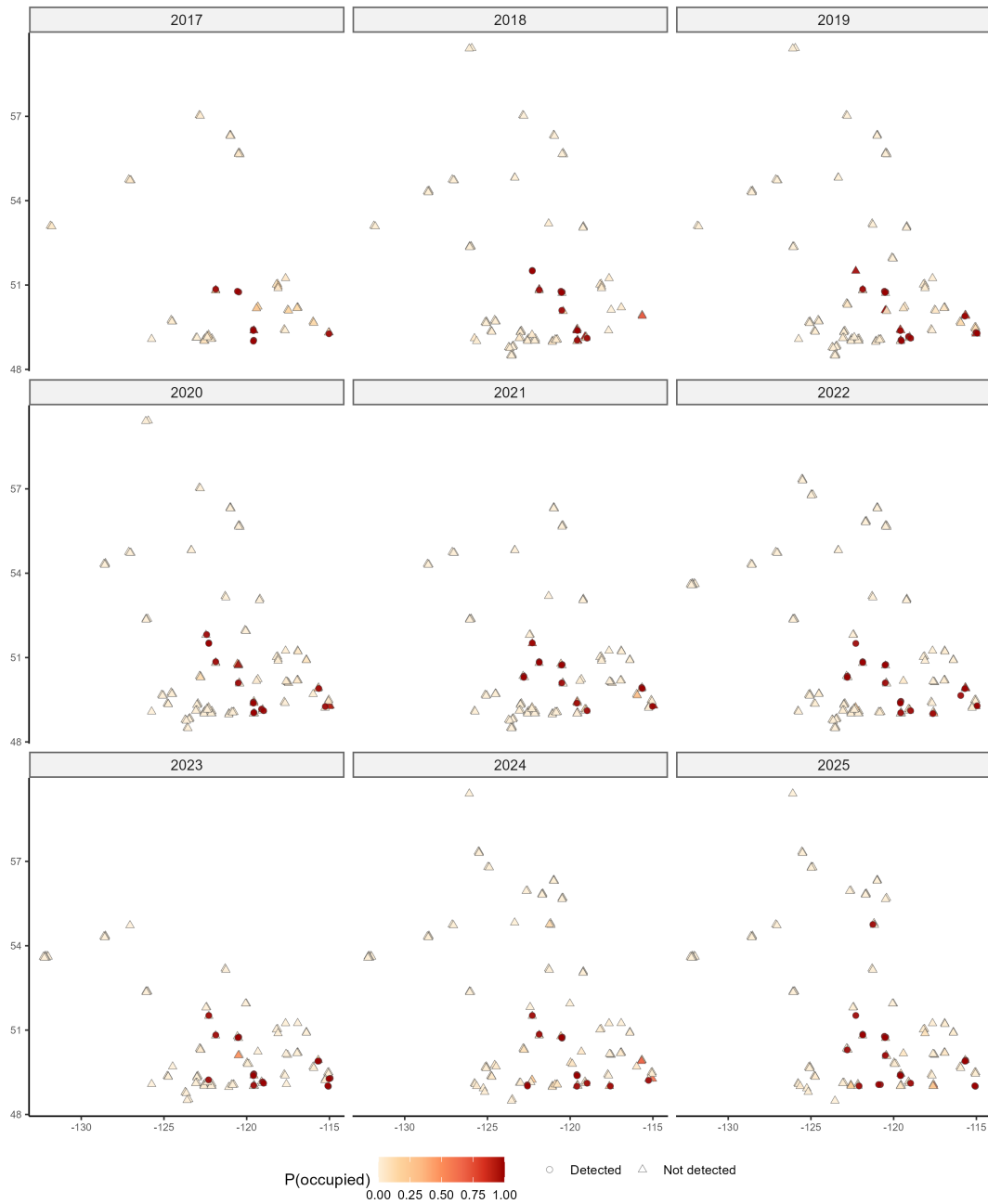
MYTH — Fringed myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

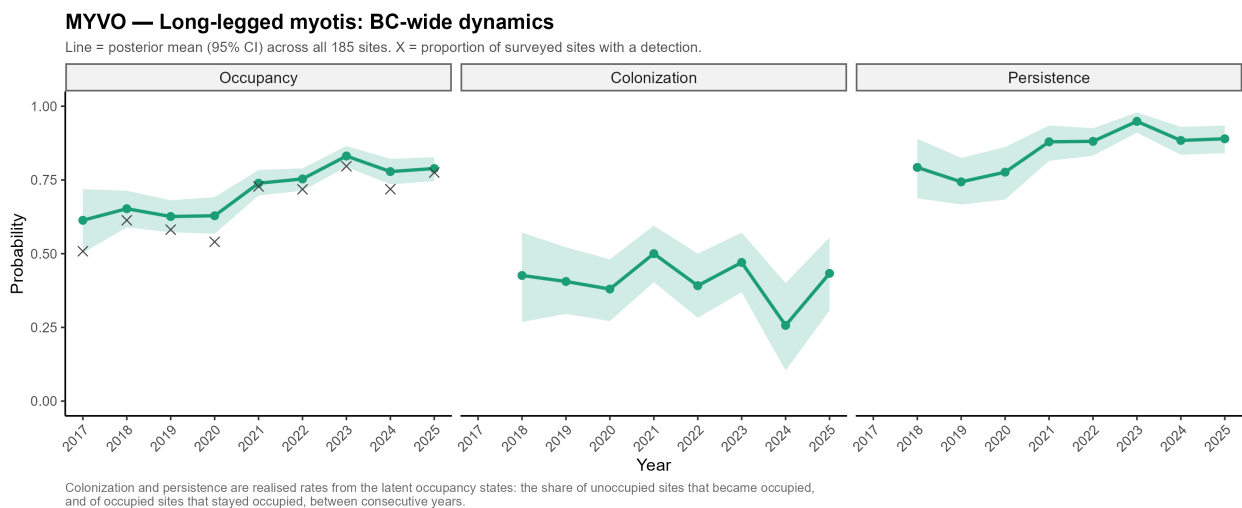
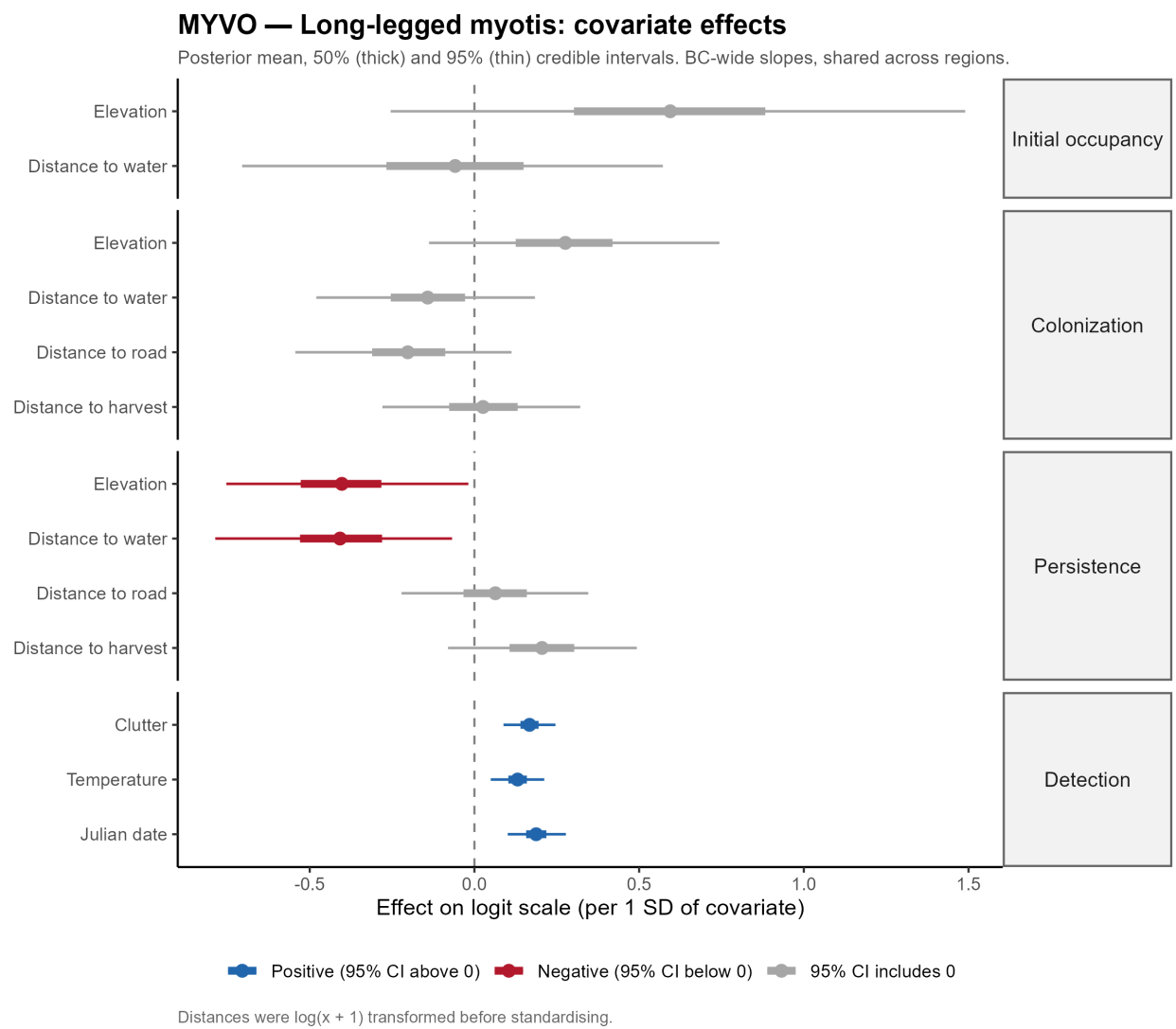


MYTH — Fringed myotis: P(occupancy) by site and year

Surveyed site-years only. Triangles = not detected that year; their colour is model inference.

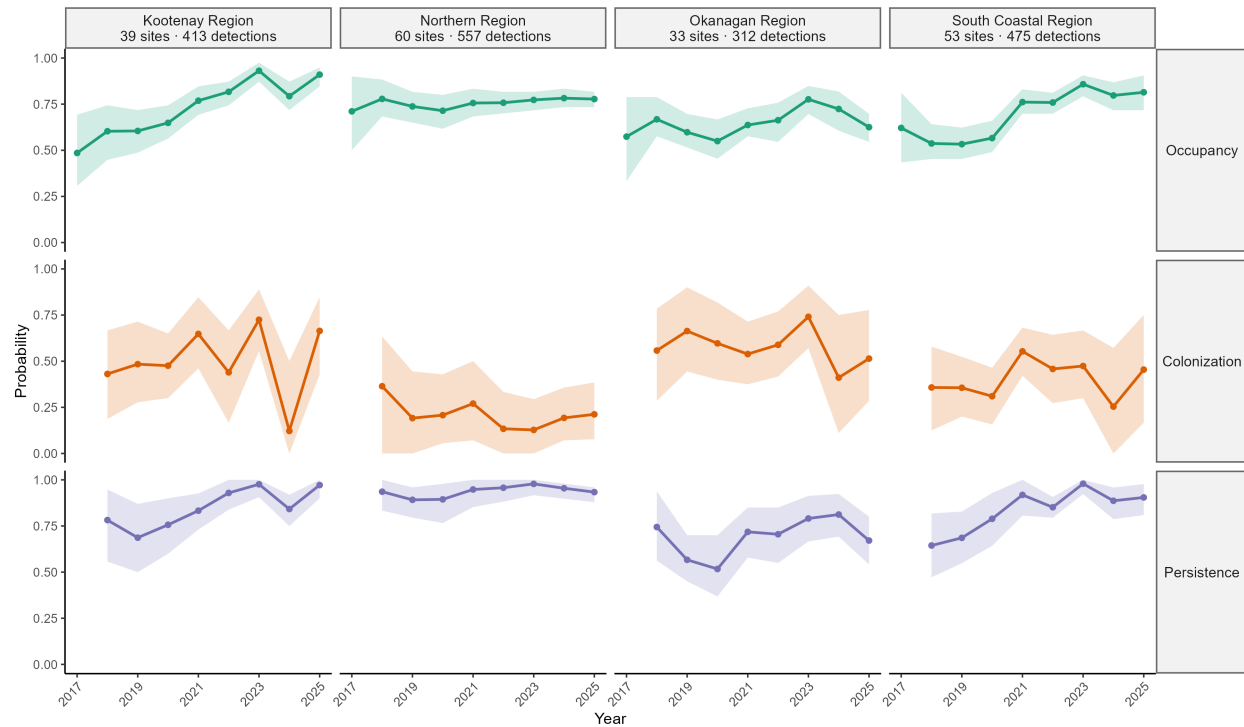


7.14 Long-legged Myotis (*Myotis volans*)



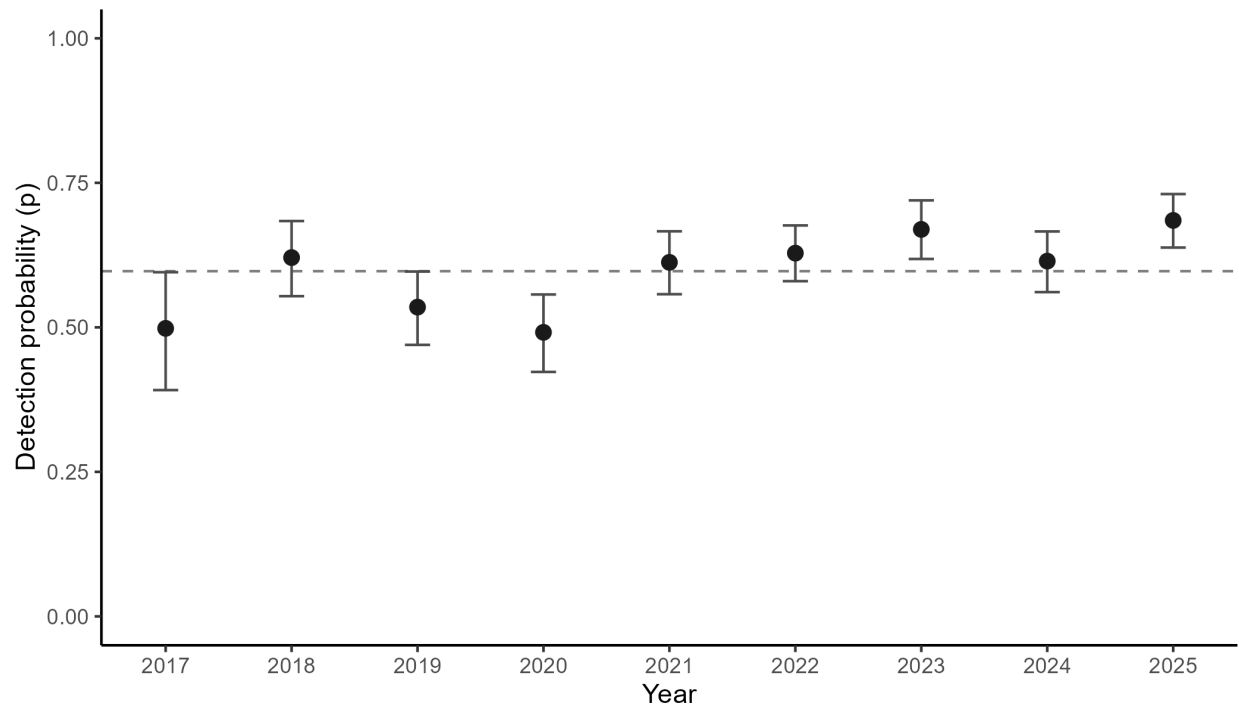
MYVO — Long-legged myotis: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



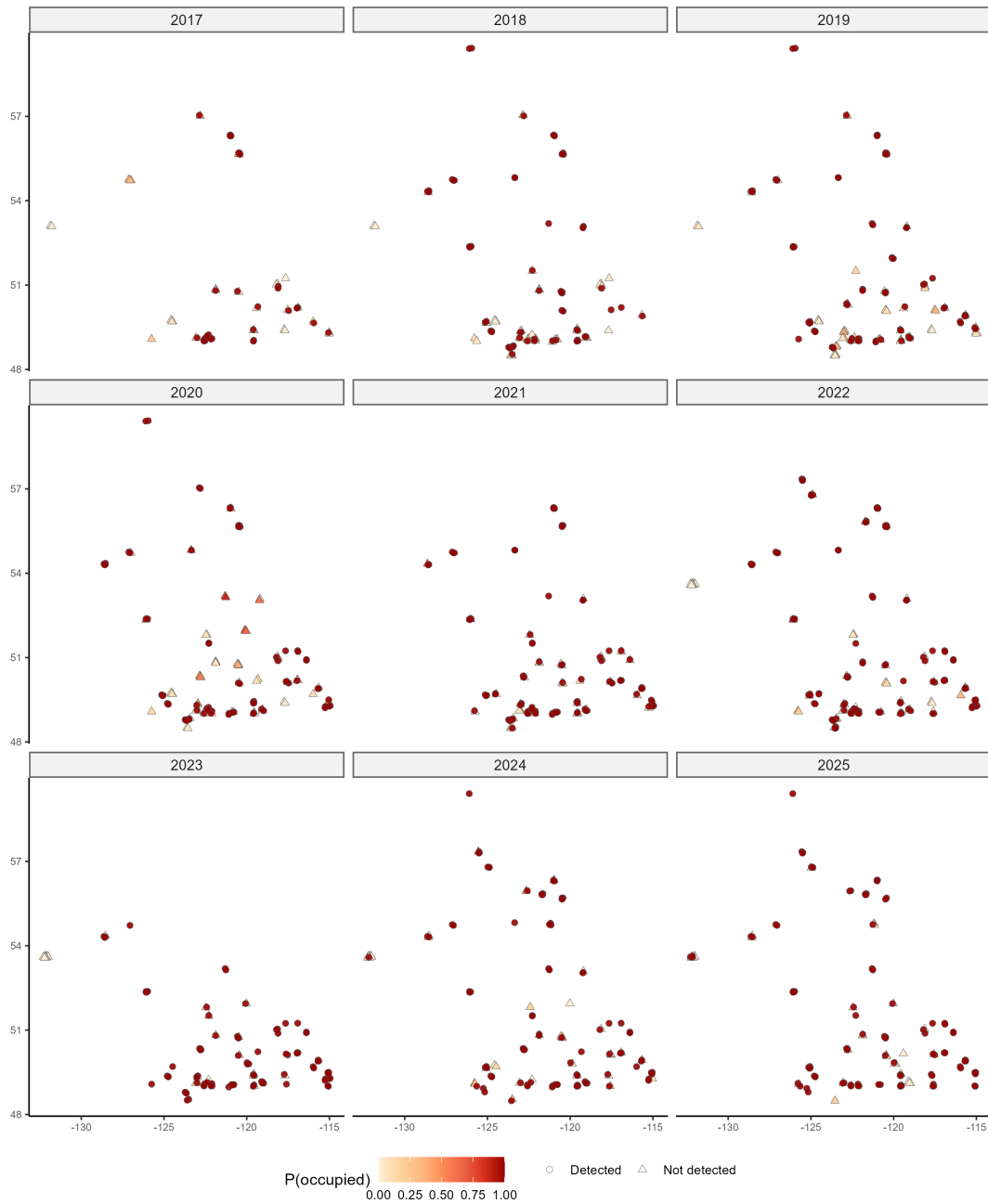
MYVO — Long-legged myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

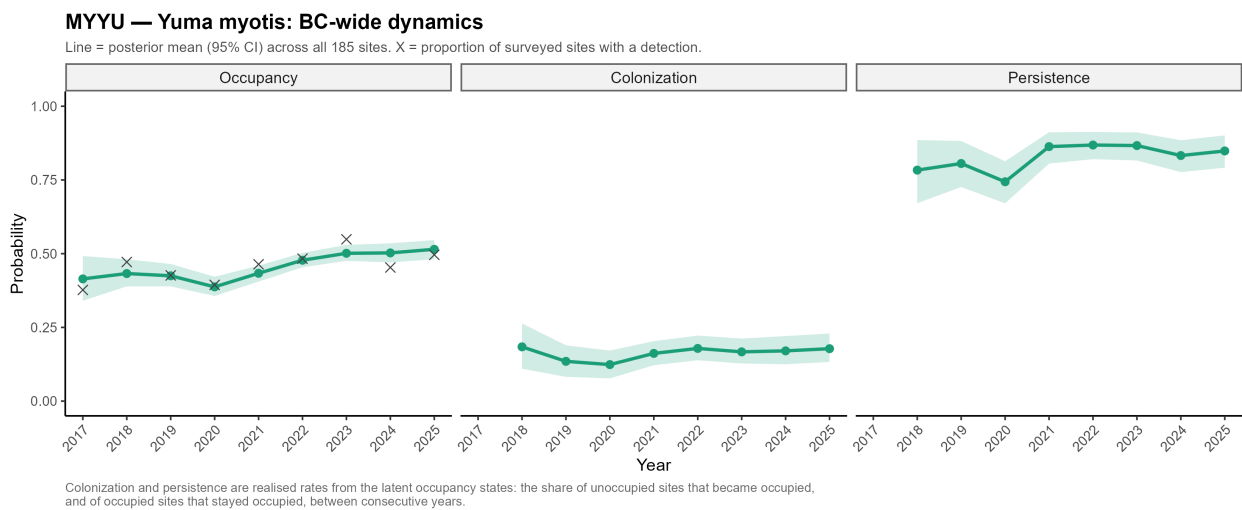
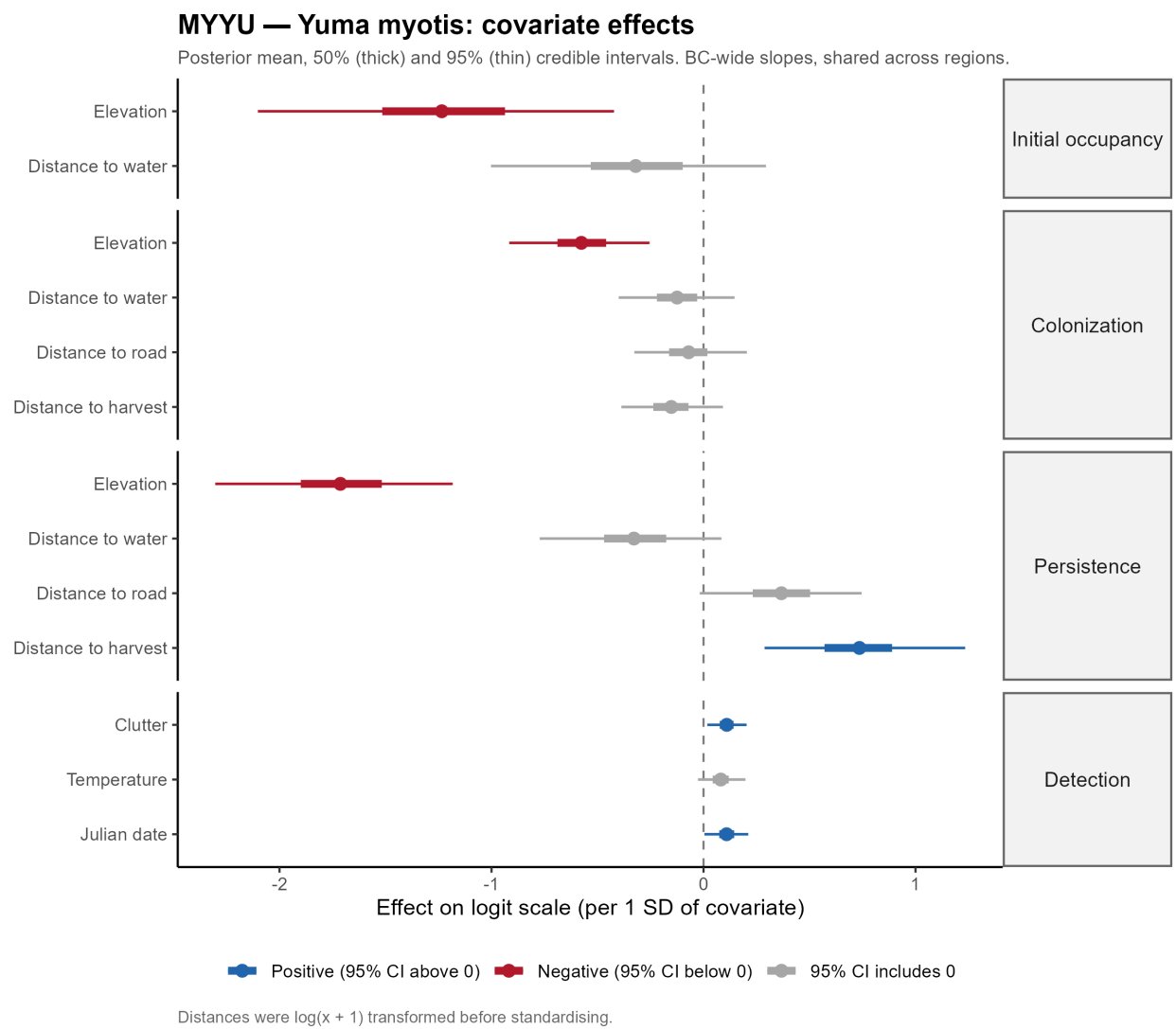


MYVO — Long-legged myotis: P(occupancy) by site and year

Surveyed site-years only. Triangles = not detected that year; their colour is model inference.

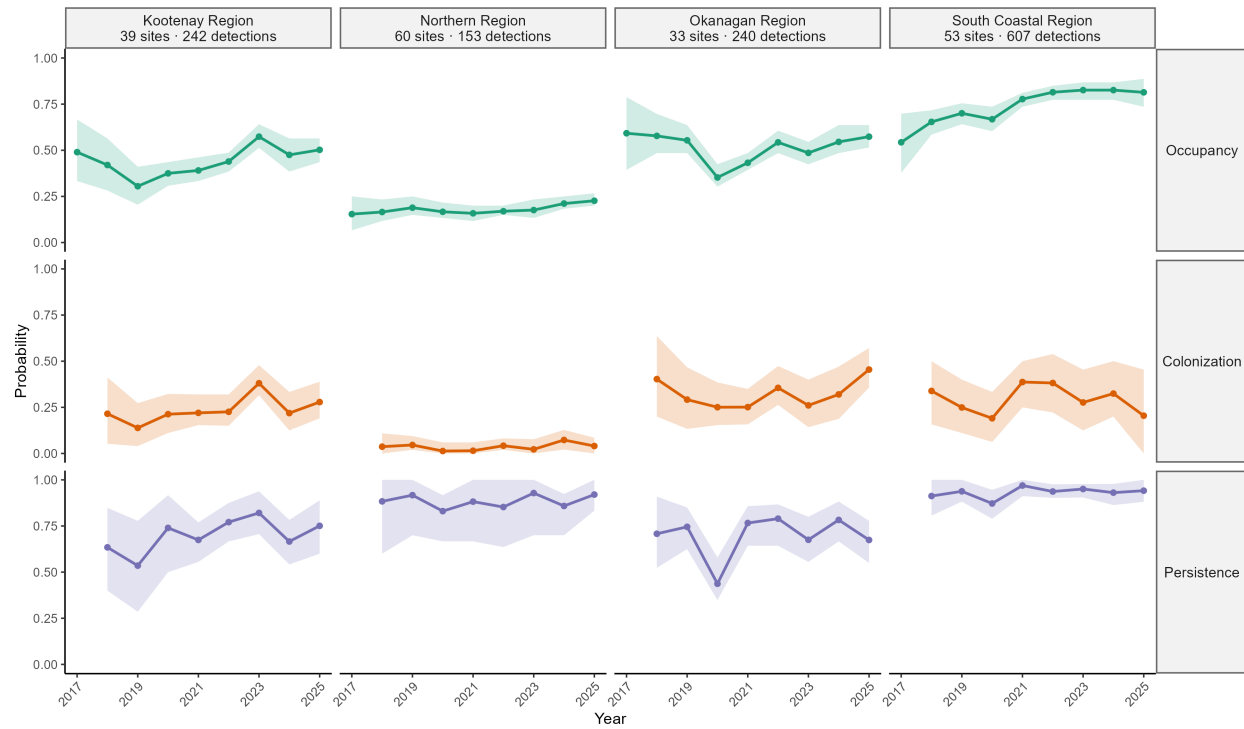


7.15 Yuma Myotis (*Myotis yumanensis*)



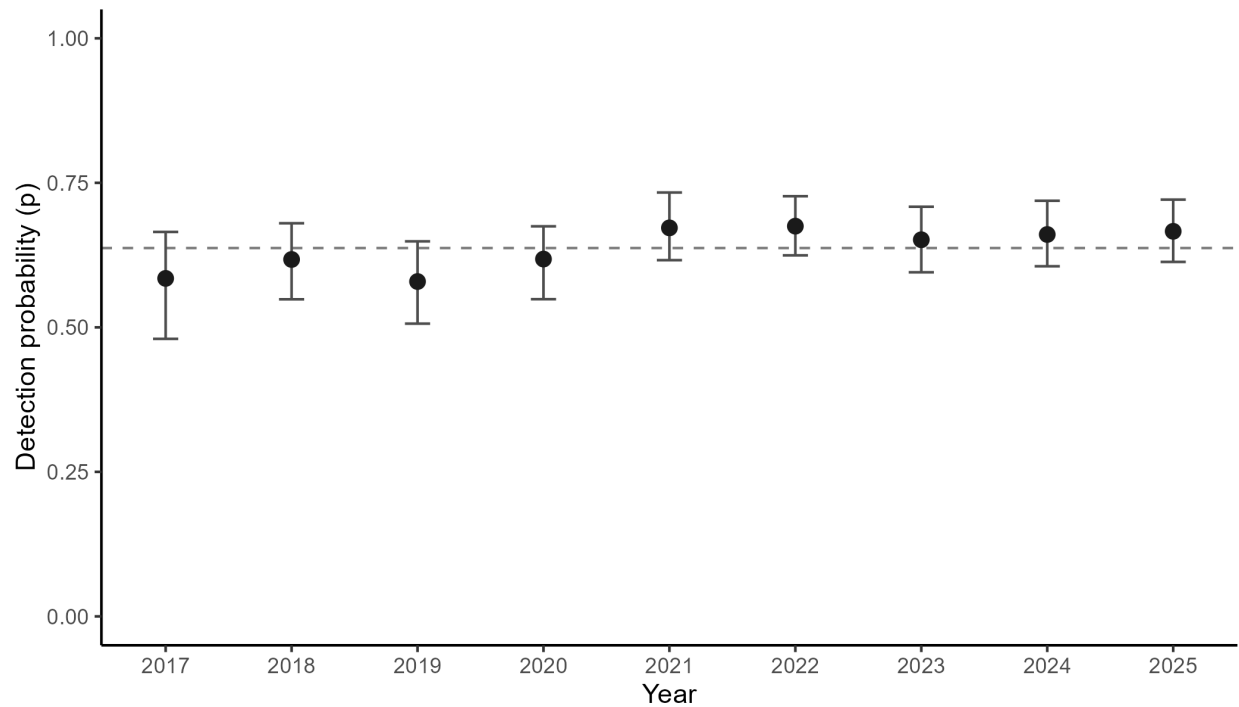
MYYU — Yuma myotis: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



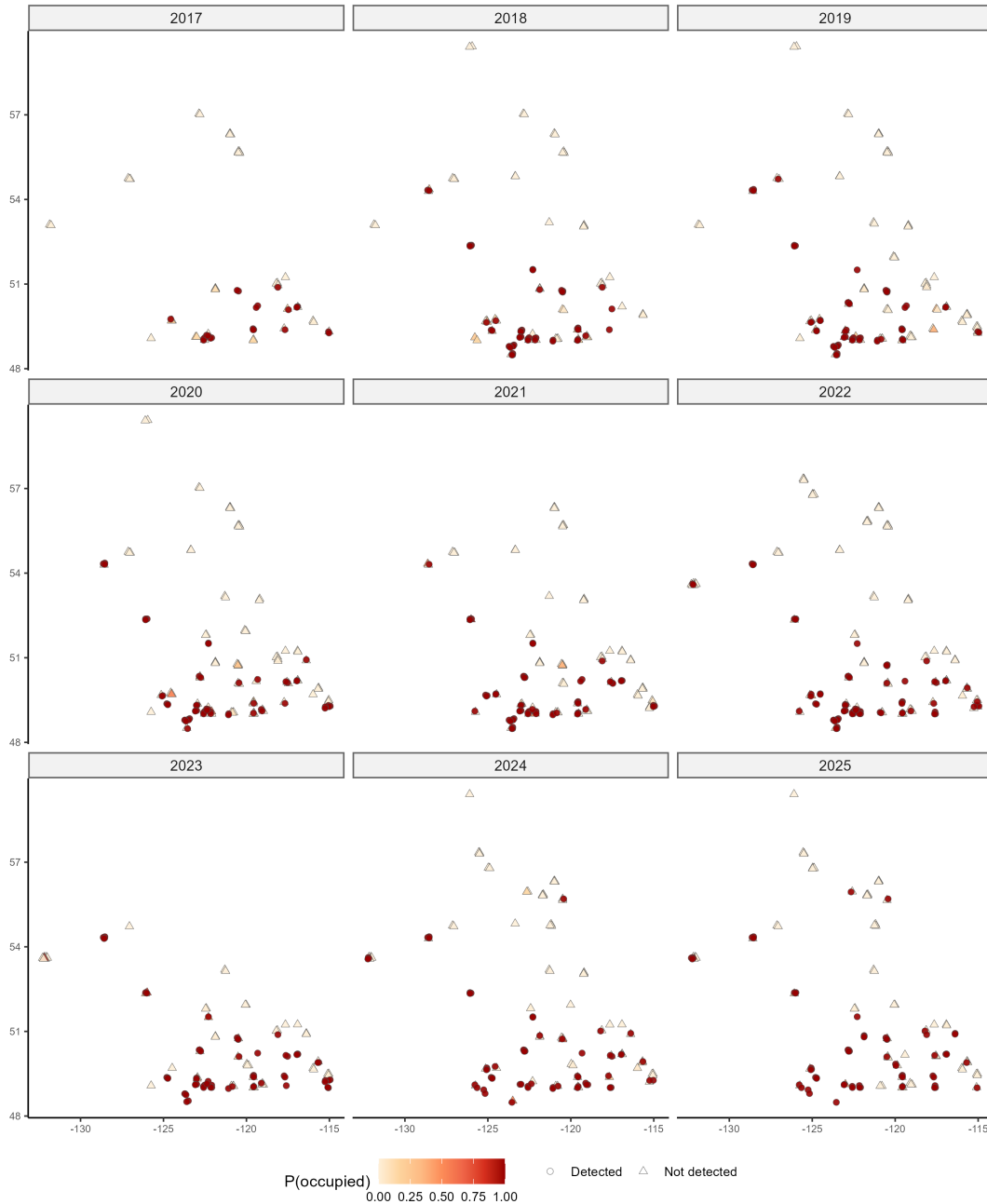
MYYU — Yuma myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



MYYU — Yuma myotis: P(occupancy) by site and year

Surveyed site-years only. Triangles = not detected that year; their colour is model inference.



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